BIRATIONALLY SUPERRIGID FANO VARIETIES WITH SMALL ALPHA INVARIANTS

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Abstract. We prove that there exist birationally superrigid Fano varieties with arbitrarily small alpha invariants, which provides another negative answer to a question of Stibitz and Zhuang whether the alpha invariant of a birationally superrigid Fano variety is always greater than $1/2$. Our examples are hypersurfaces in weighted projective spaces, among them one of dimension 15 whose alpha invariant is at most $1/3$. We also prove that a smooth quadric threefold admits a faithful action of a finite group of order 7200 such that the quadric is equivariantly birationally superrigid and has equivariant alpha invariant at most $1/3$, which answers an equivariant question of Cheltsov, Sarikyan, and Zhuang in the negative in dimension three.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

A Fano variety X is called *birationally superrigid* if X is terminal and $\mathbb{Q}$-factorial, the Picard number of X is one, and any birational map from X to a Mori fiber space is an isomorphism (see Definition 2.6). Let X be a Fano variety with terminal $\mathbb{Q}$-factorial singularities such that $\operatorname{rank}\operatorname{Pic}(X) = 1$, and assume that X is birationally rigid, that is, any birational map from X to a Mori fiber space $Y \to Z$ has Z a point and $Y \cong X$. Kim, Okada, and Won conjectured that X is K-stable, and that X admits a Kähler–Einstein metric if moreover X has only quotient singularities [KOW18, Conjecture 1.9]. We refer the reader to [Xu21] for the definitions of K-stability, K-semistability, and K-polystability. The Kim–Okada–Won conjecture is known for many classes of Fano varieties. See, for example, [Zhu20, KOW23, CO24, Oka26]. In [Liu26, Theorem 1.4 and Corollary 1.5], we constructed a birationally superrigid Fano variety of dimension 25 which is K-semistable but not K-stable. In particular, the Kim–Okada–Won conjecture does not hold in general [Liu26, Corollary 1.5].

The alpha invariant $\alpha(X)$ [Tia87] of a Fano variety X is the infimum of $\operatorname{lct}(X; D)$ over $\mathbb{Q}$-divisors $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$ (see Definition 2.2). It is closely related to the K-stability of birationally superrigid Fano varieties. By [SZ19, Theorem 1.2 and Corollary 3.2], a birationally

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superrigid Fano variety X with $\alpha(X) \geq \frac{1}{2}$ is K-stable. Thus a positive answer to the following question of Stibitz and Zhuang would imply that any birationally superrigid Fano variety is K-stable.

Question 1.1 ([SZ19, Question 1.5])**.** Let X be a birationally superrigid Fano variety. Is it true that $\alpha(X) > \frac{1}{2}$?

Cheltsov, Sarikyan, and Zhuang asked the following version of Question 1.1, with $\geq$ in place of $>$ and with birational superrigidity understood in the sense of [CSZ23, Section 1] (see the proof of Corollary 1.5).

Question 1.2 ([CSZ23, Question 1.3])**.** Is it true that $\alpha(X) \geq \frac{1}{2}$ for any birationally superrigid Fano variety X?

See also [KOW23, Question 7.4] and [DGO25, Question 81]. The variety X of [Liu26, Theorem 1.4] satisfies $\alpha(X) \leq \frac{1}{26}$, and therefore answers Question 1.1 and Question 1.2 in the negative [Liu26, Remark 1.6].

We now explain the known lower bounds for alpha invariants of birationally superrigid Fano varieties. By [SZ19, Theorem 1.6], if X is a birationally superrigid Fano variety such that $\dim X = n \geq 3$, $-K_X$ generates $\mathrm{Cl}(X)$, and $|-K_X|$ is base point free, then

$$\alpha(X) \geq \frac{1}{n+1}.$$

The assumptions that $-K_X$ generates $\mathrm{Cl}(X)$ and that $|-K_X|$ is base point free are satisfied by most known examples of birationally superrigid Fano varieties [SZ19, Section 1]. Kim, Okada, and Won showed that any quasi-smooth Fano 3-fold weighted hypersurface X of index one, that is, $a_0 + \cdots + a_4 - d = 1$ for $X = X_d \subset \mathbb{P}(a_0, \ldots, a_4)$, satisfies $\alpha(X) \geq \frac{1}{2}$ [KOW23, Theorem 1.8]. The birationally superrigid Fano varieties known at the time of [Xu21, Section 9] all satisfy $\alpha(X) > \frac{1}{2}$. By [CSZ23, Theorem 1.4], the bound $\frac{1}{2}$ in Question 1.2 cannot be improved: for any real number $\epsilon > 0$, there exists a birationally superrigid Fano variety X (in the sense of [CSZ23, Section 1], which here agrees with Definition 2.6, since a Fano variety with terminal singularities and $\operatorname{rank}\mathrm{Cl}(X) = 1$ is $\mathbb{Q}$-factorial of Picard number one, $\mathrm{Cl}(X) \otimes \mathbb{Q}$ being spanned by K_X, and the composition of a birational map from X to a Mori fiber space Y with an isomorphism $Y \cong X$ is an element of $\mathrm{Bir}(X) = \mathrm{Aut}(X)$) such that $\operatorname{rank}\mathrm{Cl}(X) = 1$ and

$$\frac{1}{2} \leq \alpha(X) \leq \frac{1}{2} + \epsilon.$$

The varieties in [CSZ23, Theorem 1.4] are quasi-smooth well-formed hypersurfaces of degree $2a+1$ in $\mathbb{P}(1^{a+2}, a)$ for integers $a \geq 2$ [CSZ23, Section 2], so by adjunction [IF00, 6.14] their index is $(a+2) + a - (2a+1) = 1$. Thus the varieties in [KOW23, Theorem 1.8] and [CSZ23, Theorem 1.4] are all of index one. In dimension three, a quasi-smooth Fano 3-fold weighted hypersurface with terminal singularities is birationally rigid if and only if it has index one [ACP21, Corollary 1.3].

We now turn to the equivariant setting. Let X be a Fano variety and $G \subset \mathrm{Aut}(X)$ a finite subgroup. We refer the reader to Definition 2.7 for the G-birational superrigidity of X, and to Definition 2.2 for the G-equivariant alpha invariant $\alpha_G(X)$, which is the infimum of $\mathrm{lct}(X; D)$ over G-invariant $\mathbb{Q}$-divisors $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$. If

$$\alpha_G(X) > \frac{\dim X}{\dim X + 1},$$

then X is G-equivariantly K-stable [OS12, Theorem 1.10] and K-polystable [Zhu21, Corollary 1.3], and X admits a Kähler–Einstein metric if moreover X is smooth [Tia87, Theorem 4.1]. Here G-equivariant K-stability is defined in [OS12, Section 2.2], and the invariant in the hypothesis of [OS12, Theorem 1.10] is $\alpha_G(X)$ by [OS12, Section 1, (2)]. Since G is finite, G is reductive and the G-alpha invariant of [Zhu21, Definition 2.4] equals $\alpha_G(X)$ (see the paragraph after Definition 2.2).

The alpha invariant in [Tia87, Theorem 4.1] is the analytic one, which equals $\alpha_G(X)$ for smooth X by [CS08, Appendix A]. Cheltsov, Sarikyan, and Zhuang asked the following equivariant version of Question 1.2.

Question 1.3 ([CSZ23, Question 1.5])**.** Is it true that $\alpha_G(X) \geq \frac{1}{2}$ for any Fano variety X and any finite subgroup $G \subset \mathrm{Aut}(X)$ such that X is G-birationally superrigid?

The following cases of Question 1.3 are known.

(1) When $\dim X = 2$, the answer is positive. More precisely, $\alpha_G(X) \geq \frac{2}{3}$ (see [CSZ23, Exercise 1.6] and the references therein).
(2) In high dimensions, the answer is negative, and $\alpha_G(X)$ can be arbitrarily small [CSZ23, Theorem 1.8]. In [CSZ23, Section 5], X is a complete intersection of Fermat hypersurfaces in a projective space, and $-K_X$ is $\mathbb{Q}$-linearly equivalent to c times the G-invariant divisor cut out on X by one more Fermat hypersurface, where c is a positive rational number. Thus $\alpha_G(X) \leq \frac{1}{c}$ [CSZ23, Section 5]. The G-birational superrigidity of X is proved in [CSZ23, Theorem 5.2] under the assumption that the degree of the Fermat hypersurfaces is large, so $\dim X$ is large.

In dimension three, many examples suggest a positive answer to Question 1.3 [CSZ23, Section 1]. For instance, if $G \subset \mathrm{Aut}(\mathbb{P}^3)$ is a finite subgroup and $\mathbb{P}^3$ is G-birationally superrigid, then $\alpha_G(\mathbb{P}^3) \geq \frac{1}{2}$ (see [CSZ23, Example 1.7] and the references therein). On the other hand, consider the smooth quadric threefold $(x_0^2 + \cdots + x_4^2 = 0) \subset \mathbb{P}^4$, with $\mathfrak{S}_5$ permuting the coordinates. Three times the $\mathfrak{S}_5$-invariant hyperplane section $(x_0 + \cdots + x_4 = 0)$ is an anti-canonical divisor of the quadric, so the $\mathfrak{S}_5$-equivariant alpha invariant of the quadric is at most $\frac{1}{3}$ [CSZ23, Section 3]. However, the quadric contains two $\mathfrak{S}_5$-orbits of length 5, and each of them leads to an $\mathfrak{S}_5$-birational map from the quadric to a cubic threefold with 5 nodes [CSZ23, Section 3], so the quadric is not $\mathfrak{S}_5$-birationally superrigid. For the equivariant birational geometry of smooth quadric threefolds with actions of $\mathfrak{A}_5$, we refer the reader to [PZ25]. For G-birationally rigid cubic threefolds and the G-birational rigidity of projective spaces, see [CKM26] and [CMS26], respectively.

In this paper, we construct another family of birationally superrigid Fano varieties which answer Question 1.1 and Question 1.2 in the negative, and we answer Question 1.3 in the negative in dimension three.

Theorem 1.4. *Let $r \geq 3$ and $k \geq r+2$ be integers, and let $b := r+1$ and $d := kr$. Let $X \subset \mathbb{P}(1, r, r, b^{d-1})$ be the hypersurface defined by*

$$x^{bd} + y_1^{kb} + y_2^{kb} + z_1^d + \cdots + z_{d-1}^d = 0,$$

where $x, y_1, y_2, z_1, \ldots, z_{d-1}$ are the coordinates of weights $1, r, r, b, \ldots, b$. Let H be the class of the divisorial sheaf $\mathcal{O}_X(1)$, and let $S := (x = 0) \subset X$. Then X is a birationally superrigid Fano variety of dimension d, $\mathrm{Cl}(X) = \mathbb{Z}[H]$, $-K_X \sim rH$, and

$$\alpha(X) \leq \mathrm{lct}(X; rS) = \frac{1}{r}.$$

Suppose that the class group of a Fano variety is $\mathbb{Z}$ and that the anti-canonical class is q times a generator. The case $q = 1$ is the index assumption of [SZ19, Theorem 1.6]. Then any prime divisor is linearly equivalent to a positive integral multiple of the generator, and intersecting with a power of the anti-canonical class shows that the coefficient of a prime divisor in any $\mathbb{Q}$-divisor $D \geq 0$ that is $\mathbb{Q}$-linearly equivalent to the anti-canonical class is at most q. Hence the coefficient of a single prime divisor can force $\alpha < \frac{1}{2}$ only when $q \geq 3$. In Theorem 1.4, $q = r \geq 3$ and the prime divisor is S.

Corollary 1.5. *The following hold.*

(1) The hypersurface $X_{60} \subset \mathbb{P}(1, 3, 3, 4^{14})$ defined by $x^{60} + y_1^{20} + y_2^{20} + z_1^{15} + \cdots + z_{14}^{15} = 0$, where $x, y_1, y_2, z_1, \ldots, z_{14}$ are the coordinates of weights $1, 3, 3, 4, \ldots, 4$, is a birationally superrigid Fano variety of dimension 15 such that $\alpha(X_{60}) \leq \frac{1}{3}$.

(2) *For any real number $\epsilon > 0$, there exists a birationally superrigid Fano variety X such that $\alpha(X) < \epsilon$.*

In particular, Questions 1.1 and 1.2 have negative answers.

Part (1) of Corollary 1.5 is the case $r = 3$ and $k = 5$ of Theorem 1.4, and part (2) follows by letting r tend to infinity. By [Bir21, Theorem 1.6], for any positive integer n there exists a positive real number t_n such that $\alpha(X) \geq t_n$ for any terminal Fano variety X of dimension n (the theorem, applied to the pair $(X, 0)$, all of whose log discrepancies are at least 1 since X is terminal, bounds $\mathrm{lct}(X; D)$ from below for any $\mathbb{R}$-divisor $D \geq 0$ with $D \sim_{\mathbb{R}} -K_X$, and in particular for the $\mathbb{Q}$-divisors in the definition of $\alpha(X)$), so the dimension of the variety X in Corollary 1.5(2) necessarily tends to infinity as ϵ tends to 0. Since $\dim X = kr$ in Theorem 1.4, taking $r = 3$ and $k = \frac{d}{3}$ shows that for any integer $d \geq 15$ divisible by 3 there exists a birationally superrigid Fano variety of dimension d whose alpha invariant is at most $\frac{1}{3}$.

Unlike the variety of dimension 25 in [Liu26, Theorem 1.4], the Fermat-type weighted hypersurfaces of Theorem 1.4 give $\alpha \leq \frac{1}{r}$ for any integer $r \geq 3$, with examples of dimension 15. Moreover, the proof of birational superrigidity in this paper does not involve K-stability, whereas the proof of [Liu26, Theorem 1.4(2)] bounds log canonical thresholds through the K-semistability of log Fano pairs on finite Galois covers of projective spaces [Liu26, Proposition 7.11].

Theorem 1.6. *Let $2.\mathfrak{A}_5 \subset \mathrm{SL}_2(\mathbb{C})$ be the preimage of an icosahedral subgroup $\mathfrak{A}_5 \subset \mathrm{PGL}_2(\mathbb{C})$, and let $Q \subset \mathbb{P}(\mathrm{Mat}_2(\mathbb{C}) \oplus \mathbb{C})$ be the quadric threefold defined by $\det(M) = t^2$. Let $G := (2.\mathfrak{A}_5 \times 2.\mathfrak{A}_5)/\langle(-1,-1)\rangle$ act on Q by $(g_1, g_2) \cdot [M : t] := [g_1 M g_2^{-1} : t]$, and let $S := Q \cap (t = 0)$. Then the action of G on Q is faithful, $|G| = 7200$, Q is G-birationally superrigid, and*

$$\alpha_G(Q) \leq \mathrm{lct}(Q; 3S) = \frac{1}{3}.$$

In particular, Question 1.3 has a negative answer in dimension three.

Sketch of the proof. We focus on Theorem 1.4. Since X is terminal and $\mathbb{Q}$-factorial with $\mathrm{Cl}(X) = \mathbb{Z}[H]$ and $-K_X \sim rH$ (Proposition 3.3(1)), the Noether–Fano criterion [SZ19, Theorem 2.3] and Lemma 2.4 reduce birational superrigidity to Theorem 5.1: $(X, \frac{r}{\ell}\mathcal{M})$ is canonical for any positive integer ℓ and any mobile linear system $\mathcal{M} \subset |\ell H|$ (Definition 2.3). The canonicity of $(X, \frac{r}{\ell}\mathcal{M})$ depends only on general members of $\mathcal{M}$, whereas the bound $\alpha(X) \leq \mathrm{lct}(X; rS) = \frac{1}{r}$ uses the special member $rS = (x^r = 0)$ of the mobile linear system $|rH|$, which is spanned by x^r, y_1, and y_2 (Lemma 3.5).

Let $C := (z_1 = \cdots = z_{d-1} = 0) \subset X$. The proof of Theorem 5.1 bounds $a(F, X, 0) - \frac{r}{\ell}\mathrm{mult}_F \mathcal{M}$ from below for exceptional prime divisors F over X in three steps.

(1) The bound 1 holds unless $\mathrm{center}_X F$ is a closed point p of C (Proposition 4.1; see Propositions 4.4, 4.5, 4.7, and Lemma 4.8).
(2) If $p \in C \backslash S$, then the bound is $\frac{d-2}{2} - \frac{r}{b} \geq \frac{23}{4}$ (Proposition 5.9).
(3) If p is one of the kb quotient points of $\Sigma = C \cap S$ (Proposition 3.3(2)), then the bound is $\frac{k-1}{r} - \frac{1}{b}$ (Proposition 5.10), which is greater than 1 if and only if $k \geq r + 2$, and equals $1 + \frac{1}{r(r+1)}$ if $k = r + 2$ ($\frac{13}{12}$ for X_{60}).

Step (1) uses the bounds on smooth hypersurfaces of Theorem 2.9 and inversion of adjunction (Theorem 2.8 and Lemma 2.10), and it uses $k \geq r + 2$ only through $kr \geq 3(r + 1)$. Steps (2) and (3) use a base change of the morphism π defined by $|rH|$ outside its base locus (Set-up 3.1), which turns the family of fibers of π near p into a product with the smooth Calabi–Yau hypersurface $Y_{d,-1}$, where

$$Y_{d,c} := (c\xi_0^d + \xi_1^d + \cdots + \xi_{d-1}^d = 0) \subset \mathbb{P}^{d-1}$$

for $c \in \mathbb{C}$ (Construction 4.2). Then Lemma 5.4 reduces both bounds to estimates for plane curves (Lemma 5.2).

(a) Canonicity on Fermat-type hypersurfaces in Step (1). The proof of Step (1) restricts to $V = (x = y_1 = y_2 = 0)$ and to the hypersurfaces $Y_{d,c}$, which appear as the closures of the fibers of

π in Proposition 4.7 and as covers of open subsets of the slices $S \cap (y_2 = ty_1)$ in Proposition 4.5. Step (1) needs canonicity, whereas [Che01, Theorem 1.3] bounds log canonical thresholds, and the canonicity statement [Che01, Theorem 7.2] is proved only for smooth hypersurfaces of degree N in $\mathbb{P}^N$. For $c \neq 0$, Lemma 4.3(1) shows that $(Y_{d,c}, \lambda\Gamma)$ is canonical for any positive integer q, any $\Gamma \in |\mathcal{O}_{Y_{d,c}}(q)|$, and any $0 \leq \lambda \leq \frac{d-3}{dq}$.

Along V, S is not Cartier. Let E_V be the exceptional divisor of the weighted blow-up of X along V with weights $\frac{1}{b}(1, r, r)$ on x, y_1, y_2. Then $a(E_V, X, 0) = 1 + \frac{r}{b}$ and $\operatorname{mult}_{E_V} S = \frac{1}{b}$, so for $D \in |\ell H|$, log canonicity of $(X, S + \frac{r}{\ell}D)$ only bounds $\frac{r}{\ell}\operatorname{mult}_{E_V} D$ by $\frac{2r}{b}$, while canonicity asks for $\frac{r}{b}$. Proposition 4.4 uses instead $S + (y_1 + \gamma y_2 = 0)$, which is Cartier near V and has multiplicity one along E_V, and keeps the leading coefficient of D along V as a member Γ of $|\mathcal{O}_V(q)|$, where $q \leq \frac{\ell}{b}$ is the degree of the equation of Γ in $z_1, \dots, z_{d-1}$, so that $\frac{r}{\ell}q \leq \frac{r}{b} \leq \frac{d-3}{d}$. Theorem 2.9(1) on the smooth hypersurface $V \subset \mathbb{P}^{d-2}$ then controls all points of V at once, including the zeros of the leading coefficient.

(b) A base change at the points of C. At a closed point $p \in C$, the hypersurfaces used in Step (1) degenerate to the cone $Y_{d,0}$, which is not lc at its vertex over p, so inversion of adjunction along them says nothing at p. The $4n^2$-inequality [Puk17, Section 1, Theorem and Remark 1(iii)], with $n = \frac{\ell}{r}$, does not exclude $p \in C\backslash S$ either, since the members of $|rH|$ through p form a mobile pencil whose self-intersection, the closure of the fiber of π through p, has multiplicity $d > 4$ at p.

For $p \in C\backslash S$, the fibers of π near p are the affine hypersurfaces $w_1^d + \cdots + w_{d-1}^d = -c$, where c is an equation of $\pi(C)$, so the fiber over t is the chart $(\xi_0 \neq 0)$ of $Y_{d,c(t)}$ with $\xi_j = w_j$ (Construction-Proposition 3.2). Over the cover $(h^d = -c)$ of a neighborhood of $\pi(p)$, the substitution $w_j = h\xi_j/\xi_0$ turns the family of fibers into the product of a smooth surface and $Y_{d,-1}$, and Lemma 5.7 computes $a(F, X, 0)$ on the product. Since $Y_{d,-1}$ is smooth and $c = -1$ satisfies the hypothesis $c \neq 0$ of Lemma 4.3(1), Lemma 5.4 moves the estimate to the surface, where the members of $\mathcal{M}$ are replaced by the ideal of their coefficients in the coordinates ξ_j.

(c) The factor 2 on the plane. For $p \in C\backslash S$, after replacing $\mathcal{M}$ with $\mathcal{M}^{[r]}$ if $r \nmid \ell$ (Set-up 5.5), which does not change $\frac{r}{\ell}\operatorname{mult}_F \mathcal{M}$, we have $\ell = qr$ for a positive integer q, unrelated to the q of (a), and the bound in (b) reads (see (5.8))

$$a(F, X, 0) - \frac{r}{\ell}\operatorname{mult}_F \mathcal{M} \geq a(v_F, \mathbb{P}^2, 0) - \frac{1}{q}v_F(I) - \Big(\frac{1}{d} + \frac{1}{kb}\Big)v_F(c),$$

where v_F is the pushforward of the valuation of F to $\mathbb{P}^2$, I is the ideal at $\pi(p)$ of a linear system of plane curves of degree at most q attached to $\mathcal{M}$ (Construction-Proposition 5.3), and the last term pays for the ramification of $h^d = -c$ and the powers of h in the coefficients.

The multiplicity bound for log canonical thresholds only implies $a(v_F, \mathbb{P}^2, 0) \geq \frac{1}{q}v_F(I)$. Since $\mathcal{M}$ is mobile, the plane linear system has no fixed component, and [dFEM04, Theorem 0.1] with Bézout's theorem implies $a(v_F, \mathbb{P}^2, 0) \geq \frac{2}{q}v_F(I)$ (Lemma 5.2). Since $\pi(C)$ is smooth, $a(v_F, \mathbb{P}^2, 0) \geq v_F(c)$, so at least $(\frac{1}{2} - \frac{1}{d} - \frac{1}{kb})v_F(c)$ is left, and $v_F(c) \geq d$, since $\pi^*c = -(w_1^d + \cdots + w_{d-1}^d)$ near p by (5.4) and each w_j vanishes at p.

(d) The quotient points. At $p \in \Sigma$, the base change also extracts an r-th root x of the equation w of the line $\pi(S)$, which costs $(1 - \frac{1}{r})v_F(w)$. Both inequalities of Lemma 5.2, the second one for the line $(w = 0)$, leave $(\frac{1}{r} - \frac{1}{d} - \frac{1}{kb})v_F(c)$, and $v_F(c) \geq k$, since $\pi^*c = -\sum_j (z_j^r/y_1^b)^k$ near p and each z_j^r/y_1^b vanishes at p (proof of Proposition 5.10). The powers of x in the coefficients cost a further $\frac{r-1}{rq}v_F(w)$, which for $\ell = r$ makes the bound negative when $v_F(w) \geq v_F(c)$. Since replacing $\mathcal{M}$ with $\mathcal{M}^{[N]}$ (Set-up 5.5) changes neither $\frac{r}{\ell}\operatorname{mult}_F \mathcal{M}$ nor v_F and replaces q with Nq, letting $N \to +\infty$ removes the term $\frac{r-1}{rq}v_F(w)$.

(e) The quadric threefold. By the equivariant Noether–Fano inequality [PZ25, Theorem 2.1] (cf. [CS16, Corollary 3.3.3], where G-Mori fiber spaces are those of [CS16, Definition 1.1.5], a narrower class than in Definition 2.7), the G-birational superrigidity in Theorem 1.6 follows

once $(Q, \frac{3}{\ell}\mathcal{M})$ is canonical for any G-invariant mobile linear system $\mathcal{M} \subset |\ell H|$, where H is a hyperplane section of Q. Multiplier ideals and the Nadel–Shokurov vanishing theorem [CS08, Theorem 2.6] only detect where a pair is not klt, so we prove that $(Q, \frac{6}{\ell}\mathcal{M})$ is klt (Proposition 6.5), which implies the canonicity since $a(E, Q, 0) \geq 2$ for any exceptional prime divisor E over Q.

A G-invariant curve Γ in the non-klt locus has degree at most 72, since a general member of $\mathcal{M}$ has multiplicity at least $\frac{\ell}{6}$ along it and two general members meet in a 1-cycle of degree $2\ell^2$ (see (6.2)). If $\Gamma \subset S \cong \mathbb{P}^1 \times \mathbb{P}^1$, then its class (e_1, e_2) satisfies $\max\{e_1, e_2\} \leq 6$, since $D|_S - \frac{\ell}{6}\Gamma \geq 0$ for a general $D \in \mathcal{M}$ and $D|_S \sim \mathcal{O}(\ell, \ell)$. If the multiplier ideal defines a zero-dimensional subscheme, its length is at most

$$h^0(Q, \mathcal{O}_Q(4H)) = h^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(4)) - h^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(2)) = 55.$$

By Lemma 6.4, a G-invariant curve with no component in S has degree at least 144, one contained in S satisfies $\max\{e_1, e_2\} \geq 12$, and any G-orbit has at least 120 points (compare the $\mathfrak{S}_5$-orbits of length 5 on the quadric threefold in [CSZ23, Section 3]).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce mobile linear systems and their singularities (Definition 2.3 and Lemma 2.4), the alpha invariant and the equivariant alpha invariant, birational superrigidity and G-birational superrigidity, inversion of adjunction (Theorem 2.8 and Lemma 2.10), lower bounds for log canonical thresholds and upper bounds for multiplicities along curves on smooth hypersurfaces (Theorem 2.9), and the pushforward of divisorial valuations (Definition 2.11). In Section 3 we study the geometry of the hypersurfaces in Theorem 1.4. Section 4 reduces the proof of Theorem 5.1 to closed points of the curve $C = (z_1 = \cdots = z_{d-1} = 0) \subset X$, which is isomorphic to the Fermat curve of degree kb in $\mathbb{P}^2$ (Construction-Proposition 3.2(3)). Section 5 treats the closed points of C and proves Theorem 5.1. In Section 6 we study the quadric threefold in Theorem 1.6. In Section 7 we prove the main theorems.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and thresholds.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c\operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. We say that a pair (X, B) is *canonical* if $a(E, X, B) \geq 1$ for any exceptional prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

For any prime divisor E over X, $a(E, X, B) - 1$ is the discrepancy of E with respect to (X, B) in the sense of [KM98, Definition 2.25] and [Kol13, Definition 2.4] (for real coefficients, [Kol13, 2.20]).

Definition 2.2. Let X be a normal projective variety such that K_X is $\mathbb{Q}$-Cartier and $(X,0)$ is klt, and let $D \geq 0$ be an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X. The *log canonical threshold* of D is

$$\operatorname{lct}(X;D) := \sup\{t \geq 0 \mid (X,tD) \text{ is lc}\}.$$

We say that X is a *Fano variety* if $-K_X$ is ample. If X is a Fano variety, then the *alpha invariant* of X is

$$\alpha(X) := \inf\{\operatorname{lct}(X;D) \mid D \geq 0 \text{ is a } \mathbb{Q}\text{-divisor such that } D \sim_{\mathbb{Q}} -K_X\}.$$

If moreover $G \subset \operatorname{Aut}(X)$ is a finite subgroup, then the *G-equivariant alpha invariant* of X is

$$\alpha_G(X) := \inf\{\operatorname{lct}(X;D) \mid D \geq 0 \text{ is a } G\text{-invariant } \mathbb{Q}\text{-divisor such that } D \sim_{\mathbb{Q}} -K_X\}.$$

The invariants $\alpha(X)$ and $\alpha_G(X)$ in Definition 2.2 coincide with the alpha invariant of [SZ19, Definition 1.1] and the G-equivariant alpha invariant of [CSZ23, Section 1], respectively. The invariant $\alpha_G(X)$ also coincides with the G-alpha invariant of [Zhu21, Definition 2.4], which is defined using G-invariant linear systems $\mathcal{M} \subset |-mK_X|$, whose members are $\mathbb{Q}$-Cartier. Indeed, for D as in Definition 2.2 and m sufficiently divisible, $\{mD\}$ is such a linear system, and conversely, for such $\mathcal{M}$ and $D_0 \in \mathcal{M}$, the $\mathbb{Q}$-divisor $D' := \frac{1}{m|G|}\sum_{g\in G} g^*D_0$ is G-invariant, $D' \sim_{\mathbb{Q}} -K_X$, and $\operatorname{mult}_E D' \geq \frac{1}{m}\operatorname{mult}_E \mathcal{M}$ (Definition 2.3) for any prime divisor E over X since $g^*D_0 \in \mathcal{M}$.

Definition 2.3. Let X be a normal $\mathbb{Q}$-factorial variety. A linear system $\mathcal{M}$ of Weil divisors on X is *mobile* if $\mathcal{M}$ has no fixed component. For a linear system $\mathcal{M}$ on X and a prime divisor F over X, we define

$$\operatorname{mult}_F \mathcal{M} := \min\{\operatorname{mult}_F D \mid D \in \mathcal{M}\},$$

where $\operatorname{mult}_F D$ is the coefficient of F in h^*D for a higher model $h : X' \to X$ on which F is a prime divisor. Here h^*D is defined since X is $\mathbb{Q}$-factorial. Let $\mathcal{M}$ be a mobile linear system on X and $\lambda \geq 0$ a real number. We say that $(X, \lambda\mathcal{M})$ is *canonical* if

$$a(F,X,0) - \lambda \operatorname{mult}_F \mathcal{M} \geq 1$$

for any exceptional prime divisor F over X. We say that $(X,\lambda\mathcal{M})$ is *klt* (resp. *lc*) if $a(F,X,0) - \lambda\operatorname{mult}_F\mathcal{M} > 0$ (resp. ≥ 0) for any prime divisor F over X.

For a fixed prime divisor F over X, the members D of $\mathcal{M}$ with $\operatorname{mult}_F D > \operatorname{mult}_F \mathcal{M}$ form a proper closed subset of $\mathcal{M}$ (a projective linear subspace, since pullback of $\mathbb{Q}$-Cartier Weil divisors is additive and the order of vanishing along F of a sum of two rational functions is at least the smaller of their orders), so $\operatorname{mult}_F \mathcal{M}$ is attained by a general member of $\mathcal{M}$. Thus whether $(X,\lambda\mathcal{M})$ is canonical, klt, or lc depends only on general members of $\mathcal{M}$ (Lemma 2.4).

Lemma 2.4. *Let X be a normal $\mathbb{Q}$-factorial variety, $\mathcal{M}$ a mobile linear system on X, and $\lambda \geq 0$ a real number. Then $(X,\lambda\mathcal{M})$ is canonical (resp. klt, lc) if and only if $(X, \frac{\lambda}{s}(D_1+\cdots+D_s))$ is canonical (resp. klt, lc) for any sufficiently large integer s and general members $D_1,\dots,D_s \in \mathcal{M}$.*

Proof. For any positive integer s and any $D_1,\dots,D_s \in \mathcal{M}$, let $B := \frac{\lambda}{s}(D_1+\cdots+D_s)$. Since X is $\mathbb{Q}$-factorial, (X,B) is a pair, and

$$a(F,X,B) = a(F,X,0) - \frac{\lambda}{s}\sum_{i=1}^{s}\operatorname{mult}_F D_i \leq a(F,X,0) - \lambda\operatorname{mult}_F\mathcal{M}$$

for any prime divisor F over X, because $\operatorname{mult}_F D_i \geq \operatorname{mult}_F \mathcal{M}$. Hence if (X,B) is canonical (resp. klt, lc), then so is $(X,\lambda\mathcal{M})$.

Conversely, assume that $(X,\lambda\mathcal{M})$ is canonical (resp. klt, lc). Let $h : Y \to X$ be a log resolution of $\mathcal{M}$ and $E_1,\dots,E_k$ the exceptional prime divisors of h. Thus Y is smooth, $E_1+\cdots+E_k$ is an snc divisor, the linear system $h_*^{-1}\mathcal{M}$ is base point free, and, since $\mathcal{M}$ has no fixed component,

$$h^*D = h_*^{-1}D + \sum_{j=1}^{k}(\operatorname{mult}_{E_j}\mathcal{M})E_j$$

for a general member $D \in \mathcal{M}$. Let $s \geq 2\lambda$ be a positive integer and $D_1, \dots, D_s \in \mathcal{M}$ general members. By Bertini's theorem, each $h_*^{-1}D_i$ is smooth, and $\sum_i h_*^{-1}D_i + \sum_j E_j$ is an snc divisor. We have $K_Y + \Delta_Y = h^*(K_X + B)$, where

$$\Delta_Y := \frac{\lambda}{s}\sum_{i=1}^{s} h_*^{-1}D_i + \sum_{j=1}^{k}\big(1 - a(E_j, X, 0) + \lambda \operatorname{mult}_{E_j} \mathcal{M}\big)E_j,$$

and $a(F, X, B) = a(F, Y, \Delta_Y)$ for any prime divisor F over X by [KM98, Lemma 2.30] (for real coefficients, [Kol13, 2.20]). Write $\Delta_Y = \sum_l \delta_l P_l$, where the P_l are distinct prime divisors. The coefficient of any component of $h_*^{-1}D_i$ is $\frac{\lambda}{s} \leq \frac{1}{2}$, and the coefficient of E_j is $1 - (a(E_j, X, 0) - \lambda \operatorname{mult}_{E_j} \mathcal{M})$, which is ≤ 0 (resp. < 1, ≤ 1) by assumption. In particular $\delta_l \leq 1$ for any l, so Δ_Y is an snc sub-boundary. By [Kol13, Corollary 2.11], which holds for real coefficients by [Kol13, 2.20],

$$a(F, Y, \Delta_Y) \geq 1 + \min\Big\{1,\ \min_l(1 - \delta_l),\ \min_{l \neq l', P_l \cap P_{l'} \neq \emptyset}(1 - \delta_l - \delta_{l'})\Big\}$$

for any exceptional prime divisor F over Y, and $a(P, Y, \Delta_Y) = 1 - \operatorname{mult}_P \Delta_Y$ for any prime divisor P on Y.

In the canonical case, $\delta_l \leq \frac{1}{2}$ for any l, so $a(F, Y, \Delta_Y) \geq 1$ for any exceptional prime divisor F over Y, and $a(E_j, X, B) = 1 - \operatorname{mult}_{E_j} \Delta_Y \geq 1$ for any j. Since any exceptional prime divisor over X is either some E_j or exceptional over Y, (X, B) is canonical. In the klt (resp. lc) case, $\delta_l < 1$ (resp. ≤ 1) for any l, so $1 - \delta_l > 0$ (resp. ≥ 0) and $1 - \delta_l - \delta_{l'} > -1$ (resp. ≥ -1). Hence $a(F, X, B) > 0$ (resp. ≥ 0) for any prime divisor F over X, and (X, B) is klt (resp. lc). □

For rational λ, Lemma 2.4 shows that Definition 2.3 agrees with the convention of [SZ19, Section 2.1] for the movable boundary $\lambda\mathcal{M}$.

Definition 2.5 ([CJL20, Section 2.3])**.** Let X be a normal variety of dimension d, $p \in X$ a closed point, e a positive integer, and $a_1, \dots, a_d \in \mathbb{Z}/e$. We say that X has a *cyclic quotient singularity of type* $\frac{1}{e}(a_1, \dots, a_d)$ at p if a neighborhood of p in X is analytically isomorphic to a neighborhood of the image of the origin in $\mathbb{A}^d/\mu_e$, where μ_e acts linearly on $\mathbb{A}^d$ with weights $a_1, \dots, a_d$.

2.2. **Birational superrigidity.**

Definition 2.6. A *Mori fiber space* is a contraction $f : Y \to Z$ between normal projective varieties such that Y is terminal and $\mathbb{Q}$-factorial, $\dim Z < \dim Y$, the relative Picard number of f is one, and $-K_Y$ is f-ample. A Fano variety X is *birationally superrigid* if X is terminal and $\mathbb{Q}$-factorial, the Picard number of X is one, and any birational map from X to a Mori fiber space is an isomorphism.

The notion of birational superrigidity in Definition 2.6 is that of [SZ19, Section 2.3], and the definition of Mori fiber spaces is the usual one.

Definition 2.7. Let G be a finite group. A *G-Mori fiber space* is a G-equivariant contraction $f : Y \to Z$ between normal projective varieties with G-actions such that Y is terminal, any G-invariant Weil divisor on Y is $\mathbb{Q}$-Cartier, $\dim Z < \dim Y$, the rank of $\mathrm{Cl}(Y)^G$ minus the rank of $\mathrm{Cl}(Z)^G$ is one, and $-K_Y$ is f-ample. Let X be a Fano variety with terminal singularities and $G \subset \mathrm{Aut}(X)$ a finite subgroup such that any G-invariant Weil divisor on X is $\mathbb{Q}$-Cartier and the rank of $\mathrm{Cl}(X)^G$ is one, so that $X \to \operatorname{Spec} \mathbb{C}$ is a G-Mori fiber space. We say that X is *G-birationally superrigid* if X is not G-birational to any G-Mori fiber space other than X, and any G-birational self-map of X is biregular.

The notion of G-birational superrigidity in Definition 2.7 is that of [CSZ23, Section 1], which refers to [CS16] for G-Mori fiber spaces. In [CS16, Definition 1.1.5], G-Mori fiber spaces are defined with any G-invariant Weil divisor on Y Cartier and with the rank condition on $\mathrm{Pic}(Y)^G$

and $\mathrm{Pic}(Z)^G$. Definition 2.7 relaxes Cartier to $\mathbb{Q}$-Cartier and Pic to Cl. The definition of G-Mori fiber spaces in Definition 2.7 is the usual one.

2.3. **Adjunction and thresholds on hypersurfaces.**

Theorem 2.8 ([Kaw07, Theorem])**.** *Let X be a smooth variety, $S \subset X$ a smooth prime divisor, and $B \geq 0$ an $\mathbb{R}$-divisor on X such that $S \not\subset \operatorname{Supp} B$. Then $(X, S + B)$ is lc near S if and only if $(S, B|_S)$ is lc.*

Theorem 2.8 is the case of [Kaw07, Theorem] where X and S are smooth (here the normalization of S is S, and the different of B on S is $B|_S$, since the identity of a general surface section of X can serve as the resolution in the definition of the different in [Kaw07, p. 129]). By Theorem 2.8 for an open neighborhood of a point of S, the corresponding statement holds near that point, and we use Theorem 2.8 near points of S without further comment.

Theorem 2.9 ([Che01, Theorem 1.3 and Statement 3.3])**.** *Let $N \geq 2$ and m be positive integers, $Y \subset \mathbb{P}^N$ a smooth hypersurface of degree m, and $B \geq 0$ an $\mathbb{R}$-divisor on Y such that $B \equiv qH_Y$ for some real number $q > 0$, where H_Y is a hyperplane section of Y. Then the following hold.*

(1) $\operatorname{lct}(Y; B) \geq \min\{\frac{N-1}{qm}, \frac{1}{q}\}$.

(2) $\operatorname{mult}_\Gamma B \leq q$ *for any curve* $\Gamma \subset Y$.

Canonicity asks for one more unit of log discrepancy than log canonicity. In Lemma 2.10, the additional unit is provided by the Cartier prime divisor S containing the center. Lemma 2.10 is used three times in Section 4.

Lemma 2.10. *Let X be a smooth variety, $S \subset X$ a smooth prime divisor, $T \subset S$ a smooth prime divisor on S, $p \in T$ a closed point, and $\lambda \geq 0$ a real number. Let $D \geq 0$ be a Cartier divisor on X such that $S \not\subset \operatorname{Supp} D$. Write $D|_S = mT + D_S$, where $m \geq 0$ is an integer and $D_S \geq 0$ is a Cartier divisor on S such that $T \not\subset \operatorname{Supp} D_S$. Assume that $\lambda m \leq 1$ and $(T, \lambda D_S|_T)$ is lc near p. Then $(X, S + \lambda D)$ is lc near p. In particular, $\lambda \operatorname{mult}_F D \leq a(F, X, 0) - 1$ for any prime divisor F over X such that $p \in \operatorname{center}_X F \subset S$.*

Proof. By Theorem 2.8 applied to S, T, and λD_S, $(S, T + \lambda D_S)$ is lc near p. For any prime divisor E over S,

$$a(E, S, \lambda mT + \lambda D_S) = a(E, S, T + \lambda D_S) + (1 - \lambda m) \operatorname{mult}_E T \geq a(E, S, T + \lambda D_S),$$

since $\lambda m \leq 1$ and $\operatorname{mult}_E T \geq 0$. Thus $(S, \lambda D|_S) = (S, \lambda mT + \lambda D_S)$ is lc near p. Since $S \not\subset \operatorname{Supp} D$, by Theorem 2.8 applied to X, S, and λD, $(X, S + \lambda D)$ is lc near p.

Let F be a prime divisor over X such that $p \in \operatorname{center}_X F \subset S$. Since S is a Cartier divisor containing $\operatorname{center}_X F$, $\operatorname{mult}_F S \geq 1$. Since $(X, S + \lambda D)$ is lc near p and $p \in \operatorname{center}_X F$,

$$0 \leq a(F, X, S + \lambda D) = a(F, X, 0) - \operatorname{mult}_F S - \lambda \operatorname{mult}_F D \leq a(F, X, 0) - 1 - \lambda \operatorname{mult}_F D.$$

□

2.4. **Valuations.** In Section 5 we push arbitrary divisorial valuations, not only monomial ones, forward to the plane $\mathbb{P}^2$ and from finite covers to their bases, and compare their log discrepancies there. Lemmas 2.12 and 2.13 prepare for this.

Definition 2.11. Let X be a normal variety and $v = c \operatorname{ord}_E$ a divisorial valuation over X, where E is a prime divisor over X and $c > 0$ is a real number. For a non-zero rational function φ on X, we denote by $v(\varphi)$ the value of v at φ. For an affine open subset $U \subset X$ meeting $\operatorname{center}_X E$ and a non-zero ideal $\mathfrak{a} \subset \mathcal{O}(U)$, or for a point $x \in \operatorname{center}_X E$ and a non-zero ideal $\mathfrak{a} \subset \mathcal{O}_{X,x}$, we define

$$v(\mathfrak{a}) := \min\{v(\varphi) \mid \varphi \in \mathfrak{a}\}.$$

For an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on X, we define $v(D) := c \operatorname{mult}_E D$. For a dominant rational map $g : X \dashrightarrow T$ to a normal variety T, we define the *pushforward* g_*v of v by $g_*v(\varphi) := v(g^*\varphi)$ for any non-zero rational function φ on T.

Lemma 2.12. *Let $g : X' \dashrightarrow T$ be a dominant rational map between normal varieties and v' a divisorial valuation over X'. Assume that g_*v' is non-trivial and has a center on T. Then g_*v' is a divisorial valuation over T.*

Proof. Let $v := g_*v'$. Since g is dominant, v is a valuation on the non-zero rational functions on T, the value group of v is a subgroup of the value group of v', and the residue field of v is a subfield of the residue field of v'. Since v' is a divisorial valuation over X', the value group of v' is $c\mathbb{Z}$ for some real number $c > 0$, and the residue field of v' has transcendence degree $\dim X' - 1$ over $\mathbb{C}$. By [ZS60, Chapter VI, Section 6, Lemma 2, Corollary 1], the transcendence degree of the residue field of v' over the residue field of v is at most $\dim X' - \dim T$. Hence the residue field of v has transcendence degree at least $\dim T - 1$ over $\mathbb{C}$.

Let $V \subset T$ be the center of v on T and R the local ring of T at the generic point of V, so that $\dim R = \dim T - \dim V$ and the residue field of R has transcendence degree $\dim V$ over $\mathbb{C}$. Let d be the transcendence degree of the residue field of v over the residue field of R. Since v is non-trivial, the rational rank of v is at least one, so $d \leq \dim R - 1$ by [Abh56, Theorem 1(1)]. Thus the residue field of v has transcendence degree $d + \dim V \leq \dim T - 1$ over $\mathbb{C}$. Therefore the residue field of v has transcendence degree $\dim T - 1$ over $\mathbb{C}$, $d = \dim R - 1$, and v is real discrete by [Abh56, Theorem 1(4)]. More precisely, the value group of v is a non-zero subgroup of $c\mathbb{Z}$, hence equal to $c_0\mathbb{Z}$ for some real number $c_0 > 0$, and the valuation ring of v is a discrete valuation ring whose residue field has transcendence degree $\dim T - 1$ over $\mathbb{C}$. Since v has a center on T, [KM98, Lemma 2.45] implies that v is a divisorial valuation over T. □

Lemma 2.13. *Let $g : X' \to X$ be a finite surjective morphism between normal varieties and B an $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier. Let B' be the $\mathbb{R}$-divisor on X' such that $K_{X'} + B' = g^*(K_X + B)$. Then for any divisorial valuation v' over X', g_*v' is a divisorial valuation over X, and*

$$a(v', X', B') = a(g_*v', X, B).$$

Proof. Let $v := g_*v'$. Since v' has a center on X' and g is a finite morphism, g maps the center of v' on X' onto the center of v on X. Since g is finite, $\dim X' = \dim X$, and by [ZS60, Chapter VI, Section 6, Lemma 2, Corollary 1], the residue field of v' is algebraic over the residue field of v. Hence the residue field of v has transcendence degree $\dim X - 1 < \dim X$ over $\mathbb{C}$, so v is non-trivial. By Lemma 2.12, v is a divisorial valuation over X.

By Definition 2.11, there exist a prime divisor E over X, a prime divisor E' over X', and real numbers $c, c' > 0$ such that $v(D) = c \operatorname{mult}_E D$ for any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on X and $v'(D') = c' \operatorname{mult}_{E'} D'$ for any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D' on X'. Then $a(v, X, B) = ca(E, X, B)$ and $a(v', X', B') = c'a(E', X', B')$. Let $h : Y \to X$ be a higher model such that E is a prime divisor on Y, let Y' be the normalization of the irreducible component of $Y \times_X X'$ that is birational to X', and let $g_Y : Y' \to Y$ and $h' : Y' \to X'$ be the induced morphisms. Then g_Y is finite and surjective, h' is a higher model, and $g \circ h' = h \circ g_Y$. Since h' is proper, v' has a center P' on Y', and g_Y maps P' onto the center E of v on Y. Since g_Y is finite, $\dim P' = \dim E$, so P' is a prime divisor on Y'. Since Y' is normal, the local ring of Y' at the generic point of P' is a discrete valuation ring dominated by the valuation ring of v', hence equal to it. Thus P' is the prime divisor E' over X'.

Let e be the ramification index of g_Y along E', so that $g_Y^*E = eE'$ near the generic point of E'. Evaluating v and v' on a local equation of E near the generic point of E implies $c = ec'$. Let $K_Y + B_Y := h^*(K_X + B)$ and $K_{Y'} + B_{Y'} := h'^*(K_{X'} + B')$. Since $g \circ h' = h \circ g_Y$,

$$K_{Y'} + B_{Y'} = h'^*g^*(K_X + B) = g_Y^*h^*(K_X + B) = g_Y^*(K_Y + B_Y).$$

By the Hurwitz formula, $K_{Y'} = g_Y^*K_Y + (e-1)E'$ near the generic point of E'. Hence $\operatorname{mult}_{E'} B_{Y'} = e \operatorname{mult}_E B_Y - (e-1)$, and

$$a(v', X', B') = c'(1 - \operatorname{mult}_{E'} B_{Y'}) = c'e(1 - \operatorname{mult}_E B_Y) = ca(E, X, B) = a(g_*v', X, B).$$

□

3. Fermat-type weighted hypersurfaces

In this section, we fix Set-up 3.1 and prove the basic properties of the hypersurface X.

Set-up 3.1. Let $r \geq 3$ and $k \geq r+2$ be integers, and let $b := r+1$ and $d := kr$. Let $X \subset \mathbb{P}(1, r, r, b^{d-1})$ be the hypersurface defined by

$$x^{bd} + y_1^{kb} + y_2^{kb} + z_1^d + \cdots + z_{d-1}^d = 0,$$

where $x, y_1, y_2, z_1, \ldots, z_{d-1}$ are the coordinates of weights $1, r, r, b, \ldots, b$. Let H be the class of the divisorial sheaf $\mathcal{O}_X(1)$, and let

$$S := (x = 0), \quad V := (x = y_1 = y_2 = 0), \quad C := (z_1 = \cdots = z_{d-1} = 0), \quad \Sigma := S \cap C$$

be the corresponding closed subsets of X. Let $\pi : X \backslash V \to \mathbb{P}^2$ be the morphism defined by $\pi([x : y_1 : y_2 : z_1 : \cdots : z_{d-1}]) := [x^r : y_1 : y_2]$, where $[w : y_1 : y_2]$ are the homogeneous coordinates of $\mathbb{P}^2$, and let $C_{\mathbb{P}^2} := (w^{kb} + y_1^{kb} + y_2^{kb} = 0) \subset \mathbb{P}^2$. Since $k \geq r+2$ and $r \geq 3$, we have $d = kr \geq r^2 + 2r \geq 3b$, and hence

$$\frac{r}{b} = 1 - \frac{1}{b} \leq 1 - \frac{3}{d} = \frac{d-3}{d}.$$

By Lemma 3.5, the linear system $|rH|$ is spanned by x^r, y_1, and y_2, and π is the morphism defined by $|rH|$ outside its base locus V, with $\pi^* w = x^r$ and $\pi^* y_i = y_i$ for $i = 1, 2$. In particular, the member of $|rH|$ corresponding to the line $(w = 0)$ is $\pi^*(w = 0) = rS$. On $X \backslash S$, the fiber of π over $[1 : u_1 : u_2]$ is the affine Fermat hypersurface $w_1^d + \cdots + w_{d-1}^d = -(1 + u_1^{kb} + u_2^{kb})$ in the coordinates of Construction-Proposition 3.2(1), and it is a cone with vertex $(u_1, u_2, 0, \ldots, 0) \in C$ when $[1 : u_1 : u_2] \in C_{\mathbb{P}^2}$.

Construction-Proposition 3.2. Notation and conditions as in Set-up 3.1. Let U_1 be the open subset $(y_1 \neq 0)$ of X, let

$$\widetilde{U}_1 := (x^{bd} + 1 + y_2^{kb} + z_1^d + \cdots + z_{d-1}^d = 0) \subset \mathbb{A}^{d+1},$$

and let μ_r act on $\widetilde{U}_1$ by

$$\zeta \cdot (x, y_2, z_1, \ldots, z_{d-1}) := (\zeta x, y_2, \zeta z_1, \ldots, \zeta z_{d-1}).$$

For $1 \leq s \leq d-1$, let W_s be the open subset $(z_s \neq 0)$ of X, let

$$\widetilde{W}_s := \Big(x^{bd} + y_1^{kb} + y_2^{kb} + 1 + \sum_{j \neq s} z_j^d = 0\Big) \subset \mathbb{A}^{d+1},$$

and let μ_b act on $\widetilde{W}_s$ with weights $1, r, r$ on x, y_1, y_2 and weight 0 on z_j for $j \neq s$. Then the following hold.

(1) The open subset $X \backslash S$ is isomorphic to the smooth affine hypersurface
$$(1 + u_1^{kb} + u_2^{kb} + w_1^d + \cdots + w_{d-1}^d = 0) \subset \mathbb{A}^{d+1},$$
where $u_i = y_i / x^r$ $(i = 1, 2)$ and $w_j = z_j / x^b$ $(1 \leq j \leq d-1)$; on $X \backslash S$ we have $\pi(u_1, u_2, w_1, \ldots, w_{d-1}) = [1 : u_1 : u_2]$.

(2) The curve C is disjoint from V, and X is smooth at any point of $C \backslash \Sigma$.

(3) The morphism π restricts to an isomorphism from C onto $C_{\mathbb{P}^2}$.

(4) The curve $C_{\mathbb{P}^2}$ is smooth and intersects the line $(w = 0)$ transversally at the kb points of $\pi(\Sigma)$.

(5) The open subset U_1 is isomorphic to $\widetilde{U}_1 / \mu_r$, and $x, z_1, \ldots, z_{d-1}$ form a regular system of parameters of $\widetilde{U}_1$ at any point of $\widetilde{U}_1$ over Σ.

(6) For $1 \leq s \leq d-1$, the open subset W_s is isomorphic to $\widetilde{W}_s / \mu_b$, the variety $\widetilde{W}_s$ is smooth, and a non-trivial element of μ_b fixes only the points of $\widetilde{W}_s$ with $x = y_1 = y_2 = 0$.

(7) For any integer ℓ, the restriction of $\mathcal{O}_X(\ell)$ to $X \backslash (V \cup \Sigma)$ is an invertible sheaf isomorphic to the restriction of $\mathcal{O}_X(\ell H)$.

Proof. (1) Since x has weight 1, the open subset $(x \neq 0) \subset \mathbb{P}(1, r, r, b^{d-1})$ is isomorphic to $\mathbb{A}^{d+1}$ with coordinates $u_i = y_i/x^r$ for $i = 1, 2$ and $w_j = z_j/x^b$ for $1 \leq j \leq d-1$. Dividing the defining equation of X by x^{bd}, we see that $X \backslash S$ is the hypersurface in the statement of (1). The partial derivatives $kb \cdot u_i^{kb-1}$ and $d \cdot w_j^{d-1}$ vanish simultaneously only at the origin, which is not a point of $X \backslash S$. Thus $X \backslash S$ is smooth. On $X \backslash S$, we have $\pi = [x^r : y_1 : y_2] = [1 : u_1 : u_2]$.

(2) The coordinates x, y_1, y_2 vanish on V, and the coordinates $z_1, \ldots, z_{d-1}$ vanish on C. Since the coordinates of a point of $\mathbb{P}(1, r, r, b^{d-1})$ do not vanish simultaneously, $C \cap V = \emptyset$. Since $\Sigma = S \cap C$, we have $C \backslash \Sigma \subset X \backslash S$, and $X \backslash S$ is smooth by (1).

(3) Let $G := x^{bd} + y_1^{kb} + y_2^{kb}$ and $A := \mathbb{C}[x, y_1, y_2]/(G)$, graded by the weights $1, r, r$. Then the closed subscheme $X \cap (z_1 = \cdots = z_{d-1} = 0)$ of X is $\operatorname{Proj} A$. A monomial $x^a y_1^{\beta_1} y_2^{\beta_2}$ has weighted degree divisible by r if and only if $r \mid a$, so the r-th Veronese subring of $\mathbb{C}[x, y_1, y_2]$ is the polynomial ring $\mathbb{C}[x^r, y_1, y_2]$. Since $bd = r \cdot kb$, we have $G = (x^r)^{kb} + y_1^{kb} + y_2^{kb}$, and the r-th Veronese subring of A is $A^{(r)} = \mathbb{C}[x^r, y_1, y_2]/(G)$. By [Gro61, Proposition (2.4.7)], $\operatorname{Proj} A \cong \operatorname{Proj} A^{(r)}$. After dividing the degrees by r and writing $w = x^r$, $\operatorname{Proj} A^{(r)}$ is $C_{\mathbb{P}^2}$, and the isomorphism $\operatorname{Proj} A \to C_{\mathbb{P}^2}$ is $[x : y_1 : y_2] \to [x^r : y_1 : y_2]$, which is the restriction of π. The partial derivatives of $w^{kb} + y_1^{kb} + y_2^{kb}$ vanish simultaneously only at the origin, so $C_{\mathbb{P}^2}$ is smooth. Thus the scheme $X \cap (z_1 = \cdots = z_{d-1} = 0)$ is reduced, it is equal to C, and $\pi|_C : C \to C_{\mathbb{P}^2}$ is an isomorphism.

(4) We have shown that $C_{\mathbb{P}^2}$ is smooth. The restriction of $w^{kb} + y_1^{kb} + y_2^{kb}$ to the line $(w = 0)$ is $y_1^{kb} + y_2^{kb}$, which has kb simple zeros $[0 : 1 : t]$ with $t^{kb} = -1$. Thus $C_{\mathbb{P}^2}$ intersects $(w = 0)$ transversally at kb points. Since $\pi^* w = x^r$, we have $C \cap \pi^{-1}(w = 0) = C \cap S = \Sigma$, and by (3), $\pi(\Sigma) = C_{\mathbb{P}^2} \cap (w = 0)$.

(5) Let $F := x^{bd} + y_1^{kb} + y_2^{kb} + z_1^d + \cdots + z_{d-1}^d$. The coordinate ring of U_1 is the degree 0 part of $(\mathbb{C}[x, y_1, y_2, z_1, \ldots, z_{d-1}]/(F))[y_1^{-1}]$. For a weighted homogeneous polynomial f of degree rm, consider the map

$$f/y_1^m \to f(x, 1, y_2, z_1, \ldots, z_{d-1}).$$

It sends monomials as follows:

$$x^a y_1^{\beta_1} y_2^{\beta_2} z_1^{\gamma_1} \cdots z_{d-1}^{\gamma_{d-1}} \to x^a y_2^{\beta_2} z_1^{\gamma_1} \cdots z_{d-1}^{\gamma_{d-1}},$$

and $\zeta \in \mu_r$ acts on the image by $\zeta^{a+\gamma_1+\cdots+\gamma_{d-1}}$. Since $b \equiv 1 \pmod r$, we have

$$a + \gamma_1 + \cdots + \gamma_{d-1} \equiv a + r\beta_2 + b(\gamma_1 + \cdots + \gamma_{d-1}) \pmod r,$$

so a monomial in $x, y_2, z_1, \ldots, z_{d-1}$ is μ_r-invariant if and only if the weighted degree of the monomial is divisible by r. Thus the map is an isomorphism from the degree 0 part of $\mathbb{C}[x, y_1, y_2, z_1, \ldots, z_{d-1}][y_1^{-1}]$ onto the μ_r-invariant part of $\mathbb{C}[x, y_2, z_1, \ldots, z_{d-1}]$, and it sends F/y_1^{kb} to the μ_r-invariant polynomial $F(x, 1, y_2, z_1, \ldots, z_{d-1})$ defining $\widetilde{U}_1$. Since taking μ_r-invariants is exact, the map induces an isomorphism from the coordinate ring of U_1 onto the μ_r-invariant part of the coordinate ring of $\widetilde{U}_1$. Hence $U_1 \cong \widetilde{U}_1/\mu_r$, where the quotient map sends $(x, y_2, z_1, \ldots, z_{d-1})$ to $[x : 1 : y_2 : z_1 : \cdots : z_{d-1}]$.

Let $\tilde{p} \in \widetilde{U}_1$ be a point over Σ. Then $x(\tilde{p}) = z_1(\tilde{p}) = \cdots = z_{d-1}(\tilde{p}) = 0$, so $y_2(\tilde{p})^{kb} = -1$. Thus the partial derivative kby_2^{kb-1} of $F(x, 1, y_2, z_1, \ldots, z_{d-1})$ with respect to y_2 is non-zero at $\tilde{p}$. By the implicit function theorem, $x, z_1, \ldots, z_{d-1}$ form a regular system of parameters of $\widetilde{U}_1$ at $\tilde{p}$.

(6) A monomial in x, y_1, y_2 and z_j for $j \neq s$ is μ_b-invariant if and only if the weighted degree of the monomial is divisible by b. Since taking μ_b-invariants is exact, the map $f/z_s^m \to f|_{z_s=1}$ for weighted homogeneous polynomials f of degree bm is an isomorphism from the coordinate ring of W_s onto the μ_b-invariant part of the coordinate ring of $\widetilde{W}_s$. Thus $W_s \cong \widetilde{W}_s/\mu_b$. The partial derivatives of the equation of $\widetilde{W}_s$ vanish simultaneously only at the origin of $\mathbb{A}^{d+1}$, which is not a point of $\widetilde{W}_s$, so $\widetilde{W}_s$ is smooth ($\widetilde{W}_s$ is irreducible, since the polynomial defining $\widetilde{W}_s$ is irreducible by Eisenstein's criterion in the variable z_j for some $j \neq s$). Since $\gcd(r, b) = 1$, any

$\zeta \in \mu_b \backslash \{1\}$ satisfies $\zeta \neq 1$ and $\zeta^r \neq 1$, so the fixed points of ζ in $\widetilde{W}_s$ are exactly the points of $\widetilde{W}_s$ with $x = y_1 = y_2 = 0$.

(7) Let $U := X \backslash (V \cup \Sigma)$. A point of U satisfies $x \neq 0$, or satisfies $y_i \neq 0$ and $z_j \neq 0$ for some i and j. Thus U is covered by the open subsets $(x \neq 0)$ and $(y_i z_j \neq 0)$ of X, and on them the homogeneous elements x and z_j/y_i of degree 1 (note that $b - r = 1$) are invertible. Hence $\mathcal{O}_X(\ell)$ is free on $(x \neq 0)$ and on $(y_i z_j \neq 0)$, generated by x^ℓ and $(z_j/y_i)^\ell$, respectively, so $\mathcal{O}_X(\ell)|_U = (\mathcal{O}_X(1)|_U)^{\otimes \ell}$ is invertible for any integer ℓ, as for the open subset of the weighted projective space in [Dol82, 1.5.5]. Since H is the class of $\mathcal{O}_X(1)$, we have $\mathcal{O}_X(\ell)|_U \cong \mathcal{O}_U(\ell H)$. □

Proposition 3.3. *Notation and conditions as in Set-up 3.1. Then the following hold.*

(1) X is a normal projective variety of dimension d with only cyclic quotient singularities. Moreover, X is terminal and $\mathbb{Q}$-factorial, $\mathrm{Cl}(X) = \mathbb{Z}[H]$, and $-K_X \sim rH$. In particular, X is a Fano variety of Picard number one.

(2) The singular locus of X is $V \cup \Sigma$. The subvariety V is isomorphic to the smooth hypersurface $(z_1^d + \cdots + z_{d-1}^d = 0) \subset \mathbb{P}^{d-2}$, and X has a cyclic quotient singularity of type $\frac{1}{b}(1, r, r, 0, \ldots, 0)$ at each point of V. The set Σ consists of the kb points $[0 : 1 : t : 0 : \cdots : 0]$ with $t^{kb} = -1$, and X has a cyclic quotient singularity of type $\frac{1}{r}(1, \ldots, 1)$ at each point of Σ.

(3) S is a prime divisor, $S \sim H$, (X, S) is plt, and $\mathrm{lct}(X; rS) = \frac{1}{r}$.

Proof. We first describe V and Σ. Setting $x = y_1 = y_2 = 0$, we see that V is the hypersurface $(z_1^d + \cdots + z_{d-1}^d = 0) \subset \mathbb{P}(b^{d-1})$. Since all the weights of $\mathbb{P}(b^{d-1})$ are equal to b, [Gro61, Proposition (2.4.7)] implies that $\mathbb{P}(b^{d-1}) \cong \mathbb{P}^{d-2}$ with homogeneous coordinates $z_1, \ldots, z_{d-1}$. The partial derivatives $d \cdot z_j^{d-1}$ vanish simultaneously only at the origin, so V is isomorphic to the smooth hypersurface $(z_1^d + \cdots + z_{d-1}^d = 0) \subset \mathbb{P}^{d-2}$, and $\dim V = d - 3$. By Construction-Proposition 3.2(3)(4), $\pi|_C$ is injective and $\pi(\Sigma)$ is the set of the kb points $[0 : 1 : t]$ with $t^{kb} = -1$. Since $[0 : 1 : t : 0 : \cdots : 0] \in C$ and $\pi([0 : 1 : t : 0 : \cdots : 0]) = [0 : 1 : t]$, Σ consists of the kb points $[0 : 1 : t : 0 : \cdots : 0]$ with $t^{kb} = -1$.

Recall that X is *quasismooth* if the affine cone over X is smooth outside the vertex, and X is *well formed* if the greatest common divisor of any $d + 1$ of the weights $1, r, r, b, \ldots, b$ is 1 and $X \cap \mathrm{Sing}\, \mathbb{P}(1, r, r, b^{d-1})$ has codimension at least 2 in X, see [PS20, Section 2]. The partial derivatives of the defining polynomial of X vanish simultaneously only at the origin, so X is quasismooth. Since $\gcd(r, b) = 1$ and each of r and b is the weight of at least two coordinates, any $d + 1$ of the weights contain 1 or contain both r and b, and hence have greatest common divisor 1. For $1 \leq s \leq d - 1$, the open subsets $(x \neq 0)$, $(y_1 \neq 0)$, and $(z_s \neq 0)$ of $\mathbb{P}(1, r, r, b^{d-1})$ are isomorphic to $\mathbb{A}^{d+1}$, to the quotient of $\mathbb{A}^{d+1}$ by μ_r with weights $1, 0, 1, \ldots, 1$ on $x, y_2, z_1, \ldots, z_{d-1}$, and to the quotient of $\mathbb{A}^{d+1}$ by μ_b with weights $1, r, r, 0, \ldots, 0$ on x, y_1, y_2 and z_j for $j \neq s$, respectively [IF00, 5.3]. Since $\gcd(r, b) = 1$, a non-trivial element of μ_r fixes only points with $x = z_1 = \cdots = z_{d-1} = 0$, and a non-trivial element of μ_b fixes only points with $x = y_1 = y_2 = 0$. Together with the symmetry between y_1 and y_2, this implies that

$$\mathrm{Sing}\, \mathbb{P}(1, r, r, b^{d-1}) \subset (x = z_1 = \cdots = z_{d-1} = 0) \cup (x = y_1 = y_2 = 0),$$

and so $X \cap \mathrm{Sing}\, \mathbb{P}(1, r, r, b^{d-1}) \subset \Sigma \cup V$. Since $\dim V = d - 3$ and Σ is finite, X is well formed. By [PS20, Lemma 2.2, Proposition 2.10, Theorem 2.12, Theorem 2.15], X is irreducible and normal with only quotient singularities,

$$\omega_X \cong \mathcal{O}_X(bd - 1 - 2r - (d - 1)b) = \mathcal{O}_X(-r),$$

and $\mathrm{Cl}(X) = \mathbb{Z}[H]$ since $d \geq 3$.

Next we describe the local structure of X. The partial derivatives of the equation of $\widetilde{U}_1$ vanish simultaneously only at the origin of $\mathbb{A}^{d+1}$, which is not a point of $\widetilde{U}_1$, so $\widetilde{U}_1$ is smooth. For $1 \leq s \leq d - 1$, $W_s \cong \widetilde{W}_s/\mu_b$ and $\widetilde{W}_s$ is smooth by Construction-Proposition 3.2(6).

Let $p \in X\backslash(V \cup \Sigma)$. If $p \notin S$, then X is smooth at p by Construction-Proposition 3.2(1). If $p \in S$, then $y_i(p) \neq 0$ and $z_j(p) \neq 0$ for some i and j, and by the symmetry between y_1 and y_2, we may assume that $i = 1$. By Construction-Proposition 3.2(5), $U_1 \cong \widetilde{U}_1/\mu_r$, and a preimage $\tilde{p} \in \widetilde{U}_1$ of p satisfies $z_j(\tilde{p}) \neq 0$. Thus no non-trivial element of μ_r fixes $\tilde{p}$, the quotient map $\widetilde{U}_1 \to U_1$ is étale at $\tilde{p}$, and X is smooth at p.

Let $p \in \Sigma$. By Construction-Proposition 3.2(5), p has a unique preimage $\tilde{p} \in \widetilde{U}_1$, namely $\tilde{p} = (0, t, 0, \dots, 0)$, and $x, z_1, \dots, z_{d-1}$ form a regular system of parameters of $\widetilde{U}_1$ at $\tilde{p}$, on which μ_r acts linearly with weights $1, \dots, 1$. By the inverse function theorem, the μ_r-equivariant map $\widetilde{U}_1 \to \mathbb{A}^d$ given by $(x, z_1, \dots, z_{d-1})$ is biholomorphic on a μ_r-invariant neighborhood of $\tilde{p}$. Taking quotients, X has a cyclic quotient singularity of type $\frac{1}{r}(1, \dots, 1)$ at p in the sense of Definition 2.5.

Let $p \in V$, and choose s such that $z_s(p) \neq 0$. Then p has a unique preimage $\tilde{p} \in \widetilde{W}_s$, and $x(\tilde{p}) = y_1(\tilde{p}) = y_2(\tilde{p}) = 0$. Since $\sum_{j \neq s} z_j(\tilde{p})^d = -1$, there exists $j_0 \neq s$ such that $z_{j_0}(\tilde{p}) \neq 0$, so the partial derivative $d \cdot z_{j_0}^{d-1}$ of the equation of $\widetilde{W}_s$ is non-zero at $\tilde{p}$. By the implicit function theorem, x, y_1, y_2 and the $d-3$ functions $z_j - z_j(\tilde{p})$ for $j \notin \{s, j_0\}$ form a regular system of parameters of $\widetilde{W}_s$ at $\tilde{p}$, on which μ_b acts linearly with weights $1, r, r, 0, \dots, 0$. By the inverse function theorem, the μ_b-equivariant coordinate map $\widetilde{W}_s \to \mathbb{A}^d$ is biholomorphic on a μ_b-invariant neighborhood of $\tilde{p}$. Taking quotients, X has a cyclic quotient singularity of type $\frac{1}{b}(1, r, r, 0, \dots, 0)$ at p.

We show that X is terminal at the points of $V \cup \Sigma$. In the sense of [CJL20, Section 2.3], the action of μ_r with weights $1, \dots, 1$ is small, and so is the action of μ_b with weights $1, r, r, 0, \dots, 0$, since after removing any one of the weights, the remaining weights contain 1 or r, and $\gcd(r, b) = 1$. For $1 \leq j \leq r-1$, we have

$$\sum_{i=1}^{d} \left\{\frac{j}{r}\right\} = \frac{dj}{r} = kj \geq 2 > 1.$$

For $1 \leq j \leq b-1$, since $jr \equiv -j \pmod{b}$, we have

$$\left\{\frac{j}{b}\right\} + 2\left\{\frac{jr}{b}\right\} = \frac{j}{b} + 2\left(1 - \frac{j}{b}\right) = 2 - \frac{j}{b} > 1.$$

By [CJL20, Lemma 2.4], X is terminal at any point of $V \cup \Sigma$.

We show that $V \cup \Sigma \subset \operatorname{Sing} X$ by computing embedding dimensions (the local ring and the analytic local ring of X at a point have isomorphic completions [Ser56, Proposition 3], and taking completions commutes with taking invariants of a finite group). At $p \in \Sigma$, the completed local ring of X is the ring of μ_r-invariants of $\mathbb{C}[[x_1, \dots, x_d]]$ with weights $1, \dots, 1$. The monomials of degree r in $x_1, \dots, x_d$ are invariant, and none of them is a product of two invariant monomials of positive degree, so the embedding dimension of X at p is at least

$$\binom{d+r-1}{r} \geq \binom{d+1}{2} > d.$$

At $p \in V$, the completed local ring of X is the ring of μ_b-invariants of $\mathbb{C}[[x, y_1, y_2, t_1, \dots, t_{d-3}]]$ with weights $1, r, r, 0, \dots, 0$. Since $1 + r = b$ and $\gcd(r, b) = 1$, the invariant monomials $t_1, \dots, t_{d-3}$, xy_1, xy_2, x^b, and $y_1^i y_2^{b-i}$ for $0 \leq i \leq b$ are not products of two invariant monomials of positive degree. Thus the embedding dimension of X at p is at least $(d-3) + (b+4) > d$. Hence X is singular along $V \cup \Sigma$, and $\operatorname{Sing} X = V \cup \Sigma$. This proves (2).

(1) The variety X is a hypersurface in the projective variety $\mathbb{P}(1, r, r, b^{d-1})$ of dimension $d+1$, so X is projective and $\dim X = d$. We have shown that X is normal, $\operatorname{Cl}(X) = \mathbb{Z}[H]$, and X has only cyclic quotient singularities by (2) and is terminal. By [KM98, Proposition 5.15], X is $\mathbb{Q}$-factorial.

Let $U := X\backslash(V \cup \Sigma)$. Since $\dim V = d-3$ and Σ is finite, $X\backslash U$ has codimension at least 2 in X. Since $\omega_X \cong \mathcal{O}_X(-r)$, Construction-Proposition 3.2(7) implies that

$$\omega_X|_U \cong \mathcal{O}_X(-r)|_U \cong \mathcal{O}_U(-rH),$$

so $K_X|_U \sim -rH|_U$. By [Har77, Chapter II, Proposition 6.5(b)], $\mathrm{Cl}(X) \to \mathrm{Cl}(U)$ is an isomorphism, hence $-K_X \sim rH$.

The zero divisor $\mathrm{div}(x)$ of the section x of $\mathcal{O}_X(1)$ is a non-zero divisor $\mathrm{div}(x) \geq 0$ in the class H. Let A be an ample Cartier divisor on X. Since $\mathrm{Cl}(X) = \mathbb{Z}[H]$, we have $A \sim aH$ for some integer a, and

$$(A^d) = a(A^{d-1} \cdot \mathrm{div}(x)), \quad (A^d) > 0, \quad (A^{d-1} \cdot \mathrm{div}(x)) > 0.$$

Thus $a > 0$ and H is ample. Since X is $\mathbb{Q}$-factorial and $\mathrm{Cl}(X) = \mathbb{Z}[H]$, the Picard number of X is one, and $-K_X \sim rH$ is ample. Thus X is a Fano variety of Picard number one.

(3) The affine cone over $S \subset \mathbb{P}(r, r, b^{d-1})$ is the hypersurface

$$(y_1^{kb} + y_2^{kb} + z_1^d + \cdots + z_{d-1}^d = 0) \subset \mathbb{A}^{d+1},$$

which is smooth outside the vertex since the partial derivatives vanish simultaneously only at the origin. Since $\dim S = d - 1 \geq 1$, S is irreducible by [PS20, Lemma 2.2], so S is a prime divisor. The support of $\mathrm{div}(x)$ is S, so $\mathrm{div}(x) = cS$ for some integer $c \geq 1$. Write $S \sim mH$ for some integer m. Then $H \sim cS \sim cmH$, and since $\mathrm{Cl}(X) = \mathbb{Z}[H]$, we have $cm = 1$. Thus $c = m = 1$ and $S \sim H$.

We show that (X, S) is plt. The variety X is covered by $X \backslash S$, U_1, $(y_2 \neq 0)$, and W_s for $1 \leq s \leq d - 1$. Since $X \backslash S$ is smooth by Construction-Proposition 3.2(1), and by the symmetry between y_1 and y_2, it suffices to show that $(U_1, S|_{U_1})$ and $(W_s, S|_{W_s})$ are plt. Let $g : \widetilde{U}_1 \to U_1$ and $g_s : \widetilde{W}_s \to W_s$ be the quotient maps. The non-trivial elements of μ_r fix only points of $\widetilde{U}_1$ over Σ, and the non-trivial elements of μ_b fix only points of $\widetilde{W}_s$ over V (Construction-Proposition 3.2(6)), so g and g_s are étale in codimension one. Thus $K_{\widetilde{U}_1} + (x = 0) = g^*(K_{U_1} + S|_{U_1})$ and $K_{\widetilde{W}_s} + (x = 0) = g_s^*(K_{W_s} + S|_{W_s})$. The divisors

$$(x = 0) = (1 + y_2^{kb} + z_1^d + \cdots + z_{d-1}^d = 0) \subset \widetilde{U}_1 \quad \text{and} \quad (x = 0) = \Big(y_1^{kb} + y_2^{kb} + 1 + \sum_{j \neq s} z_j^d = 0\Big) \subset \widetilde{W}_s$$

are smooth since the partial derivatives of the equations vanish simultaneously only at the origin of $\mathbb{A}^d$. Since $\widetilde{U}_1$ and $\widetilde{W}_s$ are smooth, $(\widetilde{U}_1, (x = 0))$ and $(\widetilde{W}_s, (x = 0))$ are plt. By [Kol+92, 20.4.1] ($\widetilde{U}_1$ and $\widetilde{W}_s$ are irreducible, since their defining polynomials are irreducible by Eisenstein's criterion), $(U_1, S|_{U_1})$ and $(W_s, S|_{W_s})$ are plt. Hence (X, S) is plt.

Since (X, S) is plt, (X, tS) is lc for any $0 \leq t \leq 1$. For $t > 1$, $a(S, X, tS) = 1 - t < 0$, so (X, tS) is not lc. Thus $\mathrm{lct}(X; rS) = \frac{1}{r}$. □

Lemma 3.4. *Let $X \subset \mathbb{A}^n$ be the hypersurface defined by a non-constant weighted homogeneous polynomial with positive weights. Assume that the singular locus of X has codimension at least 2 in X. Then X is integral and normal.*

Proof. Since X is a hypersurface in the smooth variety $\mathbb{A}^n$ and is regular in codimension one, X is normal by [Har77, Chapter II, Proposition 8.23(b)]. The $\mathbb{C}^*$-action given by the weights preserves X, and hence each irreducible component of X since $\mathbb{C}^*$ is connected. Since the weights are positive, the closure of each orbit contains the origin, so each irreducible component of X contains the origin. Since X is normal, the local ring of X at the origin is a domain, so only one irreducible component of X passes through the origin, and hence X is irreducible, reduced since it is normal, and thus integral. □

Lemma 3.5. *Notation and conditions as in Set-up 3.1. Let*

$$R := \mathbb{C}[x, y_1, y_2, z_1, \ldots, z_{d-1}]/(x^{bd} + y_1^{kb} + y_2^{kb} + z_1^d + \cdots + z_{d-1}^d)$$

be graded by the weights of the coordinates, and let R_ℓ be the degree ℓ part of R. Then $H^0(X, \mathcal{O}_X(\ell H)) = R_\ell$ for any integer $\ell \geq 0$. Moreover, any element of R_ℓ is the image of a unique weighted homogeneous polynomial of degree ℓ whose degree in z_{d-1} is less than d, and the image of a unique weighted homogeneous polynomial of degree ℓ whose degree in y_2 is less than kb.

Proof. The affine scheme $\operatorname{Spec} R$ is the hypersurface

$$(x^{bd} + y_1^{kb} + y_2^{kb} + z_1^d + \cdots + z_{d-1}^d = 0) \subset \mathbb{A}^{d+2},$$

and the partial derivatives of the equation vanish simultaneously only at the origin, which has codimension $d+1 \geq 2$ in $\operatorname{Spec} R$. By Lemma 3.4, R is a normal domain.

Let $U := X \backslash (V \cup \Sigma)$. By Construction-Proposition 3.2(7), $\mathcal{O}_X(\ell)|_U \cong \mathcal{O}_U(\ell H)$ for any integer ℓ. By Proposition 3.3(2), $X \backslash U$ has codimension at least 3 in X, so

$$H^0(X, \mathcal{O}_X(\ell H)) = H^0(U, \mathcal{O}_X(\ell H)) = H^0(U, \mathcal{O}_X(\ell)).$$

Let $q : \operatorname{Spec} R \backslash \{0\} \to X$ be the quotient by the $\mathbb{C}^*$-action given by the weights. For any homogeneous element $f \in R$ of positive degree, $H^0(D_+(f), \mathcal{O}_X(\ell))$ is the degree ℓ part of

$$R[f^{-1}] = H^0(q^{-1}(D_+(f)), \mathcal{O}_{\operatorname{Spec} R}).$$

Hence $H^0(U, \mathcal{O}_X(\ell))$ is the degree ℓ part of $H^0(q^{-1}(U), \mathcal{O}_{\operatorname{Spec} R})$. The complement of $q^{-1}(U)$ in $\operatorname{Spec} R$ is the cone over $V \cup \Sigma$ together with the vertex, and it has codimension at least 3 in $\operatorname{Spec} R$ since $\dim V = d-3$. Since R is a normal domain, any regular function on $q^{-1}(U)$ belongs to $R_{\mathfrak{p}}$ for any prime ideal $\mathfrak{p}$ of height one, so $H^0(q^{-1}(U), \mathcal{O}_{\operatorname{Spec} R}) = R$ by [Har77, Chapter II, Proposition 6.3A]. Thus $H^0(X, \mathcal{O}_X(\ell H)) = R_\ell$.

Let $F := x^{bd} + y_1^{kb} + y_2^{kb} + z_1^d + \cdots + z_{d-1}^d$. Any element of R_ℓ is the image of a weighted homogeneous polynomial P of degree ℓ. We regard F as a monic polynomial of degree d in z_{d-1} and divide P by F with remainder. Taking the weighted homogeneous parts of degree $\ell - bd$ and ℓ of the quotient and the remainder, we may assume that the remainder is weighted homogeneous of degree ℓ. The degree of the remainder in z_{d-1} is less than d. If P_1 and P_2 are two such remainders whose images in R_ℓ are equal, then $P_1 - P_2 = QF$ for a polynomial Q, and if $Q \neq 0$, the degree of QF in z_{d-1} is at least d, which is not possible. Thus $Q = 0$ and $P_1 = P_2$. Dividing P by F regarded as a monic polynomial of degree kb in y_2, the argument above with y_2 in place of z_{d-1} and kb in place of d implies the existence and uniqueness of a weighted homogeneous representative of degree ℓ whose degree in y_2 is less than kb. □

4. Reduction to points of a Fermat curve

In this section, we prove Proposition 4.1.

Proposition 4.1. *Notation and conditions as in Set-up 3.1. Let ℓ be a positive integer, $\mathcal{M} \subset |\ell H|$ a mobile linear system, and F a prime divisor over X such that*

$$\frac{r}{\ell} \operatorname{mult}_F \mathcal{M} > a(F, X, 0) - 1.$$

Then $\operatorname{center}_X F$ *is a closed point of C.*

Using the threshold bounds of Theorem 2.9 and inversion of adjunction (Theorem 2.8), we confine the centers of the prime divisors F in Proposition 4.1 to closed points of C. The proof of Proposition 4.1 is at the end of this section.

4.1. Fermat hypersurfaces.

In this subsection, we bound the thresholds of divisors on Fermat-type hypersurfaces.

Construction 4.2. Let $N \geq 5$ be an integer and $c \in \mathbb{C}$, and let

$$Y_{N,c} := (c\xi_0^N + \xi_1^N + \cdots + \xi_{N-1}^N = 0) \subset \mathbb{P}^{N-1},$$

where $[\xi_0 : \cdots : \xi_{N-1}]$ are the homogeneous coordinates of $\mathbb{P}^{N-1}$.

If $c = 0$, then $Y_{N,0}$ is the cone over the smooth hypersurface $(\xi_1^N + \cdots + \xi_{N-1}^N = 0) \subset \mathbb{P}^{N-2}$ with vertex $[1 : 0 : \cdots : 0]$. In Propositions 4.5 and 4.7, the vertex of $Y_{d,0}$ corresponds to a point of C, and Proposition 4.1 allows centers at closed points of C. Thus it suffices that Lemma 4.3(2) treats $Y_{N,0}$ away from the vertex.

Since $Y_{N,c}$ is a hypersurface of degree N in $\mathbb{P}^{N-1}$, the canonicity bound [Che01, Theorem 7.2] for hypersurfaces of degree N in $\mathbb{P}^N$ does not apply, and Lemma 4.3 deduces the bound $\frac{N-3}{Nq}$ from Theorem 2.9 and inversion of adjunction (Theorem 2.8). In the applications in this section, $N = d$ and the numbers λ and q of Lemma 4.3 satisfy $\lambda q \le \frac{r}{b} \le \frac{d-3}{d}$ (Set-up 3.1).

Lemma 4.3. *Notation and conditions as in Construction 4.2. Let q be a positive integer, $\Gamma \in |\mathcal{O}_{Y_{N,c}}(q)|$, and λ a real number such that $0 \le \lambda \le \frac{N-3}{Nq}$. Then the following hold.*

(1) If $c \neq 0$, then $Y_{N,c}$ is smooth and $(Y_{N,c}, \lambda\Gamma)$ is canonical.
(2) If $c = 0$, then $(Y_{N,0}, \lambda\Gamma)$ is lc on $Y_{N,0}\backslash\{[1:0:\cdots:0]\}$.

Proof. (1) Since $c \neq 0$, the partial derivatives $Nc\xi_0^{N-1}, N\xi_1^{N-1}, \dots, N\xi_{N-1}^{N-1}$ have no common zero on $\mathbb{P}^{N-1}$. Thus $Y_{N,c}$ is smooth. Since any two irreducible components of a hypersurface in $\mathbb{P}^{N-1}$ intersect [Har77, Chapter I, Theorem 7.2], and a hypersurface is singular at the points of such an intersection, $Y_{N,c}$ is irreducible.

Let F be an exceptional prime divisor over $Y_{N,c}$, and let $W := \mathrm{center}_{Y_{N,c}}(F)$. We show that $a(F, Y_{N,c}, \lambda\Gamma) \ge 1$. Note that $\lambda < \frac{1}{q}$, since $N - 3 < N$.

Assume first that $\dim W \ge 1$. Let $Z \subset W$ be a curve, where $Z = W$ if $\dim W = 1$. By the upper semicontinuity of multiplicities and Theorem 2.9(2) applied to the smooth hypersurface $Y_{N,c} \subset \mathbb{P}^{N-1}$ of degree N,

$$\mathrm{mult}_W \Gamma \le \mathrm{mult}_Z \Gamma \le q.$$

Since [Kol96, 3.14 Exercise] considers $\mathbb{Q}$-divisors and λ is a real number, we fix a rational number λ' such that $\lambda \le \lambda' < \frac{1}{q}$. Then $\lambda' \mathrm{mult}_W \Gamma < 1$. By the upper semicontinuity of multiplicities,

$$U_W := \{y \in Y_{N,c} \mid \lambda' \mathrm{mult}_y \Gamma \le 1\}$$

is an open subset of $Y_{N,c}$, and $W \cap U_W \neq \emptyset$. By [Kol96, 3.14 Exercise, (3.14.1)], $(U_W, \lambda'\Gamma|_{U_W})$ is canonical. Since $\mathrm{center}_{Y_{N,c}}(F) \cap U_W \neq \emptyset$ and F is exceptional,

$$a(F, Y_{N,c}, \lambda\Gamma) = a(F, U_W, \lambda'\Gamma|_{U_W}) + (\lambda' - \lambda)\,\mathrm{mult}_F \Gamma \ge 1.$$

Assume now that W is a closed point p. The hyperplane sections of $Y_{N,c}$ through p form a linear system with base locus $\{p\}$. By [Har77, Chapter III, Corollary 10.9 and Remark 10.9.2], for a general hyperplane $L \subset \mathbb{P}^{N-1}$ through p, the intersection $Y_{N,c} \cap L$ is smooth away from p. Moreover, $Y_{N,c} \cap L$ is smooth at p when L is not the embedded tangent hyperplane of $Y_{N,c}$ at p. Thus for a general hyperplane L through p, $Y_{N,c} \cap L$ is a smooth hypersurface of degree N in $L \cong \mathbb{P}^{N-2}$. As in the first paragraph, $Y_{N,c} \cap L$ is irreducible. Since Γ has finitely many irreducible components, we may take L such that $Y_{N,c} \cap L \not\subset \mathrm{Supp}\,\Gamma$ (otherwise the irreducible variety $Y_{N,c} \cap L$ of dimension $N - 3$ is an irreducible component of $\mathrm{Supp}\,\Gamma$ contained in L, while a general hyperplane through p contains no irreducible component of $\mathrm{Supp}\,\Gamma$). Then $\Gamma|_{Y_{N,c}\cap L} \ge 0$ is numerically equivalent to q times a hyperplane section of $Y_{N,c} \cap L$. By Theorem 2.9(1) applied to the smooth hypersurface $Y_{N,c} \cap L \subset \mathbb{P}^{N-2}$ of degree N,

$$\mathrm{lct}(Y_{N,c} \cap L; \Gamma|_{Y_{N,c}\cap L}) \ge \min\left\{\frac{(N-2)-1}{qN}, \frac{1}{q}\right\} = \frac{N-3}{qN} \ge \lambda.$$

Thus $(Y_{N,c} \cap L, \lambda\Gamma|_{Y_{N,c}\cap L})$ is lc. By Theorem 2.8, $(Y_{N,c}, Y_{N,c} \cap L + \lambda\Gamma)$ is lc near $Y_{N,c} \cap L$. Since $Y_{N,c} \cap L$ is a Cartier divisor which contains $p = \mathrm{center}_{Y_{N,c}}(F)$, we have $\mathrm{mult}_F(Y_{N,c} \cap L) \ge 1$. Therefore

$$a(F, Y_{N,c}, \lambda\Gamma) = a(F, Y_{N,c}, Y_{N,c} \cap L + \lambda\Gamma) + \mathrm{mult}_F(Y_{N,c} \cap L) \ge 1.$$

(2) Let $o := [1:0:\cdots:0]$. The partial derivatives $N\xi_1^{N-1}, \dots, N\xi_{N-1}^{N-1}$ have no common zero on $Y_{N,0}\backslash\{o\}$, so $Y_{N,0}\backslash\{o\}$ is smooth. Fix a closed point $p \in Y_{N,0}\backslash\{o\}$. For $\gamma \in \mathbb{C}^{N-1}$ with coordinates $\gamma_1, \dots, \gamma_{N-1}$, let

$$L_\gamma := \left(\xi_0 = \sum_{j=1}^{N-1} \gamma_j \xi_j\right) \subset \mathbb{P}^{N-1}.$$

Then $o \notin L_\gamma$, and the projection $[\xi_0 : \cdots : \xi_{N-1}] \to [\xi_1 : \cdots : \xi_{N-1}]$ maps L_γ isomorphically onto $\mathbb{P}^{N-2}$ and $Y_{N,0} \cap L_\gamma$ onto the smooth hypersurface $(\xi_1^N + \cdots + \xi_{N-1}^N = 0) \subset \mathbb{P}^{N-2}$ of degree N. Since $p \neq o$, we have $\xi_j(p) \neq 0$ for some $j \geq 1$. Hence the set of γ such that $p \in L_\gamma$ is an affine hyperplane of $\mathbb{C}^{N-1}$, which is an infinite set. Since Γ has finitely many irreducible components, there exists γ such that $p \in L_\gamma$ and $Y_{N,0} \cap L_\gamma \not\subset \operatorname{Supp} \Gamma$ (as in (1), $Y_{N,0} \cap L_\gamma$ is irreducible of dimension $N-3$, so it is contained in $\operatorname{Supp} \Gamma$ only if it is an irreducible component of $\operatorname{Supp} \Gamma$, and it determines γ since it is not contained in a hyperplane of L_γ). By Theorem 2.9(1) applied to the smooth hypersurface $Y_{N,0} \cap L_\gamma \subset L_\gamma \cong \mathbb{P}^{N-2}$ of degree N,

$$\operatorname{lct}(Y_{N,0} \cap L_\gamma; \Gamma|_{Y_{N,0} \cap L_\gamma}) \geq \min\left\{\frac{(N-2)-1}{qN}, \frac{1}{q}\right\} = \frac{N-3}{qN} \geq \lambda.$$

Since $Y_{N,0} \cap L_\gamma$ is a smooth prime divisor on the smooth variety $Y_{N,0} \backslash \{o\}$, Theorem 2.8 implies that $(Y_{N,0} \backslash \{o\}, Y_{N,0} \cap L_\gamma + \lambda\Gamma)$ is lc near $Y_{N,0} \cap L_\gamma$. Thus $(Y_{N,0}, \lambda\Gamma)$ is lc near p. Since $p \in Y_{N,0} \backslash \{o\}$ is arbitrary, (2) holds. □

4.2. **Centers contained in S.** In this subsection, we bound the multiplicities of divisors in $|\ell H|$ along prime divisors over X whose centers are contained in S but not in Σ.

Proposition 4.4 is used in the proof of Proposition 4.5, and Proposition 4.5 is used in the proof of Proposition 4.1 for a general member of $\mathcal{M}$, which does not contain S since $\mathcal{M}$ is mobile.

Along V, the variety X has cyclic quotient singularities of type $\frac{1}{b}(1, r, r, 0, \ldots, 0)$, and S is not Cartier. Part (a) of the sketch of the proof in Section 1 explains why we use the divisor $S + (y_1 + \gamma y_2 = 0)$ for $\gamma \in \mathbb{C}$, which is Cartier near V, instead of S. We apply inversion of adjunction three times on a smooth cover of degree b of a neighborhood of a point of V.

Proposition 4.4. *Notation and conditions as in Set-up 3.1. Let ℓ be a positive integer, $D \in |\ell H|$ a divisor with $S \not\subset \operatorname{Supp} D$, and F a prime divisor over X such that $\operatorname{center}_X F \subset V$. Then*

$$\frac{r}{\ell} \operatorname{mult}_F D \leq a(F, X, 0) - 1.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct a smooth cover of an open subset of X which contains the generic point of $\operatorname{center}_X F$.

By Proposition 3.3(2), V is isomorphic to a smooth hypersurface in $\mathbb{P}^{d-2}$, and hence V is irreducible. Since $x = y_1 = y_2 = 0$ on V, there exists $s \in \{1, \ldots, d-1\}$ such that z_s does not vanish at the generic point of $\operatorname{center}_X F$. Let $U := W_s$ and $X' := \widetilde{W}_s$, and let $h : X' \to U$ be the quotient morphism of Construction-Proposition 3.2(6). By Construction-Proposition 3.2(6), X' is smooth, and the fixed locus of any $\zeta \in \mu_b \backslash \{1\}$ is

$$V' := (x = y_1 = y_2 = 0) \subset X',$$

since μ_b acts trivially on the coordinates z_j, $j \neq s$. Moreover, $\operatorname{codim}(V', X') = 3$. Thus h is étale outside V', and in particular h is étale in codimension one. Moreover, $V' = (1 + \sum_{j \neq s} z_j^d = 0) \subset \mathbb{A}^{d-2}$. Since μ_b acts trivially on V', h maps V' isomorphically onto the affine chart $V \cap U$ of V. In particular, V' is smooth and irreducible.

Step 2. In this step, we define the polynomial Q_γ and the divisor Γ on V.

By Lemma 3.5, D is the divisor of a section of $\mathcal{O}_X(\ell H)$ represented by a unique weighted homogeneous polynomial f of degree ℓ whose degree in z_{d-1} is less than d. Write

$$f = \sum_{t,i,j \geq 0} x^t y_1^i y_2^j P_{tij}(z),$$

where $z = (z_1, \ldots, z_{d-1})$ and P_{tij} are homogeneous polynomials whose degrees in z_{d-1} are less than d, such that

$$t + r(i+j) + b \deg P_{tij} = \ell \quad \text{whenever } P_{tij} \neq 0.$$

If $P_{0ij}=0$ for all i and j, then x divides f and $S\subset \operatorname{Supp} D$, a contradiction. Let

$$m:=\min\{i+j\mid P_{0ij}\neq 0\}\quad\text{and}\quad q:=\frac{\ell-rm}{b}.$$

Then $rm\leq\ell$, and $q=\deg P_{0ij}$ is a non-negative integer for any i,j such that $i+j=m$ and $P_{0ij}\neq 0$. For $\gamma\in\mathbb{C}$, let

$$Q_\gamma(z):=\sum_{i+j=m}(-\gamma)^i P_{0ij}(z).$$

As a polynomial in γ with coefficients in $\mathbb{C}[z]$, Q_γ has coefficients $(-1)^iP_{0,i,m-i}$, $0\leq i\leq m$, and at least one of them is non-zero. Thus there exists $\gamma\in\mathbb{C}$ such that $Q_\gamma\neq 0$, and we fix such γ. The degree of Q_γ in z_{d-1} is less than d. Thus Q_γ is not divisible by $z_1^d+\cdots+z_{d-1}^d$. Since V is smooth and irreducible, $z_1^d+\cdots+z_{d-1}^d$ is irreducible and generates the homogeneous ideal of V. Hence $Q_\gamma|_V\neq 0$. Since Q_γ is homogeneous of degree q in z,

$$\Gamma:=(Q_\gamma|_V=0)\in|\mathcal{O}_V(q)|,$$

where $\mathcal{O}_V(1)$ is the hyperplane bundle of V under the isomorphism of Proposition 3.3(2) from V onto the smooth hypersurface $(z_1^d+\cdots+z_{d-1}^d=0)\subset\mathbb{P}^{d-2}$. The sheaf $\mathcal{O}_V(1)$ corresponds to $\mathcal{O}_X(b)|_V$ and not to $\mathcal{O}_X(1)|_V$. If $q=0$, then Q_γ is a non-zero constant and $\Gamma=0$.

On X', let

$$S':=(x=0),\quad T':=(y_1+\gamma y_2=0),\quad W':=S'\cap T'.$$

In the coordinates y_1, y_2, and z_j, $j\neq s$,

$$S'=\Big(y_1^{kb}+y_2^{kb}+1+\sum_{j\neq s}z_j^d=0\Big)\subset\mathbb{A}^d,$$

and, substituting $y_1=-\gamma y_2$, in the coordinates x, y_2, and z_j, $j\neq s$,

$$T'=\Big(x^{bd}+\big((-\gamma)^{kb}+1\big)y_2^{kb}+1+\sum_{j\neq s}z_j^d=0\Big)\subset\mathbb{A}^d.$$

In the coordinates y_2 and z_j, $j\neq s$,

$$W'=\Big(\big((-\gamma)^{kb}+1\big)y_2^{kb}+1+\sum_{j\neq s}z_j^d=0\Big)\subset\mathbb{A}^{d-1},$$

and $V'=W'\cap(y_2=0)$. Each of the equations of S', T', and W' is the sum of 1 and constant multiples of powers v^e, $e\geq 2$, of single coordinates v. At a common zero of the partial derivatives, each such term with a non-zero coefficient vanishes and the equation takes the value 1. Thus S', T', and W' are smooth. In particular, the scheme-theoretic intersection $S'\cap T'$ is the smooth divisor W' on S', and V' is the divisor $(y_2=0)$ on W'.

The variety X' is the slice $(z_s=1)$ of the affine cone over X, h is the restriction of the projection from the cone to U, and D is cut out by f on the cone. Hence the restrictions of h^*D and of the divisor of the function $f|_{z_s=1}$ to $X'\backslash V'$, where h is étale, are equal. Both are Weil divisors on the smooth variety X', and V' has codimension 3 in X', so the pullback $D':=h^*D$ is the divisor of $f|_{z_s=1}$. Since $x=0$ and $y_1=-\gamma y_2$ on W',

$$f|_{W'}=\sum_{i,j\geq 0}(-\gamma)^i y_2^{i+j}P_{0ij}(z)|_{z_s=1}=y_2^m g,$$

where g is a polynomial in y_2 and z_j, $j\neq s$, such that $g(0,z)=Q_\gamma(z)|_{z_s=1}$. Under the isomorphism $h|_{V'}:V'\to V\cap U$, the divisor of $Q_\gamma|_{z_s=1}$ on V' corresponds to $\Gamma|_{V\cap U}$.

Step 3. In this step, we show that $(V,\lambda\Gamma)$ is lc, where $\lambda:=\frac{r}{\ell}$.

Since $rm\leq\ell$, we have $\lambda m\leq 1$ and

$$\lambda q=\frac{r(\ell-rm)}{b\ell}\leq\frac{r}{b}\leq\frac{d-3}{d},$$

where the last inequality is the one in Set-up 3.1. If $q = 0$, then $\Gamma = 0$ and $(V, \lambda\Gamma)$ is lc since V is smooth. If $q > 0$, then by Theorem 2.9(1) applied to $V \subset \mathbb{P}^{d-2}$, a smooth hypersurface of degree d,

$$\operatorname{lct}(V;\Gamma) \geq \min\left\{\frac{(d-2)-1}{qd}, \frac{1}{q}\right\} = \frac{d-3}{qd} \geq \lambda.$$

Thus $(V, \lambda\Gamma)$ is lc.

Step 4. In this step, we show that $(X', S' + T' + \lambda D')$ is lc near V' by three inversions of adjunction.

Fix a closed point $p' \in V'$. We work on an open neighborhood of p' in X' on which S', T', W', and V' are irreducible, and denote the restrictions of S', T', W', V', and D' to it again by S', T', W', V', and D'. Since V' is irreducible and $Q_\gamma|_{z_s=1}$ is a non-zero regular function on V', we have $g|_{V'} \neq 0$. In particular, $f|_{W'} \neq 0$, so $S' \not\subset \operatorname{Supp} D'$ and $W' \not\subset \operatorname{Supp}(D'|_{S'})$. Since $f|_{W'} = y_2^m g$,

$$D'|_{W'} = mV' + D_{W'},$$

where $D_{W'} \geq 0$ is the divisor of g on W', and $V' \not\subset \operatorname{Supp} D_{W'}$. By **Step 2**, $D_{W'}|_{V'}$ corresponds to $\Gamma|_{V\cap U}$ under $h|_{V'}$, so $(V', \lambda D_{W'}|_{V'})$ is lc by **Step 3**. Since $\lambda m \leq 1$, Lemma 2.10 applied to $(X, S, T, D) := (S', W', V', D'|_{S'})$ and the point p' implies that $(S', W' + \lambda D'|_{S'})$ is lc near p'. Since $T' \neq S'$, $S' \not\subset \operatorname{Supp} D'$, and $T'|_{S'} = W'$, Theorem 2.8 applied to X', S', and $T' + \lambda D'$ implies that $(X', S' + T' + \lambda D')$ is lc near p'. Since $p' \in V'$ is arbitrary, there exists an open neighborhood X'_0 of V' in X' such that $(X'_0, S' + T' + \lambda D')$ is lc.

Step 5. In this step, we descend to U and conclude.

Let $S_U := S \cap U$ and $T_U := (y_1 + \gamma y_2 = 0) \cap U$. Since h is finite, $U_0 := U \backslash h(X' \backslash X'_0)$ is open in U, and $V \cap U \subset U_0$ since $h^{-1}(V \cap U) = V' \subset X'_0$. Since h is étale in codimension one, $h^* S_U = S'$, $h^* T_U = T'$, and

$$K_{X'} + S' + T' + \lambda D' = h^*(K_U + S_U + T_U + \lambda D|_U).$$

By [KM98, Proposition 5.20(4)], $(U_0, S_U + T_U + \lambda D|_{U_0})$ is lc.

The rational function $x(y_1 + \gamma y_2)/z_s$ on X is regular on U, since $x(y_1 + \gamma y_2)$ and z_s both have degree $1 + r = b$. The pullback of $x(y_1 + \gamma y_2)/z_s$ to X' is $x(y_1 + \gamma y_2)$, whose divisor is $S' + T' = h^*(S_U + T_U)$. Thus $S_U + T_U$ is the divisor of $x(y_1 + \gamma y_2)/z_s$ on U, and $S_U + T_U$ is a Cartier divisor on U which contains $V \cap U$. By the choice of s, the generic point of $\operatorname{center}_X F$ is contained in $V \cap U \subset U_0$. Since $\operatorname{center}_X F \subset V$, we have $\operatorname{mult}_F(S_U + T_U) \geq 1$. Since $(U_0, S_U + T_U + \lambda D|_{U_0})$ is lc and U_0 contains the generic point of $\operatorname{center}_X F$,

$$0 \leq a(F, U_0, S_U + T_U + \lambda D|_{U_0}) = a(F, X, 0) - \operatorname{mult}_F(S_U + T_U) - \frac{r}{\ell}\operatorname{mult}_F D \leq a(F, X, 0) - 1 - \frac{r}{\ell}\operatorname{mult}_F D.$$

□

On $S \backslash (V \cup \Sigma)$, both X and S are smooth, but $D|_S$ is a divisor on the weighted hypersurface S, so the bound of Theorem 2.9(1) cannot be used directly. We slice S by the divisors $(y_2 = ty_1)$, $t \in \mathbb{C}$, regrade each slice by a Veronese subring into a hypersurface of $\mathbb{P}(r, 1^{d-1})$, and cover the hypersurface near a given point by $Y_{d,1+t^{kb}}$ via an étale morphism. The cone $Y_{d,0}$ appears exactly when the slice contains a point of Σ, and then the vertex of $Y_{d,0}$ is mapped to that point, so Proposition 4.5 excludes the centers contained in Σ.

Proposition 4.5. *Notation and conditions as in Set-up 3.1. Let ℓ be a positive integer, $D \in |\ell H|$ a divisor with $S \not\subset \operatorname{Supp} D$, and F a prime divisor over X such that $\operatorname{center}_X F \subset S$ and $\operatorname{center}_X F \not\subset \Sigma$. Then*

$$\frac{r}{\ell}\operatorname{mult}_F D \leq a(F, X, 0) - 1.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we take a closed point $p \in \operatorname{center}_X F$ near which X and S are smooth.

If $\operatorname{center}_X F \subset V$, then the inequality follows from Proposition 4.4. Thus we may assume that $\operatorname{center}_X F \not\subset V$. Since $\operatorname{center}_X F \not\subset \Sigma$ and Σ is finite, there exists a closed point p of $\operatorname{center}_X F$

with $p \notin V \cup \Sigma$. Since $p \in S \backslash V$, we have $x(p) = 0$ and $(y_1(p), y_2(p)) \neq (0,0)$. Since $p \notin \Sigma = S \cap C$, we have $z_j(p) \neq 0$ for some j. Interchanging y_1 and y_2 preserves X, H, S, V, C, and Σ, so we may assume that $y_1(p) \neq 0$, that is, $p \in U_1$.

By Construction-Proposition 3.2(5), $U_1 \cong \widetilde{U}_1/\mu_r$. Let $\tilde{p} \in \widetilde{U}_1$ be a point over p. Since μ_r acts on z_j by multiplication and $z_j(\tilde{p}) \neq 0$, the stabilizer of $\tilde{p}$ in μ_r is trivial. Thus the quotient morphism $\widetilde{U}_1 \to U_1$ is étale near $\tilde{p}$. The preimage of S in $\widetilde{U}_1$ is

$$\widetilde{S} := (x = 0) = \Big(1 + y_2^{kb} + \sum_{j=1}^{d-1} z_j^d = 0\Big) \subset \mathbb{A}^d,$$

and the partial derivatives of the equations of $\widetilde{U}_1$ and $\widetilde{S}$ have no common zero on $\widetilde{U}_1$ and on $\widetilde{S}$, respectively. Hence X and S are smooth near p, and S is a Cartier divisor near p.

Step 2. In this step, we slice S through p and factor the restriction of D to S.

Let $t := y_2(p)/y_1(p)$, which is well defined since $y_1(p) \neq 0$ and $\deg y_1 = \deg y_2 = r$, and let

$$c_t := 1 + t^{kb}, \qquad T := S \cap (y_2 - t y_1 = 0).$$

Then $p \in T$. Let

$$R_S := \mathbb{C}[y_1, y_2, z_1, \dots, z_{d-1}] / \Big(y_1^{kb} + y_2^{kb} + \sum_{j=1}^{d-1} z_j^d\Big)$$

be graded by the weights of the coordinates. Then

$$R_S/(y_2 - t y_1) \cong \mathbb{C}[y, z_1, \dots, z_{d-1}] / \Big(c_t y^{kb} + \sum_{j=1}^{d-1} z_j^d\Big), \quad y := y_1.$$

The affine hypersurface $A_t := (c_t y^{kb} + \sum_{j=1}^{d-1} z_j^d = 0) \subset \mathbb{A}^d$ is defined by a weighted homogeneous polynomial with positive weights. The partial derivatives show that the singular locus of A_t is contained in $A_t \cap (z_1 = \cdots = z_{d-1} = 0)$, whose codimension in A_t is at least $d - 2 \geq 2$. By Lemma 3.4, A_t is integral. Hence $R_S/(y_2 - t y_1)$ is a domain, and T is integral.

By Lemma 3.5, D is the divisor of a section of $\mathcal{O}_X(\ell H)$ represented by a weighted homogeneous polynomial f of degree ℓ. Let f_0 be the image of $f(0, y_1, y_2, z)$ in R_S. Since $S \not\subset \operatorname{Supp} D$, we have $f_0 \neq 0$. If $f_0 = (y_2 - t y_1)^{m'} g'$ for an integer $m' \geq 0$ and a homogeneous element $g' \in R_S$, then $\deg g' = \ell - r m' \geq 0$. Thus there exist a largest integer $m \geq 0$ and a homogeneous element $g \in R_S$ of degree $\ell' := \ell - rm$ such that

$$f_0 = (y_2 - t y_1)^m g,$$

and $g \notin (y_2 - t y_1)$. Let $\lambda := \frac{r}{\ell}$. Then $\lambda m \leq 1$.

The preimage of T in $\widetilde{U}_1$ is

$$\widetilde{T} := (x = 0,\ y_2 = t) = \Big(c_t + \sum_{j=1}^{d-1} z_j^d = 0\Big) \subset \mathbb{A}^{d-1},$$

which is smooth near $\tilde{p}$ since $z_j(\tilde{p}) \neq 0$ for some j. Thus T is smooth near p, and near p, T is the Cartier divisor on S whose pullback to $\widetilde{S}$ is $(y_2 - t = 0)$. The pullback of D to $\widetilde{U}_1$ is the divisor of $f(x, 1, y_2, z)$, and $f(0, 1, y_2, z) = (y_2 - t)^m g(1, y_2, z)$ on $\widetilde{S}$. Thus near p,

$$D|_S = mT + D_S,$$

where $D_S \geq 0$ is the Cartier divisor on S near p whose pullback to $\widetilde{S}$ is the divisor of $g(1, y_2, z)$. Since the image of g in the domain $R_S/(y_2 - t y_1)$ is non-zero and T is integral, $T \not\subset \operatorname{Supp} D_S$.

Step 3. In this step, we identify T with a hypersurface $W_t \subset \mathbb{P}(r, 1^{d-1})$ and describe $D_S|_T$ near p.

Let $\bar{g}$ be the image of g in $R_S/(y_2 - ty_1)$, a non-zero homogeneous element of degree ℓ'. Any monomial $y^i z_1^{e_1} \cdots z_{d-1}^{e_{d-1}}$ of degree ℓ' satisfies

$$ri + b(e_1 + \cdots + e_{d-1}) = \ell'.$$

Since $r \equiv -1 \pmod{b}$, we have $i \equiv -\ell' \pmod{b}$. Let $s \in \{0, \ldots, b-1\}$ be the integer such that $s \equiv -\ell' \pmod{b}$. Then $i \geq s$ for any such monomial, and

$$\bar{g} = y^s G(y^b, z_1, \ldots, z_{d-1}),$$

where $G(\eta, \xi_1, \ldots, \xi_{d-1})$ is a weighted homogeneous polynomial of degree

$$q := \frac{\ell' - rs}{b}$$

with respect to the weights $\deg \eta = r$ and $\deg \xi_j = 1$, and q is a non-negative integer.

Since $\gcd(r, b) = 1$, a monomial $y^i z_1^{e_1} \cdots z_{d-1}^{e_{d-1}}$ has degree divisible by b if and only if $b \mid i$. Thus the b-th Veronese subring of $R_S/(y_2 - ty_1)$, with degrees divided by b, is

$$\mathbb{C}[\eta, \xi_1, \ldots, \xi_{d-1}]/\Big(c_t \eta^k + \sum_{j=1}^{d-1} \xi_j^d\Big), \quad \eta := y^b,\ \xi_j := z_j,$$

where $\deg \eta = r$ and $\deg \xi_j = 1$. By [Gro61, Proposition (2.4.7)],

$$T \cong W_t := (c_t \eta^k + \xi_1^d + \cdots + \xi_{d-1}^d = 0) \subset \mathbb{P}(r, 1^{d-1}).$$

Near p, $D_S|_T$ is the divisor of $\bar{g}$, and $\eta = y_1^b$ does not vanish near p. Thus near p, $rbD_S|_T$ is the divisor of the degree-zero element $\bar{g}^{rb}/\eta^{\ell'}$. Since $\bar{g}^b = \eta^s G^b$ and $\ell' - rs = bq$, we have $\bar{g}^{rb}/\eta^{\ell'} = (G^r/\eta^q)^b$, which is an element of the degree-zero localization of the Veronese subring at η. Since $\deg \eta = rb$ is divisible by b, any degree-zero element ϕ/η^n of the localization of $R_S/(y_2 - ty_1)$ at η has $\deg \phi = nrb$, so ϕ is an element of the Veronese subring. Thus the degree-zero localization of $R_S/(y_2 - ty_1)$ at η equals the degree-zero localization of the Veronese subring at η. Hence near p, $D_S|_T$ is the divisor $(G = 0)$ on W_t. Moreover,

$$\lambda q = \frac{r(\ell - rm - rs)}{b\ell} \leq \frac{r}{b} \leq \frac{d-3}{d},$$

where the last inequality is the one in Set-up 3.1.

Step 4. In this step, we show that $(T, \lambda D_S|_T)$ is lc near p.

If $q = 0$, then G is a non-zero constant, so $D_S|_T = 0$ near p. Since T is smooth near p, $(T, \lambda D_S|_T)$ is lc near p.

Assume that $q > 0$. Since $d = kr$, the morphism

$$h : Y_{d,c_t} \to W_t, \quad [\xi_0 : \xi_1 : \cdots : \xi_{d-1}] \to [\xi_0^r : \xi_1 : \cdots : \xi_{d-1}]$$

is well defined and surjective. Let

$$W_t^\circ := \big(W_t \cap (\eta \neq 0)\big) \backslash \{[1 : 0 : \cdots : 0]\}.$$

Since $\eta(p) \neq 0$ and $\xi_j(p) = z_j(p) \neq 0$ for some j, W_t° is an open neighborhood of p in W_t. Moreover,

$$h^{-1}(W_t^\circ) = \big(Y_{d,c_t} \cap (\xi_0 \neq 0)\big) \backslash \{[1 : 0 : \cdots : 0]\}.$$

In the affine coordinates ξ_j/ξ_0 on $(\xi_0 \neq 0)$ and under the isomorphism $(\eta \neq 0) \cong \mathbb{A}^{d-1}/\mu_r$, where μ_r acts by multiplication [IF00, 5.3], h sends $u := (\xi_j/\xi_0)_j$ to the class of u in $\mathbb{A}^{d-1}/\mu_r$, since

$$[\xi_0^r : \xi_1 : \cdots : \xi_{d-1}] = [1 : \xi_1/\xi_0 : \cdots : \xi_{d-1}/\xi_0] \quad \text{in } \mathbb{P}(r, 1^{d-1}).$$

Thus the restriction of h to $h^{-1}(W_t^\circ)$ is the quotient of

$$\Big(c_t + \sum_{j=1}^{d-1} \xi_j^d = 0\Big) \backslash \{0\} \subset \mathbb{A}^{d-1}$$

by μ_r, and μ_r acts freely on $\mathbb{A}^{d-1}\setminus\{0\}$. Thus $h^{-1}(W_t^\circ) \to W_t^\circ$ is finite and étale.

Let Γ_t be the divisor of $G(\xi_0^r, \xi_1, \dots, \xi_{d-1})$ on Y_{d,c_t}. If $c_t \neq 0$, then Y_{d,c_t} is a smooth, hence irreducible, hypersurface in $\mathbb{P}^{d-1}$ by Lemma 4.3(1). If $c_t = 0$, then Y_{d,c_t} is the cone over the smooth, hence irreducible, hypersurface $(\xi_1^d + \cdots + \xi_{d-1}^d = 0) \subset \mathbb{P}^{d-2}$, and hence irreducible. Since Y_{d,c_t} is irreducible, h is surjective, and G does not vanish identically on W_t (since $\bar{g} = y^s G(y^b, z_1, \dots, z_{d-1}) \neq 0$), the polynomial $G(\xi_0^r, \xi_1, \dots, \xi_{d-1})$ does not vanish identically on Y_{d,c_t}. Thus $\Gamma_t \in |\mathcal{O}_{Y_{d,c_t}}(q)|$, and over W_t°,

$$K_{Y_{d,c_t}} + \lambda\Gamma_t = h^*\big(K_{W_t} + \lambda(G=0)\big).$$

Since $d \geq 5$ and $\lambda \leq \frac{d-3}{dq}$, Lemma 4.3 with $N = d$ and $c = c_t$ implies the following. If $c_t \neq 0$, then $(Y_{d,c_t}, \lambda\Gamma_t)$ is canonical, hence lc [KM98, Corollary 2.31(1)]. If $c_t = 0$, then $(Y_{d,0}, \lambda\Gamma_t)$ is lc on $Y_{d,0}\setminus\{[1:0:\cdots:0]\}$. In both cases, $(h^{-1}(W_t^\circ), \lambda\Gamma_t)$ is lc. By [KM98, Proposition 5.20(4)], $(W_t^\circ, \lambda(G=0))$ is lc. By **Step 3**, $(T, \lambda D_S|_T)$ is lc near p.

Step 5. In this step, we conclude by Lemma 2.10.

By **Steps 1 and 2**, there exists an open neighborhood X_p of p in X on which X, S, and T are smooth, S and T are irreducible, D is Cartier, and $D|_S = mT + D_S$, where $D_S \geq 0$ is Cartier and $T \not\subset \operatorname{Supp} D_S$. Moreover, $S \not\subset \operatorname{Supp} D$, $\lambda m \leq 1$, and $(T, \lambda D_S|_T)$ is lc near p by **Step 4**. Since $p \in \operatorname{center}_X F \subset S$, by Lemma 2.10 applied to X_p, the restrictions of S, T, D, and D_S to X_p, the point p, and the numbers λ and m,

$$\frac{r}{\ell}\operatorname{mult}_F D \leq a(F, X, 0) - 1.$$

□

4.3. **Centers not contained in S.** In this subsection, we treat the prime divisors over X whose centers are not contained in S, and we prove Proposition 4.1. The next lemma bounds from below the log discrepancies of prime divisors whose centers dominate the base of a smooth morphism, by restricting to general closed fibers.

Lemma 4.6. *Let $f : X \to T$ be a smooth surjective morphism between smooth varieties and $B \geq 0$ an $\mathbb{R}$-divisor on X. For a closed point $t \in T$, let $X_t := f^{-1}(t)$. Then the following hold.*

1. *If $(X', B|_{X'})$ is lc for a general closed point $t \in T$ and any connected component X' of X_t, then there exists a non-empty open subset $T^\circ \subset T$ such that $(f^{-1}(T^\circ), B|_{f^{-1}(T^\circ)})$ is lc. In particular, $a(F, X, B) \geq 0$ for any prime divisor F over X such that $f(\operatorname{center}_X F)$ is dense in T.*
2. *If $(X', B|_{X'})$ is canonical for a general closed point $t \in T$ and any connected component X' of X_t, then $a(F, X, B) \geq 1$ for any prime divisor F over X such that F is exceptional over X and $f(\operatorname{center}_X F)$ is dense in T.*

Proof. Let $h : \widetilde{X} \to X$ be a log resolution of (X, B). In (2), we choose h so that F is a prime divisor on $\widetilde{X}$. Let $E_1, \dots, E_n$ be the prime divisors on $\widetilde{X}$ that are exceptional over X or components of $h_*^{-1}B$, and let

$$\Delta := \sum_{i=1}^n \big(1 - a(E_i, X, B)\big)E_i.$$

Then $K_{\widetilde{X}} + \Delta = h^*(K_X + B)$ and $\sum_i E_i$ is an snc divisor. For a subset $I \subset \{1, \dots, n\}$, let $E_I := \bigcap_{i\in I} E_i$, where $E_\emptyset := \widetilde{X}$. We apply generic smoothness [Har77, Chapter III, Corollary 10.7] to the restrictions of $f \circ h$ to the components of the smooth varieties E_I that dominate T, and we remove from T the closures of the images of the components of the E_I, of $h(\operatorname{Exc}(h))$, and of $\operatorname{Supp} B$ that do not dominate T. Since X is smooth, $h(\operatorname{Exc}(h))$ has codimension at least 2 in X, and X_t has pure dimension $\dim X - \dim T$ for any closed point $t \in T$. By [Har77, Chapter II, Exercise 3.22(e)] for the components of $h(\operatorname{Exc}(h))$ and of $\operatorname{Supp} B$ that dominate T, we obtain a non-empty open subset $T_1 \subset T$ such that for any closed point $t \in T_1$, the following hold:

- $\widetilde{X}_t := (f \circ h)^{-1}(t)$ is smooth, and $\sum_i E_i|_{\widetilde{X}_t}$ is a reduced snc divisor on $\widetilde{X}_t$;
- $h(\mathrm{Exc}(h)) \cap X_t$ has codimension at least 2 in X_t, and $X_t \cap \operatorname{Supp} B$ has codimension at least 1 in X_t.

Fix a closed point $t \in T_1$, and let $h_t := h|_{\widetilde{X}_t} : \widetilde{X}_t \to X_t$. Then h_t is an isomorphism over a dense open subset of any connected component of X_t, the restriction $B|_{X_t}$ is defined, and h_t is a log resolution of $(X_t, B|_{X_t})$. Since the normal bundles of $\widetilde{X}_t$ in $\widetilde{X}$ and of X_t in X are trivial, we have $K_{\widetilde{X}_t} = K_{\widetilde{X}}|_{\widetilde{X}_t}$ and $K_{X_t} = K_X|_{X_t}$. Restricting $K_{\widetilde{X}} + \Delta = h^*(K_X + B)$ to $\widetilde{X}_t$, we obtain

$$K_{\widetilde{X}_t} + \Delta|_{\widetilde{X}_t} = h_t^*(K_{X_t} + B|_{X_t}).$$

Thus for any i and any component E_t of $E_i \cap \widetilde{X}_t$,

$$a(E_t, X_t, B|_{X_t}) = a(E_i, X, B), \tag{4.1}$$

where the left-hand side is computed on the connected component of X_t containing $h_t(E_t)$.

(1) By [Kol13, Corollary 2.13(1)], which holds for real coefficients by [Kol13, 2.20], for an open subset $U \subset X$, the pair $(U, B|_U)$ is lc if and only if $a(E_i, X, B) \geq 0$ for any i such that $E_i \cap h^{-1}(U) \neq \emptyset$. Let I_- be the set of indices i such that $a(E_i, X, B) < 0$. Suppose that E_i dominates T for some $i \in I_-$. Then for a general closed point $t \in T_1$, the intersection $E_i \cap \widetilde{X}_t$ is non-empty, and (4.1) shows that $(X', B|_{X'})$ is not lc for the connected component X' of X_t containing the image of a component of $E_i \cap \widetilde{X}_t$, a contradiction. Thus no E_i with $i \in I_-$ dominates T, and

$$T^\circ := T \setminus \bigcup_{i \in I_-} \overline{f(h(E_i))}$$

is a non-empty open subset of T. Since $E_i \cap h^{-1}(f^{-1}(T^\circ)) = \emptyset$ for any $i \in I_-$, the pair $(f^{-1}(T^\circ), B|_{f^{-1}(T^\circ)})$ is lc. If $f(\operatorname{center}_X F)$ is dense in T, then $\operatorname{center}_X F \cap f^{-1}(T^\circ) \neq \emptyset$, and hence $a(F, X, B) = a(F, f^{-1}(T^\circ), B|_{f^{-1}(T^\circ)}) \geq 0$.

(2) Since F is exceptional over X, we have $F = E_{i_0}$ for some i_0. Moreover, $h(F) = \operatorname{center}_X F$ is contained in $h(\mathrm{Exc}(h))$ and dominates T. Let $t \in T_1$ be a general closed point. Then $F \cap \widetilde{X}_t \neq \emptyset$, and for any component F_t of $F \cap \widetilde{X}_t$, the image $h_t(F_t) \subset h(\mathrm{Exc}(h)) \cap X_t$ has codimension at least 2 in X_t. Thus F_t is exceptional over X_t. Since the connected components of X_t are canonical with the restricted boundary, (4.1) implies that $a(F, X, B) = a(F_t, X_t, B|_{X_t}) \geq 1$. □

The fibers of π over the affine plane $(w \neq 0) \subset \mathbb{P}^2$ are affine pieces of the hypersurfaces $Y_{d,c}$, and π is smooth on $X \setminus (S \cup C)$. We divide the centers according to the dimension of their images in $(w \neq 0)$: when the image is dense, the canonical bound on the fibers (Lemma 4.3(1)) suffices, and when the image is a curve or a point, we apply inversion of adjunction once along a prime divisor P, or twice along P and a fiber. Since $\mathcal{M}$ has no fixed component, a general member of $\mathcal{M}$ does not contain P.

Proposition 4.7. *Notation and conditions as in Set-up 3.1. Let ℓ be a positive integer, $\mathcal{M} \subset |\ell H|$ a mobile linear system, and F a prime divisor over X such that $\operatorname{center}_X F \not\subset S \cup C$. Then*

$$\frac{r}{\ell} \operatorname{mult}_F \mathcal{M} \leq a(F, X, 0) - 1.$$

Proof. Let $\lambda := r/\ell$ and $W := \operatorname{center}_X F$. If F is a prime divisor on X, then a general member of $\mathcal{M}$ does not contain F, since $\mathcal{M}$ has no fixed component, and hence $\operatorname{mult}_F \mathcal{M} = 0 = a(F, X, 0) - 1$. Thus we may assume that F is exceptional over X.

We take general members of $\mathcal{M}$ as follows. In $\mathcal{M}$, the members D with $\operatorname{mult}_F D > \operatorname{mult}_F \mathcal{M}$ form a proper linear subspace, and for any prime divisor P on X, the members containing P form a proper linear subspace, since $\mathcal{M}$ has no fixed component (Definition 2.3). Hence for finitely many prime divisors $P_1, \ldots, P_s$ on X, any member D of $\mathcal{M}$ outside finitely many proper linear subspaces satisfies $\operatorname{mult}_F D = \operatorname{mult}_F \mathcal{M}$ and $P_i \not\subset \operatorname{Supp} D$ for any i.

We regard $\mathbb{A}^2 := (w \neq 0) \subset \mathbb{P}^2$ as the affine plane with the coordinates u_1, u_2, where $[1 : u_1 : u_2] \in \mathbb{P}^2$. By Construction-Proposition 3.2(1), $\pi(X\backslash S) \subset \mathbb{A}^2$, and $\pi|_{X\backslash S}$ is the restriction of the projection $\mathbb{A}^{d+1} \to \mathbb{A}^2$ to the coordinates u_1, u_2. Let

$$c(u_1, u_2) := 1 + u_1^{kb} + u_2^{kb}, \qquad X^\circ := X\backslash(S \cup C),$$

and for a closed point $t = (t_1, t_2) \in \mathbb{A}^2$, let $X_t := \pi^{-1}(t) \cap (X\backslash S)$. Then

$$X_t \cong \big(c(t_1, t_2) + w_1^d + \cdots + w_{d-1}^d = 0\big) \subset \mathbb{A}^{d-1}.$$

Since $d \geq 3b \geq 12$, the hypersurface $Y_{d,c(t_1,t_2)}$ of Construction 4.2 is defined. Under $\xi_0 = 1$ and $\xi_j = w_j$ for $1 \leq j \leq d-1$, the fiber X_t is isomorphic to the open subset $(\xi_0 \neq 0)$ of $Y_{d,c(t_1,t_2)}$, and $X_t \cap C = (w_1 = \cdots = w_{d-1} = 0) \cap X_t$ is mapped to the point $[1 : 0 : \cdots : 0]$, which belongs to $Y_{d,c(t_1,t_2)}$ if and only if $c(t_1, t_2) = 0$. If $c(t_1, t_2) \neq 0$, then $Y_{d,c(t_1,t_2)}$ is a smooth, hence irreducible, hypersurface in $\mathbb{P}^{d-1}$ (Lemma 4.3(1)). If $c(t_1, t_2) = 0$, then $Y_{d,0}$ is the cone with vertex $[1 : 0 : \cdots : 0]$ over the smooth, hence irreducible, hypersurface $(\xi_1^d + \cdots + \xi_{d-1}^d = 0) \subset \mathbb{P}^{d-2}$, and hence irreducible. Thus the variety $X_t\backslash C$ is non-empty and irreducible. Moreover, $X\backslash S$ is smooth, and at any point of X°, we have $w_j \neq 0$ for some j, so that

$$\frac{\partial}{\partial w_j}\Big(c(u_1, u_2) + \sum_{i=1}^{d-1} w_i^d\Big) = d \cdot w_j^{d-1} \neq 0.$$

By the Jacobian criterion, $\pi|_{X^\circ} : X^\circ \to \mathbb{A}^2$ is a smooth surjective morphism whose fiber over t is $X_t\backslash C$.

Let $D \in \mathcal{M}$, and let $f \in H^0(X, \mathcal{O}_X(\ell H))$ with $D = (f = 0)$, viewed as a weighted homogeneous polynomial of degree ℓ (Lemma 3.5). On $X\backslash S$, the divisor D is the divisor of the polynomial $f(1, u_1, u_2, w_1, \ldots, w_{d-1})$. Any monomial of weight ℓ has degree at most ℓ/b in $z_1, \ldots, z_{d-1}$, and hence $f(1, u_1, u_2, w_1, \ldots, w_{d-1})$ has degree at most $\lfloor \ell/b \rfloor$ in $w_1, \ldots, w_{d-1}$. By Set-up 3.1,

$$\lambda\Big\lfloor \frac{\ell}{b} \Big\rfloor \leq \frac{r}{b} \leq \frac{d-3}{d}. \tag{4.2}$$

We show the following.

$(*)$ Let $t = (t_1, t_2) \in \mathbb{A}^2$ be a closed point, and let g be a polynomial in $w_1, \ldots, w_{d-1}$ of degree q with $\lambda q \leq (d-3)/d$ which does not vanish identically on X_t. Let Γ_t be the divisor of g on $X_t\backslash C$. Then $(X_t\backslash C, \lambda\Gamma_t)$ is lc, and it is canonical if $c(t_1, t_2) \neq 0$.

If $q \geq 1$, then $\xi_0^q g(\xi_1/\xi_0, \ldots, \xi_{d-1}/\xi_0)$ does not vanish identically on $Y_{d,c(t_1,t_2)}$, since X_t is dense in the irreducible variety $Y_{d,c(t_1,t_2)}$. Thus it defines a member $\Gamma \in |\mathcal{O}_{Y_{d,c(t_1,t_2)}}(q)|$ whose restriction to $X_t\backslash C$ is Γ_t, and $\lambda \leq \frac{d-3}{dq}$. If $c(t_1, t_2) \neq 0$, then $(Y_{d,c(t_1,t_2)}, \lambda\Gamma)$ is canonical by Lemma 4.3(1), and hence lc [KM98, Corollary 2.31(1)]. If $c(t_1, t_2) = 0$, then $(Y_{d,0}, \lambda\Gamma)$ is lc on $Y_{d,0}\backslash\{[1 : 0 : \cdots : 0]\}$ by Lemma 4.3(2). Restricting to $X_t\backslash C$ proves $(*)$ when $q \geq 1$. If $q = 0$, then g is a non-zero constant and $\Gamma_t = 0$. Lemma 4.3 with $q = 1$ and $\lambda = 0$ shows that $Y_{d,c(t_1,t_2)}$ is canonical if $c(t_1, t_2) \neq 0$, and that $Y_{d,0}$ is lc on $Y_{d,0}\backslash\{[1 : 0 : \cdots : 0]\}$. Thus $(*)$ holds.

Since $W \not\subset S \cup C$ and W is irreducible, $W \cap X^\circ$ is a dense open subset of W, and the closure of $\pi(W \cap X^\circ)$ in $\mathbb{A}^2$ is irreducible. Thus one of the following three cases occurs.

Case 1. $\pi(W \cap X^\circ)$ is dense in $\mathbb{A}^2$. Take a general member D of $\mathcal{M}$, so that $\operatorname{mult}_F D = \operatorname{mult}_F \mathcal{M}$, and let $f \in H^0(X, \mathcal{O}_X(\ell H))$ with $D = (f = 0)$. For a general closed point $t = (t_1, t_2) \in \mathbb{A}^2$, we have $c(t_1, t_2) \neq 0$, and $f(1, t_1, t_2, w_1, \ldots, w_{d-1})$ does not vanish identically on X_t. By $(*)$ and (4.2), the pair $(X_t\backslash C, \lambda D|_{X_t\backslash C})$ is canonical, and $X_t\backslash C$ is irreducible. Since F is exceptional over X° and $\pi(\operatorname{center}_{X^\circ} F) = \pi(W \cap X^\circ)$ is dense in $\mathbb{A}^2$, by Lemma 4.6(2) for the smooth surjective morphism $\pi|_{X^\circ} : X^\circ \to \mathbb{A}^2$ and the boundary $\lambda D|_{X^\circ}$, we have

$$\lambda \operatorname{mult}_F \mathcal{M} = a(F, X, 0) - a(F, X^\circ, \lambda D|_{X^\circ}) \leq a(F, X, 0) - 1.$$

Case 2. The closure T of $\pi(W \cap X^\circ)$ is a curve. Let T_1 be the smooth locus of T. Since $T \backslash T_1$ is finite,

$$X_1 := X^\circ \backslash \pi^{-1}(T \backslash T_1)$$

is an open subset of X. Let $P_1 := \pi^{-1}(T) \cap X_1 = \pi^{-1}(T_1) \cap X^\circ$. Since $\pi|_{X^\circ}$ is smooth, $\pi|_{P_1} : P_1 \to T_1$ is smooth and surjective, and P_1 is a smooth divisor on X_1. Since smooth morphisms are open, any connected component of P_1 dominates T_1, and hence intersects $X_t \backslash C$ in a non-empty open subset for a general closed point $t \in T_1$. Since $X_t \backslash C$ is irreducible and the connected components of P_1 are disjoint, P_1 is irreducible. Let P be the closure of P_1 in X. Then P is a prime divisor on X with $P \cap X_1 = P_1$. Since π maps the generic point of W to the generic point of T, we have $W \cap X_1 \neq \emptyset$ and $W \cap X_1 \subset P_1$.

Take a general member D of $\mathcal{M}$, so that $\operatorname{mult}_F D = \operatorname{mult}_F \mathcal{M}$ and $P \not\subset \operatorname{Supp} D$, and let $f \in H^0(X, \mathcal{O}_X(\ell H))$ with $D = (f = 0)$. Then $f(1, u_1, u_2, w_1, \dots, w_{d-1})$ does not vanish identically on P_1, and hence for a general closed point $t = (t_1, t_2) \in T_1$, it does not vanish identically on X_t. By $(*)$ and (4.2), the pair $(X_t \backslash C, \lambda D|_{X_t \backslash C})$ is lc, both when $c(t_1, t_2) \neq 0$ and when $T \subset (c(u_1, u_2) = 0)$. By Lemma 4.6(1) for $\pi|_{P_1} : P_1 \to T_1$ and the boundary $\lambda D|_{P_1}$, there exists a non-empty open subset $T^\circ \subset T_1$ such that the pair

$$(P_1 \cap \pi^{-1}(T^\circ), \lambda D|_{P_1 \cap \pi^{-1}(T^\circ)})$$

is lc. Let

$$X_2 := X_1 \backslash \pi^{-1}(T_1 \backslash T^\circ).$$

Then X_2 is an open subset of X, $P \cap X_2 = P_1 \cap \pi^{-1}(T^\circ)$ is a smooth prime divisor on the smooth variety X_2, and $W \cap X_2 \neq \emptyset$ since $\pi(W \cap X^\circ)$ is dense in T. By Theorem 2.8, $(X_2, P|_{X_2} + \lambda D|_{X_2})$ is lc on an open neighborhood U of $P \cap X_2$ in X_2. Since $\operatorname{center}_U F = W \cap U$ is non-empty and contained in the Cartier divisor $P \cap U$, we have $\operatorname{mult}_F P \geq 1$. Since

$$0 \leq a(F, U, P|_U + \lambda D|_U) = a(F, X, 0) - \operatorname{mult}_F P - \lambda \operatorname{mult}_F D,$$

we obtain

$$\lambda \operatorname{mult}_F \mathcal{M} = \lambda \operatorname{mult}_F D \leq a(F, X, 0) - \operatorname{mult}_F P \leq a(F, X, 0) - 1.$$

Case 3. $\pi(W \cap X^\circ)$ is a closed point $t = (t_1, t_2) \in \mathbb{A}^2$. Let $P := (y_2 - t_2 x^r = 0) \subset X$. On $X \backslash S$, we have $P \cap (X \backslash S) = (u_2 = t_2)$. Since $r \cdot kb = bd$, substituting $y_2 = t_2 x^r$ shows that the graded ring of P is

$$R_P := \mathbb{C}[x, y_1, z_1, \dots, z_{d-1}] / \big((1 + t_2^{kb}) x^{bd} + y_1^{kb} + z_1^d + \cdots + z_{d-1}^d\big),$$

with the weights $1, r, b, \dots, b$. We show that R_P is a domain. The affine cone $\operatorname{Spec} R_P \subset \mathbb{A}^{d+1}$ is a hypersurface, and the singular locus of $\operatorname{Spec} R_P$ is the origin if $1 + t_2^{kb} \neq 0$, and the x-axis $(y_1 = z_1 = \cdots = z_{d-1} = 0)$ if $1 + t_2^{kb} = 0$. In both cases the singular locus has codimension at least 2 in $\operatorname{Spec} R_P$, so R_P is a domain by Lemma 3.4, and P is a prime divisor on X. Substituting $y_1 = t_1 x^r$, we have

$$R_P / (y_1 - t_1 x^r) \cong Q := \mathbb{C}[x, z_1, \dots, z_{d-1}] / \big(c(t_1, t_2) x^{bd} + z_1^d + \cdots + z_{d-1}^d\big).$$

The singular locus of $\operatorname{Spec} Q \subset \mathbb{A}^d$ is the origin if $c(t_1, t_2) \neq 0$, and the x-axis if $c(t_1, t_2) = 0$, which has codimension $d - 2 \geq 2$ in $\operatorname{Spec} Q$. By Lemma 3.4, Q is a domain, that is, $y_1 - t_1 x^r$ generates a prime ideal of R_P. Moreover, $x \neq 0$ in Q.

Take a general member D of $\mathcal{M}$, so that $\operatorname{mult}_F D = \operatorname{mult}_F \mathcal{M}$ and $P \not\subset \operatorname{Supp} D$, and let $f \in H^0(X, \mathcal{O}_X(\ell H))$ with $D = (f = 0)$. Then the image $\bar{f}$ of f in R_P is non-zero. Write

$$\bar{f} = (y_1 - t_1 x^r)^m g,$$

where $m \geq 0$ is the largest integer such that $(y_1 - t_1 x^r)^m$ divides $\bar{f}$, and $g \in R_P$ is homogeneous of weight $\ell - rm \geq 0$. The integer m exists since $rm \leq \ell$ for any such m. In particular, $\lambda m = rm/\ell \leq 1$. Since $g \notin (y_1 - t_1 x^r)$, the image $\bar{g}$ of g in Q is non-zero. Since Q is a domain

and $x \neq 0$ in Q, the element $\bar{g}/x^{\ell-rm}$ is a non-zero element of the degree zero part of $Q[x^{-1}]$, which is the coordinate ring

$$\mathbb{C}[w_1, \dots, w_{d-1}]/\big(c(t_1, t_2) + w_1^d + \cdots + w_{d-1}^d\big)$$

of X_t. Thus $g(1, t_1, w_1, \dots, w_{d-1})$ does not vanish identically on X_t. Any monomial in $x, y_1, z_1, \dots, z_{d-1}$ of weight $\ell - rm$ has degree at most $(\ell - rm)/b$ in $z_1, \dots, z_{d-1}$, and hence $g(1, t_1, w_1, \dots, w_{d-1})$ has degree $q \leq \lfloor(\ell - rm)/b\rfloor$, so that $\lambda q \leq (d-3)/d$ by (4.2).

Since $\pi|_{X^\circ}$ is smooth, $P \cap X^\circ = \pi^{-1}((u_2 = t_2)) \cap X^\circ$ is smooth and is smooth over the line $(u_2 = t_2) \subset \mathbb{A}^2$, and $P \cap X^\circ$ is a prime divisor on X° as a non-empty open subset of P. Thus $X_t \backslash C$ is a smooth prime divisor on $P \cap X^\circ$, and it is the divisor of $u_1 - t_1$ on $P \cap X^\circ$. Setting $x = 1$ in $\bar{f} = (y_1 - t_1 x^r)^m g$, we obtain

$$D|_{P \cap X^\circ} = m(X_t \backslash C) + D_P,$$

where D_P is the divisor of $g(1, u_1, w_1, \dots, w_{d-1})$ on $P \cap X^\circ$ and $X_t \backslash C \not\subset \operatorname{Supp} D_P$. The restriction $D_P|_{X_t \backslash C}$ is the divisor of $g(1, t_1, w_1, \dots, w_{d-1})$, and hence $(X_t \backslash C, \lambda D_P|_{X_t \backslash C})$ is lc by $(*)$. Let p be a closed point of $W \cap X^\circ$. Then

$$\operatorname{center}_{X^\circ} F = W \cap X^\circ \subset X_t \backslash C \subset P \cap X^\circ.$$

Since $\lambda m \leq 1$, by Lemma 2.10 for the smooth variety X°, the smooth prime divisor $P \cap X^\circ$, the smooth prime divisor $X_t \backslash C$ on $P \cap X^\circ$, the point p, and the Cartier divisor $D|_{X^\circ}$, we have

$$\lambda \operatorname{mult}_F \mathcal{M} = \lambda \operatorname{mult}_F D \leq a(F, X^\circ, 0) - 1 = a(F, X, 0) - 1.$$

□

For the curve C itself, we do not use that the linear system is mobile: the equation of D given by Lemma 3.5, whose degree in y_2 is less than kb, bounds the multiplicity of D at a general point of C by ℓ/b, and hence, by [Kol96, 3.14 Exercise, (3.14.1)], the pair $(X, \frac{r}{\ell} D)$ is canonical near such a point. The closed points of C are treated in Section 5.

Lemma 4.8. *Notation and conditions as in Set-up 3.1. Let ℓ be a positive integer, $D \in |\ell H|$, and F a prime divisor over X such that $\operatorname{center}_X F = C$. Then*

$$\frac{r}{\ell} \operatorname{mult}_F D \leq a(F, X, 0) - 1.$$

Proof. Let $\lambda := r/\ell$ and $c(u_1, u_2) := 1 + u_1^{kb} + u_2^{kb}$. By Construction-Proposition 3.2(3)(4), C is isomorphic to the smooth, hence irreducible, plane curve $C_{\mathbb{P}^2}$. By Construction-Proposition 3.2(1), we have

$$C \backslash \Sigma = C \cap (X \backslash S) = \big(c(u_1, u_2) = w_1 = \cdots = w_{d-1} = 0\big),$$

and π maps $C \backslash \Sigma$ isomorphically onto the affine curve $(c(u_1, u_2) = 0) \subset \mathbb{A}^2$. Since $(c(u_1, u_2) = 0) \subset \mathbb{A}^2$ is smooth, c has no multiple factor, and hence any polynomial in u_1, u_2 vanishing on $(c(u_1, u_2) = 0)$ is divisible by c.

By Lemma 3.5, $D = (f = 0)$ for the image f of a unique weighted homogeneous polynomial of degree ℓ whose degree in y_2 is less than kb. Write

$$f = \sum_J z^J P_J(x, y_1, y_2),$$

where $J = (j_1, \dots, j_{d-1}) \in \mathbb{Z}_{\geq 0}^{d-1}$, $z^J := z_1^{j_1} \cdots z_{d-1}^{j_{d-1}}$, $|J| := j_1 + \cdots + j_{d-1}$, and P_J is weighted homogeneous of degree $\ell - b|J|$ with $\deg_{y_2} P_J < kb$. Let J be an index with $P_J \neq 0$. A monomial $x^e y_1^{\alpha_1} y_2^{\alpha_2}$ of weight $\ell - b|J|$ is determined by (α_1, α_2), and hence $P_J(1, u_1, u_2) \neq 0$ and $\deg_{u_2} P_J(1, u_1, u_2) < kb$. Since c is monic of degree kb in u_2, any non-zero multiple of c has degree at least kb in u_2. Thus $P_J(1, u_1, u_2)$ does not vanish identically on $C \backslash \Sigma$. Since C is irreducible, a general closed point $p \in C$ satisfies $p \notin \Sigma$, $u_2(p) \neq 0$, and $P_J(1, u_1(p), u_2(p)) \neq 0$ for any J with $P_J \neq 0$. Let

$$m := \min\{|J| \mid P_J \neq 0\}.$$

Since P_J has weight $\ell - b|J| \geq 0$ when $P_J \neq 0$, we have $m \leq \ell/b$, and hence $\lambda m \leq r/b < 1$.

Fix such a point p. By Construction-Proposition 3.2(1)(2), X is smooth at p, and $X \backslash S$ is the hypersurface $(G = 0) \subset \mathbb{A}^{d+1}$ with $G := c(u_1, u_2) + \sum_j w_j^d$. At p, we have $w_1 = \cdots = w_{d-1} = 0$ and

$$\frac{\partial G}{\partial u_2}(p) = kbu_2(p)^{kb-1} \neq 0.$$

Thus $u_1 - u_1(p), w_1, \ldots, w_{d-1}$ form a system of local coordinates of X at p. Near p, the divisor D is defined by

$$f(1, u_1, u_2, w_1, \ldots, w_{d-1}) = \sum_{|J| \geq m} w^J P_J(1, u_1, u_2),$$

where $w^J := w_1^{j_1} \cdots w_{d-1}^{j_{d-1}}$ and the functions $P_J(1, u_1, u_2)$ are regular at p. The terms with $|J| > m$ have order greater than m at p, and the homogeneous part of degree m of the Taylor expansion at p is

$$\sum_{|J|=m} P_J(1, u_1(p), u_2(p)) w^J \neq 0.$$

Thus $\operatorname{mult}_p D = m$.

Let $e := \lfloor 1/\lambda \rfloor$. Near a smooth point of X at which D is defined by a regular function φ, the set $\{p' \mid \operatorname{mult}_{p'} D \geq e+1\}$ is the common zero locus of φ and the partial derivatives of φ of order at most e, and hence it is closed. Since $\operatorname{mult}_{p'} D$ is an integer, $\operatorname{mult}_{p'}(\lambda D) \leq 1$ if and only if $\operatorname{mult}_{p'} D \leq e$. Thus

$$U := \{p' \in X_{\mathrm{reg}} \mid \operatorname{mult}_{p'}(\lambda D) \leq 1\}$$

is an open subset of X. Since $m < 1/\lambda$, we have $m \leq e$ and $p \in U$. Since $\lambda = r/\ell$ is rational, $\lambda D|_U \geq 0$ is a $\mathbb{Q}$-divisor on the smooth variety U, and hence $(U, \lambda D|_U)$ is canonical by [Kol96, 3.14 Exercise, (3.14.1)]. Since C has codimension $d - 1 \geq 2$ in X, the divisor F is exceptional over X, and $\operatorname{center}_U F = C \cap U \neq \emptyset$. Therefore

$$a(F, X, 0) - \lambda \operatorname{mult}_F D = a(F, U, \lambda D|_U) \geq 1.$$

□

Proof of Proposition 4.1. Let $W := \operatorname{center}_X F$. Since $\mathcal{M}$ has no fixed component, a general member D of $\mathcal{M}$ satisfies $S \not\subset \operatorname{Supp} D$ and $\operatorname{mult}_F D = \operatorname{mult}_F \mathcal{M}$, and hence

$$\frac{r}{\ell} \operatorname{mult}_F D > a(F, X, 0) - 1.$$

Suppose first that $W \subset S$. If $W \not\subset \Sigma$, then Proposition 4.5 implies that $\frac{r}{\ell} \operatorname{mult}_F D \leq a(F, X, 0) - 1$, a contradiction. Thus $W \subset \Sigma$, and since $\Sigma = S \cap C$ is a finite set by Construction-Proposition 3.2(3)(4), W is a closed point of C. Suppose next that $W \not\subset S$. If $W \not\subset C$, then, since $\operatorname{mult}_F D = \operatorname{mult}_F \mathcal{M}$, Proposition 4.7 implies that $\frac{r}{\ell} \operatorname{mult}_F D \leq a(F, X, 0) - 1$, a contradiction. Thus $W \subset C$. Since C is an irreducible curve, either $W = C$ or W is a closed point of C. If $W = C$, then Lemma 4.8 implies that $\frac{r}{\ell} \operatorname{mult}_F D \leq a(F, X, 0) - 1$, a contradiction. Thus W is a closed point of C in both cases. □

5. Points of the Fermat curve

In this section, we prove Theorem 5.1.

Theorem 5.1. *Notation and conditions as in Set-up 3.1. Let ℓ be a positive integer and $\mathcal{M} \subset |\ell H|$ a mobile linear system. Then $(X, \frac{r}{\ell}\mathcal{M})$ is canonical.*

By Proposition 4.1, it suffices to consider exceptional prime divisors over X whose centers on X are closed points of C.

Since the fiber of π through a closed point p of C is a cone with vertex p, Subsections 5.3 and 5.4 compute on the product of a finite cover of a neighborhood of $\pi(p)$ in $\mathbb{P}^2$ with $Y_{d,-1}$, and Subsections 5.1 and 5.2 prove Lemmas 5.2 and 5.4, which are used there.

5.1. **A plane linear system.** In this subsection, we prove two estimates for linear systems of plane curves, and we attach to any mobile linear system in $|qrH|$ a linear system of plane curves without fixed components.

Lemma 5.2. *Let Λ be a linear system of plane curves of degree $q_0 \geq 0$ without fixed components, $p \in \mathbb{P}^2$ a closed point, $I \subset \mathcal{O}_{\mathbb{P}^2,p}$ the ideal generated by the local equations at p of the members of Λ, and $q \geq q_0$ a positive real number. Then*

$$a(v, \mathbb{P}^2, 0) \geq \frac{2}{q} v(I)$$

for any divisorial valuation v over $\mathbb{P}^2$ such that $\operatorname{center}_{\mathbb{P}^2} v = p$. Moreover, if $L \subset \mathbb{P}^2$ is a line through p and h is a local equation of L at p, then

$$a(v, \mathbb{P}^2, 0) - \frac{1}{q} v(I) \geq v(h)$$

for any such v.

Proof. Let v be a divisorial valuation over $\mathbb{P}^2$ such that $\operatorname{center}_{\mathbb{P}^2} v = p$.

We first prove the first inequality. If p is not a base point of Λ, then $I = \mathcal{O}_{\mathbb{P}^2,p}$ and $v(I) = 0$, while $a(v, \mathbb{P}^2, 0) > 0$ by [KM98, Corollary 2.31(2)]. Thus we may assume that p is a base point of Λ. In particular, $q_0 \geq 1$. Since Λ has no fixed components, the base locus of Λ is a finite set, so I is $\mathfrak{m}_p$-primary. Let $D_1, D_2 \in \Lambda$ be general members, and let $f_1, f_2 \in I$ be local equations of D_1, D_2 at p. Any irreducible component of D_1 is not contained in the base locus of Λ, so the general member D_2 does not contain it. Thus D_1 and D_2 have no common component, and $D_1 \cap D_2$ is a finite set containing p. Hence $(f_1, f_2) \subset I$ is an $\mathfrak{m}_p$-primary ideal generated by two elements, that is, a parameter ideal of the regular local ring $\mathcal{O}_{\mathbb{P}^2,p}$ of dimension two. Let $e(\cdot)$ denote the Samuel multiplicity of an $\mathfrak{m}_p$-primary ideal of $\mathcal{O}_{\mathbb{P}^2,p}$. By [Mat86, Formula 14.4], [Mat86, Theorem 17.11], and Bézout's theorem [Har77, Chapter V, Proposition 1.4 and Example 1.4.2], we have

$$e(I) \leq e((f_1, f_2)) = \operatorname{length}\big(\mathcal{O}_{\mathbb{P}^2,p}/(f_1, f_2)\big) = (D_1 \cdot D_2)_p \leq (D_1 \cdot D_2) = q_0^2,$$

where $(D_1 \cdot D_2)_p$ is the local intersection multiplicity of D_1 and D_2 at p.

Let $\operatorname{lct}(I)$ be the log canonical threshold of I in the sense of [dFEM04]. Let $\mathfrak{b} \subset \mathcal{O}_{\mathbb{P}^2}$ be the ideal sheaf generated by the equations of the members of Λ, so that $\mathfrak{b}_p = I$ and p is an irreducible component of the closed subscheme $V(\mathfrak{b})$. Following [dFEM04, p. 605], $\operatorname{lct}(I)$ is the maximum of the log canonical thresholds of U along $U \cap V(\mathfrak{b})$ for the open neighborhoods U of p. Take U attaining the maximum. For any $0 \leq c < \operatorname{lct}(I)$, the pair $(U, c(U \cap V(\mathfrak{b})))$ is lc, that is, $a(v', U, 0) \geq c v'(\mathfrak{b})$ for any divisorial valuation v' over U. Letting c tend to $\operatorname{lct}(I)$, we have $a(v', \mathbb{P}^2, 0) \geq \operatorname{lct}(I) v'(I)$ for any divisorial valuation v' over $\mathbb{P}^2$ such that $\operatorname{center}_{\mathbb{P}^2} v' = p$. By [dFEM04, Theorem 0.1] in the case $n = 2$ (the case $n = 2$ is due to Corti, see [dFEM04, p. 603]), we have

$$e(I) \geq \frac{4}{\operatorname{lct}(I)^2}.$$

Thus $\operatorname{lct}(I) \geq \frac{2}{q_0} \geq \frac{2}{q}$, and the inequality $a(v', \mathbb{P}^2, 0) \geq \operatorname{lct}(I) v'(I)$ above with $v' = v$ implies

$$a(v, \mathbb{P}^2, 0) \geq \operatorname{lct}(I) v(I) \geq \frac{2}{q} v(I).$$

We prove the second inequality. Let $D \in \Lambda$ be a general member and f_D a local equation of D at p. Since Λ has no fixed components, $L \not\subset \operatorname{Supp} D$. Since I is generated by the local equations at p of finitely many members of Λ spanning Λ, and the set of members whose local equations f' satisfy $v(f') > v(I)$ corresponds to a proper linear subspace, we have $v(f_D) = v(I)$. Since $L \cong \mathbb{P}^1$ and $(D \cdot L) = q_0$, the divisor $D|_L$ on L has degree q_0, so any coefficient of $\frac{1}{q} D|_L$ is at most $\frac{q_0}{q} \leq 1$. Thus $(L, \frac{1}{q} D|_L)$ is lc. By Theorem 2.8, $(\mathbb{P}^2, L + \frac{1}{q} D)$ is lc near L. Since $p \in L$, we have

$$0 \leq a\Big(v, \mathbb{P}^2, L + \frac{1}{q} D\Big) = a(v, \mathbb{P}^2, 0) - v(h) - \frac{1}{q} v(f_D) = a(v, \mathbb{P}^2, 0) - v(h) - \frac{1}{q} v(I),$$

which is the second inequality. □

Both inequalities of Lemma 5.2 are equalities when Λ is the linear system spanned by the unions of q lines through p, so that $q_0 = q$ and $I = \mathfrak{m}_p^q$, and v is the valuation given by the exceptional curve of the blow-up of $\mathbb{P}^2$ at p (here $a(v, \mathbb{P}^2, 0) = 2$ by [KM98, Lemma 2.29], while $v(I) = q$ and $v(h) = 1$). The hypothesis that Λ has no fixed components cannot be dropped: if $\Lambda = \{qL\}$ for a line L through p with local equation h at p, and v is the monomial valuation in local coordinates s, h at p with $v(s) = 1$ and $v(h) = n$ for an integer $n \geq 2$, then the first inequality fails (here $a(v, \mathbb{P}^2, 0) = 1 + n$, the sum of the weights of v, while $\frac{2}{q}v(I) = 2n$).

Construction-Proposition 5.3. Notation and conditions as in Set-up 3.1. Let q be a positive integer, $\mathcal{M} \subset |qrH|$ a mobile linear system, and $W_{\mathcal{M}} \subset H^0(X, \mathcal{O}_X(qrH))$ the subspace of sections defining the members of $\mathcal{M}$. Then the following hold.

(1) Any $f \in H^0(X, \mathcal{O}_X(qrH))$ can be written uniquely as

$$f = \sum_J x^{s_J} z^J P_{f,J}(x^r, y_1, y_2),$$

where $J = (j_1, \ldots, j_{d-1})$ runs over $\mathbb{Z}_{\geq 0}^{d-1}$ with $j_{d-1} < d$,

$$z^J := z_1^{j_1} \cdots z_{d-1}^{j_{d-1}}, \qquad |J| := j_1 + \cdots + j_{d-1},$$

$s_J \in \{0, 1, \ldots, r-1\}$ is the integer such that r divides $s_J + b|J|$, and $P_{f,J} \in \mathbb{C}[w, y_1, y_2]$ is either zero or a homogeneous polynomial of degree $q - \frac{s_J + b|J|}{r}$.

(2) For any f and J as in (1) with $P_{f,J} \neq 0$, we have $b|J| \leq qr$.

(3) The non-zero polynomials among $P_{f,J}$, where $f \in W_{\mathcal{M}}$ and J is as in (1), have no common factor of positive degree.

(4) For any closed point $p \in \mathbb{P}^2$ and any linear form A on $\mathbb{P}^2$ with $A(p) \neq 0$, the ideal $I_p \subset \mathcal{O}_{\mathbb{P}^2,p}$ generated by $P_{f,J}/A^{\deg P_{f,J}}$ for all $f \in W_{\mathcal{M}}$ and J with $P_{f,J} \neq 0$ does not depend on A, and there exists a linear system Λ_p of plane curves of some degree $q_p \leq q$ without fixed components such that I_p is generated by the local equations at p of the members of Λ_p.

Proof. We prove (1). By Lemma 3.5, $H^0(X, \mathcal{O}_X(qrH)) = R_{qr}$, and any $f \in R_{qr}$ is the image of a unique weighted homogeneous polynomial $\tilde{f}$ of degree qr whose degree in z_{d-1} is less than d. Write

$$\tilde{f} = \sum_J z^J \tilde{f}_J(x, y_1, y_2),$$

where J runs over $\mathbb{Z}_{\geq 0}^{d-1}$ with $j_{d-1} < d$, and $\tilde{f}_J \in \mathbb{C}[x, y_1, y_2]$ is either zero or weighted homogeneous of degree $qr - b|J|$. For any monomial $x^i y_1^{i_1} y_2^{i_2}$ of $\tilde{f}_J$, we have $i + r(i_1 + i_2) = qr - b|J|$, so $i \equiv s_J \pmod r$. Since $0 \leq s_J \leq r - 1$, we have $i \geq s_J$ and r divides $i - s_J$. Thus $\tilde{f}_J = x^{s_J} P_{f,J}(x^r, y_1, y_2)$ for a unique $P_{f,J} \in \mathbb{C}[w, y_1, y_2]$, and $P_{f,J}$ is either zero or homogeneous of degree

$$\frac{qr - b|J| - s_J}{r} = q - \frac{s_J + b|J|}{r}.$$

Conversely, for any expression of f as in (1), the polynomial $\sum_J x^{s_J} z^J P_{f,J}(x^r, y_1, y_2)$ is weighted homogeneous of degree qr with degree in z_{d-1} less than d, so it is equal to $\tilde{f}$ by the uniqueness in Lemma 3.5. Comparing the coefficients of z^J, the polynomials $P_{f,J}$ are determined by f. This proves (1).

We prove (2). If $P_{f,J} \neq 0$, then $q - \frac{s_J + b|J|}{r} = \deg P_{f,J} \geq 0$, so $b|J| \leq s_J + b|J| \leq qr$.

We prove (3). Suppose that a homogeneous polynomial $P_0 \in \mathbb{C}[w, y_1, y_2]$ of degree $e_0 > 0$ divides all the non-zero polynomials $P_{f,J}$ with $f \in W_{\mathcal{M}}$. Let $g_0 \in R_{re_0}$ be the image of $P_0(x^r, y_1, y_2)$. The polynomial $P_0(x^r, y_1, y_2)$ is non-zero, weighted homogeneous of degree re_0, and does not involve z_{d-1}. By the uniqueness in Lemma 3.5, $g_0 \neq 0$. Thus g_0 is a non-zero

section of $\mathcal{O}_X(re_0H)$, and $G_0 := \operatorname{div}(g_0)$ is a divisor such that $G_0 \geq 0$ and $G_0 \sim re_0H$. Since $re_0 > 0$ and $\operatorname{Cl}(X) = \mathbb{Z}[H]$ (Proposition 3.3(1)), we have $G_0 \neq 0$. For any $f \in W_{\mathcal{M}}$, (1) implies that

$$f = g_0 \sum_J x^{s_J} z^J \frac{P_{f,J}}{P_0}(x^r, y_1, y_2)$$

in R, where the sum runs over J with $P_{f,J} \neq 0$, and the second factor is an element of R_{qr-re_0}. Thus $\operatorname{div}(f) \geq G_0$ for any $0 \neq f \in W_{\mathcal{M}}$. Hence any prime component of G_0 is a fixed component of $\mathcal{M}$, which is not possible since $\mathcal{M}$ is mobile. This proves (3).

We prove (4). For any linear form A' on $\mathbb{P}^2$ with $A'(p) \neq 0$, we have

$$\frac{P_{f,J}}{A'^{\deg P_{f,J}}} = \frac{P_{f,J}}{A^{\deg P_{f,J}}} \cdot \left(\frac{A}{A'}\right)^{\deg P_{f,J}},$$

and A/A' is a unit of $\mathcal{O}_{\mathbb{P}^2,p}$. Thus I_p does not depend on A. For any $f \in W_{\mathcal{M}}$ and J with $P_{f,J} \neq 0$, let

$$Q_{f,J} := A^{q-\deg P_{f,J}} P_{f,J},$$

which is a homogeneous polynomial of degree q by (1). Let Q_0 be a greatest common divisor of all the $Q_{f,J}$. Any prime factor of Q_0 which is not a constant multiple of A divides all the non-zero $P_{f,J}$, which contradicts (3). Thus we may assume that $Q_0 = A^{q-q_p}$ for some integer $0 \leq q_p \leq q$. Let Λ_p be the linear system of plane curves of degree q_p defined by the linear span of all the $Q_{f,J}/A^{q-q_p}$. The polynomials $Q_{f,J}/A^{q-q_p}$ have no common factor of positive degree, so Λ_p has no fixed components. The ideal generated by the local equations at p of the members of Λ_p is generated by

$$\frac{Q_{f,J}}{A^{q-q_p}A^{q_p}} = \frac{P_{f,J}}{A^{\deg P_{f,J}}}$$

for all $f \in W_{\mathcal{M}}$ and J with $P_{f,J} \neq 0$, hence equal to I_p. This proves (4). □

5.2. **A relative estimate.** In this subsection, we prove a lower bound for log discrepancies on the product of a smooth affine surface and $Y_{N,-1}$.

Lemma 5.4 bounds from below the log discrepancy of a divisorial valuation v over $T \times Y_{N,-1}$, minus a multiple of the value of v on a linear system on $T \times Y_{N,-1}$, by the log discrepancy of $(\mathrm{pr}_1)_*v$ over T minus that multiple of the value of $(\mathrm{pr}_1)_*v$ on the ideal of $\mathcal{O}(T)$ generated by the coefficients of the sections of that linear system.

Lemma 5.4. *Notation as in Construction 4.2. Let $N \geq 5$ be an integer, T a smooth affine surface, $m \geq 0$ an integer, and $\lambda \geq 0$ a real number such that $\lambda m \leq \frac{N-3}{N}$. Let $Y := Y_{N,-1}$, and let $W \subset H^0(T \times Y, \mathrm{pr}_2^*\mathcal{O}_Y(m))$ be a non-zero linear subspace with $\dim W < +\infty$. Under the identification*

$$H^0(T \times Y, \mathrm{pr}_2^*\mathcal{O}_Y(m)) = \mathcal{O}(T) \otimes_{\mathbb{C}} H^0(Y, \mathcal{O}_Y(m)),$$

let $\mathfrak{a} \subset \mathcal{O}(T)$ be the smallest ideal such that $W \subset \mathfrak{a} \otimes_{\mathbb{C}} H^0(Y, \mathcal{O}_Y(m))$. Let v be a divisorial valuation over $T \times Y$ such that $v_T := (\mathrm{pr}_1)_*v$ is non-trivial. Then

$$a(v, T \times Y, 0) - \lambda \min_{0 \neq \sigma \in W} v(\operatorname{div}(\sigma)) \geq a(v_T, T, 0) - \lambda v_T(\mathfrak{a}).$$

Proof. The identification in the statement follows from flat base change [Har77, Chapter III, Proposition 9.3], applied to $Y \to \operatorname{Spec}\mathbb{C}$ and the flat morphism $T \to \operatorname{Spec}\mathbb{C}$, since T is affine. Let $\sigma_1, \ldots, \sigma_n$ be a basis of W and $\tau_1, \ldots, \tau_{n'}$ a basis of $H^0(Y, \mathcal{O}_Y(m))$, and write

$$\sigma_i = \sum_{j=1}^{n'} \varphi_{ij} \otimes \tau_j, \quad \varphi_{ij} \in \mathcal{O}(T).$$

Then $\mathfrak{a}$ is the ideal generated by all the φ_{ij}. By Lemma 4.3(1), Y is a smooth projective variety.

Since v is non-negative on the regular functions on $T \times Y$, v_T is non-negative on $\mathcal{O}(T)$. Thus v_T has a center on T, namely the closure of $\mathrm{pr}_1(\mathrm{center}_{T\times Y} v)$. By Lemma 2.12, v_T is a divisorial valuation over T. By [KM98, Lemma 2.45], after blowing up the center of v_T finitely many times,

we obtain a projective birational morphism $g : T' \to T$, a prime divisor E on T', and a real number $t > 0$ such that

$$v_T(\varphi) = t \operatorname{mult}_E \operatorname{div}(g^*\varphi)$$

for any non-zero rational function φ on T. Since T is a smooth surface and the blown up centers are closed points, T' is smooth. Let $g_Y := g \times \mathrm{id}_Y : T' \times Y \to T \times Y$. The center of v on $T' \times Y$ exists since g_Y is proper, and the generic point of $\operatorname{center}_{T'\times Y} v$ maps to the generic point of E, since E is the center of $(\mathrm{pr}_1)_* v = v_T$ on T'. In particular, $v(\mathrm{pr}_1^* P) = t \operatorname{mult}_E P$ for any $\mathbb{R}$-divisor P on T'.

We compare the log discrepancies over $T \times Y$ and $T' \times Y$. Write

$$K_{T'} = g^* K_T + \sum_i (a(E_i, T, 0) - 1) E_i,$$

where E_i runs over the g-exceptional prime divisors. Then

$$K_{T'\times Y} - \mathrm{pr}_1^* \sum_i (a(E_i, T, 0) - 1) E_i = g_Y^* K_{T\times Y}.$$

Since $a(E, T, 0) = 1$ when E is not g-exceptional, we have

$$a(v, T \times Y, 0) = a(v, T' \times Y, 0) + (a(E, T, 0) - 1)t. \tag{5.1}$$

We factor the pullbacks of the sections of W along E. Let

$$\rho := \frac{1}{t} v_T(\mathfrak{a}) = \min\{\operatorname{mult}_E \operatorname{div}(g^*\varphi_{ij}) \mid \varphi_{ij} \neq 0\},$$

which is a non-negative integer. Let $T'_0 \subset T'$ be an affine open subset such that $E \cap T'_0$ is a non-empty smooth curve and $E|_{T'_0} = \operatorname{div}(\psi)$ for some regular function ψ on T'_0. Since $\operatorname{div}(g^*\varphi_{ij}) - \rho E \geq 0$ on T'_0 for any $\varphi_{ij} \neq 0$ and T'_0 is normal, $\varphi'_{ij} := g^*\varphi_{ij}/\psi^\rho$ is a regular function on T'_0 for any i, j. By the definition of ρ, some φ'_{ij} does not vanish identically on $E \cap T'_0$. For any $\sigma = \sum_i c_i \sigma_i \in W$ with $c_i \in \mathbb{C}$, let

$$\bar{\sigma} := \sum_{i,j} c_i \varphi'_{ij} \otimes \tau_j \in H^0(T'_0 \times Y, \mathrm{pr}_2^* \mathcal{O}_Y(m)).$$

Then $g_Y^*\sigma = \mathrm{pr}_1^* \psi^\rho \cdot \bar{\sigma}$ on $T'_0 \times Y$. Since the generic point of $\operatorname{center}_{T'\times Y} v$ maps to the generic point of E, the center of v intersects $T'_0 \times Y$, and we may compute the values of v on $T'_0 \times Y$. Thus

$$v(\operatorname{div}(\sigma)) = \rho t + v(\operatorname{div}(\bar{\sigma}))$$

for any $0 \neq \sigma \in W$.

Let $W_1 \subset W$ be the set of $\sigma \in W$ such that $\sigma = 0$ or

$$v(\operatorname{div}(\sigma)) > \min_{0 \neq \sigma'' \in W} v(\operatorname{div}(\sigma'')).$$

Let $W_2 \subset W$ be the set of $\sigma \in W$ such that $\bar{\sigma}$ vanishes identically on $(E \cap T'_0) \times Y$. Then W_1 and W_2 are linear subspaces of W. Since the minimum is attained, $W_1 \neq W$. Since $\tau_1, \ldots, \tau_{n'}$ are linearly independent and some φ'_{ij} does not vanish identically on $E \cap T'_0$, we have $\sigma_i \notin W_2$ for some i, so $W_2 \neq W$. Thus there exists $\sigma \in W$ with $\sigma \notin W_1 \cup W_2$, and

$$\min_{0 \neq \sigma'' \in W} v(\operatorname{div}(\sigma'')) = v(\operatorname{div}(\sigma)) = \rho t + v(\operatorname{div}(\bar{\sigma})). \tag{5.2}$$

We apply inversion of adjunction along $E \times Y$. Let $E^\circ := E \cap T'_0$ and $R := \operatorname{div}(\bar{\sigma}|_{E^\circ \times Y})$, which is a divisor on $E^\circ \times Y$ with $R \geq 0$ since $\sigma \notin W_2$ and $E^\circ \times Y$ is irreducible. Write $\bar{\sigma}|_{E^\circ \times Y} = \sum_j \chi_j \otimes \tau_j$ with $\chi_j := \sum_i c_i \varphi'_{ij}|_{E^\circ}$. Some χ_j is a non-zero regular function on the irreducible curve E°, so for a general closed point $s \in E^\circ$, the restriction of $\bar{\sigma}$ to $\{s\} \times Y \cong Y$ is a non-zero section of $\mathcal{O}_Y(m)$, and $R_s := R|_{\{s\}\times Y} \in |\mathcal{O}_Y(m)|$. If $m \geq 1$, then $\lambda \leq \frac{N-3}{Nm}$, so $(Y, \lambda R_s)$ is canonical by Lemma 4.3(1) applied with $c = -1$ and $q = m$. Moreover, any prime divisor on Y contains a curve, along which R_s has multiplicity at most m by Theorem 2.9(2), so

the coefficients of λR_s are at most $\lambda m < 1$. Hence $(Y, \lambda R_s)$ is lc. If $m = 0$, then $R_s = 0$, and $(Y, \lambda R_s)$ is lc since Y is smooth. By Lemma 4.6(1) applied to $\mathrm{pr}_1 : E^\circ \times Y \to E^\circ$ and $B := \lambda R$, there exists a non-empty open subset $E^{\circ\circ} \subset E^\circ$ such that the pair

$$\left(E^{\circ\circ} \times Y, \lambda R|_{E^{\circ\circ}\times Y}\right)$$

is lc. The set $E^\circ \backslash E^{\circ\circ}$ is a finite set of closed points of T_0'. Let $T_1' := T_0' \backslash (E^\circ \backslash E^{\circ\circ})$, so that $E \cap T_1' = E^{\circ\circ}$. Then $E^{\circ\circ} \times Y$ is a smooth prime divisor on the smooth variety $T_1' \times Y$, and it is not contained in $\operatorname{Supp} \operatorname{div}(\bar{\sigma})$ since $\sigma \notin W_2$. By Theorem 2.8, the pair

$$\left(T_1' \times Y, E^{\circ\circ} \times Y + \lambda \operatorname{div}(\bar{\sigma})|_{T_1'\times Y}\right)$$

is lc near $E^{\circ\circ} \times Y$. Since the generic point of $\operatorname{center}_{T'\times Y} v$ maps to the generic point of E, the center of v intersects $T_1' \times Y$ and is contained in $E \times Y$. Since $v(E \times Y) = v(\mathrm{pr}_1^* \psi) = t$, we have

$$a(v, T' \times Y, 0) - t - \lambda v(\operatorname{div}(\bar{\sigma})) \geq 0.$$

We conclude the proof. By Definition 2.1, $a(v_T, T, 0) = ta(E, T, 0)$, and $v_T(\mathfrak{a}) = \rho t$ by the definition of ρ. By (5.1) and (5.2), we have

$$\begin{aligned} a(v, T \times Y, 0) - \lambda \min_{0\neq\sigma''\in W} v(\operatorname{div}(\sigma'')) &= a(v, T' \times Y, 0) + (a(E, T, 0) - 1)t - \lambda\rho t - \lambda v(\operatorname{div}(\bar{\sigma})) \\ &\geq t + (a(E, T, 0) - 1)t - \lambda\rho t \\ &= a(v_T, T, 0) - \lambda v_T(\mathfrak{a}). \end{aligned}$$

□

In the rest of this section, we bound log discrepancies of prime divisors over X whose centers are closed points of C, in the following set-up.

Set-up 5.5. Notation and conditions as in Set-up 3.1. Let ℓ be a positive integer and $\mathcal{M} \subset |\ell H|$ a mobile linear system. For a positive integer N, let $\mathcal{M}^{[N]} \subset |N\ell H|$ be the linear system spanned by $D_1 + \cdots + D_N$ for all $D_1, \ldots, D_N \in \mathcal{M}$. Let F be a prime divisor over X such that $p := \operatorname{center}_X F$ is a closed point of C.

The linear systems $\mathcal{M}^{[N]}$ are used twice in the proofs below: once to make the degree of the linear system divisible by r, as Construction-Proposition 5.3 assumes, and once at the points of Σ, with N arbitrarily large, to remove an error term coming from the exponents s_J of x in Construction-Proposition 5.3(1).

Definition-Lemma 5.6. Notation and conditions as in Set-up 5.5. Then the following hold.

(1) For any positive integer N, the linear system $\mathcal{M}^{[N]}$ is mobile, and $\operatorname{mult}_E \mathcal{M}^{[N]} = N \operatorname{mult}_E \mathcal{M}$ for any prime divisor E over X.
(2) The morphism π is defined on a neighborhood of p, and the pushforward v_F under π of the divisorial valuation given by F is a divisorial valuation over $\mathbb{P}^2$ with center $\pi(p)$.

Proof. (1) Let E be a prime divisor over X. The sections defining the members of $\mathcal{M}^{[N]}$ are linear combinations of products of N sections defining members of $\mathcal{M}$, and the divisor of such a product is $D_1 + \cdots + D_N$ with $D_1, \ldots, D_N \in \mathcal{M}$, whose multiplicity along E is at least $N \operatorname{mult}_E \mathcal{M}$. The multiplicity along E of the divisor of a linear combination of such products is at least the minimum of the multiplicities along E of the divisors of the products, and the N-th power of a section whose divisor D satisfies $\operatorname{mult}_E D = \operatorname{mult}_E \mathcal{M}$ defines $ND \in \mathcal{M}^{[N]}$. Thus $\operatorname{mult}_E \mathcal{M}^{[N]} = N \operatorname{mult}_E \mathcal{M}$. Since $\mathcal{M}$ has no fixed component, $\operatorname{mult}_E \mathcal{M} = 0$ for any prime divisor E on X, so $\mathcal{M}^{[N]}$ has no fixed component either.

(2) Since $p \in C$ and $C \cap V = \emptyset$ (Construction-Proposition 3.2(2)), π is a morphism on a neighborhood of p, and π is dominant by Construction-Proposition 3.2(1). The pushforward under π of the divisorial valuation given by F has center $\pi(p)$ on $\mathbb{P}^2$, and it takes positive values at the regular functions near $\pi(p)$ vanishing at $\pi(p)$. By Lemma 2.12, it is a divisorial valuation over $\mathbb{P}^2$ with center $\pi(p)$. □

Both Proposition 5.9 and Proposition 5.10 compute the log discrepancy of F on the product of a finite cover of a neighborhood of $\pi(p)$ in $\mathbb{P}^2$ with the Fermat hypersurface $Y_{d,-1}$, by means of the following identity.

Lemma 5.7. *Notation and conditions as in Set-up 5.5, and notation as in Construction 4.2. Let $T \subset \mathbb{P}^2$ be an affine open neighborhood of $\pi(p)$, $\widetilde{T}$ a smooth affine surface with a finite surjective morphism $\widetilde{T} \to T$, $\tilde{t} \in \widetilde{T}$ a closed point over $\pi(p)$, h a regular function on $\widetilde{T}$, $Y := Y_{d,-1}$, and $g : \widetilde{T} \times Y \dashrightarrow X$ a dominant rational map such that $\pi \circ g$ is the composition of the first projection pr_1 and $\widetilde{T} \to T$. Let F' be a prime divisor over $\widetilde{T} \times Y$ and e a positive integer. Assume that there exist an open neighborhood $U \subset X$ of p and a nowhere vanishing d-form Ω on the smooth locus of U such that*

$$\operatorname{div}(g^*\Omega) = (d-2)\operatorname{div}(\mathrm{pr}_1^* h)$$

on a neighborhood of $\{\tilde{t}\} \times Y$, that the center of F' on $\widetilde{T} \times Y$ is contained in $\{\tilde{t}\} \times Y$, and that g_* maps the divisorial valuation given by F' to e times the divisorial valuation given by F. Then

$$ea(F, X, 0) = a(F', \widetilde{T} \times Y, 0) + (d-2)\operatorname{mult}_{F'} \operatorname{div}(\mathrm{pr}_1^* h).$$

Proof. Choose a normal variety X' and a projective birational morphism $\varphi : X' \to \widetilde{T} \times Y$ such that F' is a prime divisor on X' and $\psi := g \circ \varphi : X' \to X$ is a morphism (such an X' exists: by [KM98, Lemma 2.45], F' is a prime divisor on the normalization X_1 of a variety given by finitely many blow-ups starting from $\widetilde{T} \times Y$, and we take for X' the normalization of the closure of the graph of the rational map $X_1 \dashrightarrow X$ induced by g). Since g_* maps the divisorial valuation given by F' to e times the divisorial valuation given by F, we have $\psi(F') = p$. Since $p \in C$ and $C \cap V = \emptyset$ (Construction-Proposition 3.2(2)), we have $p \in \pi^{-1}(T)$. Possibly shrinking U, we may assume that $U \subset \pi^{-1}(T)$.

Let $X'_U := \psi^{-1}(U)$. We show that $\psi|_{X'_U} : X'_U \to U$ is proper. Since φ is proper, the morphism $(\varphi, \psi) : X' \to (\widetilde{T} \times Y) \times X$ is proper, and so is the induced morphism $X'_U \to (\widetilde{T} \times Y) \times U$. The morphism $\pi \circ \psi|_{X'_U}$ and the composition of $\mathrm{pr}_1 \circ \varphi|_{X'_U}$ with $\widetilde{T} \to T$ coincide on a dense open subset of X'_U, hence on X'_U. Thus (φ, ψ) maps X'_U into the closed subset $(\widetilde{T} \times Y) \times_T U$ of $(\widetilde{T} \times Y) \times U$. Since $\widetilde{T} \to T$ is finite and Y is projective, the projection $(\widetilde{T} \times Y) \times_T U \to U$ is proper. Therefore $\psi|_{X'_U}$ is proper.

Since g is dominant and $\dim(\widetilde{T} \times Y) = d = \dim X$, the morphism $\psi|_{X'_U}$ is surjective and generically finite. Let

$$X'_U \xrightarrow{\psi_0} Z \xrightarrow{f} U$$

be the Stein factorization of $\psi|_{X'_U}$. Then Z is normal since $\mathcal{O}_Z = (\psi_0)_* \mathcal{O}_{X'_U}$ and X'_U is normal, and f is finite and surjective by [Har77, Chapter III, Corollary 11.5], since $\psi|_{X'_U}$ is projective (φ is projective, $\widetilde{T} \to T$ is finite, and Y is projective), and ψ_0 is birational since $\psi|_{X'_U}$ is generically finite and ψ_0 has connected fibers. We take the canonical divisors $K_U := \operatorname{div}\Omega$ of U, $K_Z := \operatorname{div}(f^*\Omega)$ of Z, and $K_{X'} := \operatorname{div}(\psi^*\Omega)$ of X'_U. Since Ω is nowhere vanishing on the smooth locus of U and U is normal, $K_U = 0$. Let $B_Z := -K_Z$. Then $K_Z + B_Z = f^* K_U = 0$. Let v_Z be the pushforward to Z of the divisorial valuation given by F'. Since $\psi(F') = p \in U$, F' is a prime divisor on X'_U. Since ψ_0 is birational and $(\psi_0)_* K_{X'} = K_Z$, we have

$$a(v_Z, Z, B_Z) = 1 + \operatorname{mult}_{F'}\big(K_{X'} - \psi_0^*(K_Z + B_Z)\big) = 1 + \operatorname{mult}_{F'} \operatorname{div}(\psi^*\Omega).$$

Since $f_* v_Z$ is the pushforward under ψ of the divisorial valuation given by F', it is e times the divisorial valuation given by F, and Lemma 2.13 implies that $a(v_Z, Z, B_Z) = ea(F, X, 0)$. Hence

$$ea(F, X, 0) = 1 + \operatorname{mult}_{F'} \operatorname{div}(\psi^*\Omega). \tag{5.3}$$

The variety Y is a smooth Fermat hypersurface, so $\widetilde{T} \times Y$ is smooth. Let $K_{\widetilde{T}\times Y} := \operatorname{div}(g^*\Omega)$. Since $\psi^*\Omega = \varphi^*(g^*\Omega)$, we have

$$a(F', \widetilde{T} \times Y, 0) = 1 + \operatorname{mult}_{F'} \operatorname{div}(\psi^*\Omega) - \operatorname{mult}_{F'} \varphi^* \operatorname{div}(g^*\Omega).$$

Combining the last equality with (5.3), we obtain

$$ea(F, X, 0) = a(F', \widetilde{T} \times Y, 0) + \operatorname{mult}_{F'} \varphi^* \operatorname{div}(g^*\Omega).$$

The center $\varphi(F')$ of F' on $\widetilde{T} \times Y$ is contained in $\{\tilde{t}\} \times Y$, and on a neighborhood of $\{\tilde{t}\} \times Y$ we have

$$\operatorname{div}(g^*\Omega) = (d-2) \operatorname{div}(\operatorname{pr}_1^* h).$$

Thus

$$\operatorname{mult}_{F'} \varphi^* \operatorname{div}(g^*\Omega) = (d-2) \operatorname{mult}_{F'} \operatorname{div}(\operatorname{pr}_1^* h),$$

which proves the lemma. □

Both Proposition 5.9 and Proposition 5.10 compute the pullback of a d-form on X to $\widetilde{T} \times Y_{d,-1}$ by means of the following residue form.

Construction-Proposition 5.8. Notation and conditions as in Construction 4.2. Let $Y := Y_{N,-1}$, let $\eta_j := \xi_j/\xi_0$ for $1 \leq j \leq N-1$, and let ω_Y be the residue along Y of the rational $(N-1)$-form on $\mathbb{P}^{N-1}$ which equals

$$\frac{d\eta_1 \wedge \cdots \wedge d\eta_{N-1}}{\eta_1^N + \cdots + \eta_{N-1}^N - 1}$$

on $(\xi_0 \neq 0)$. Then the following hold.

(1) The form ω_Y is a nowhere vanishing $(N-2)$-form on Y.

(2) For any smooth variety Z and any regular function h on Z, we have

$$d(h\eta_1) \wedge \cdots \wedge d(h\eta_{N-1}) = N \cdot h^{N-2} dh \wedge \omega_Y$$

on $Z \times \big(Y \cap (\xi_0 \neq 0)\big)$, where h, η_j, and ω_Y also denote their pullbacks to $Z \times Y$.

Proof. By Lemma 4.3(1), Y is smooth. Thus Y is irreducible: two irreducible components of Y would intersect [Har77, Chapter I, Theorem 7.2], and Y would be singular at their intersection.

(1) Since $\deg Y = N$, the rational $(N-1)$-form above has neither zeros nor poles along $(\xi_0 = 0)$, so it is a nowhere vanishing section of $\omega_{\mathbb{P}^{N-1}}(Y) \cong \mathcal{O}_{\mathbb{P}^{N-1}}$. By the adjunction formula, ω_Y is a nowhere vanishing $(N-2)$-form on Y, also along $Y \cap (\xi_0 = 0)$.

(2) On $Z \times \big(Y \cap (\xi_0 \neq 0)\big)$, we have $d(h\eta_j) = \eta_j dh + h d\eta_j$, so

$$d(h\eta_1) \wedge \cdots \wedge d(h\eta_{N-1}) = h^{N-1} d\eta_1 \wedge \cdots \wedge d\eta_{N-1} + h^{N-2} dh \wedge \theta,$$

where

$$\theta := \sum_{j=1}^{N-1} (-1)^{j-1} \eta_j d\eta_1 \wedge \cdots \wedge \widehat{d\eta_j} \wedge \cdots \wedge d\eta_{N-1}.$$

The first term vanishes because $\dim Y = N-2$. Let $G := \eta_1^N + \cdots + \eta_{N-1}^N - 1$, so that $Y \cap (\xi_0 \neq 0) = (G = 0)$. On $Y \cap (\xi_0 \neq 0) \cap (\eta_1 \neq 0)$, the equation $G = 0$ implies that

$$d\eta_1 = -\sum_{j=2}^{N-1} \Big(\frac{\eta_j}{\eta_1}\Big)^{N-1} d\eta_j.$$

Substituting the expression of $d\eta_1$ into θ, we obtain

$$\theta = \Big(\eta_1 + \sum_{j=2}^{N-1} \frac{\eta_j^N}{\eta_1^{N-1}}\Big) d\eta_2 \wedge \cdots \wedge d\eta_{N-1} = \eta_1^{1-N} d\eta_2 \wedge \cdots \wedge d\eta_{N-1}$$

on $Y \cap (\xi_0 \neq 0) \cap (\eta_1 \neq 0)$. On the other hand, near $Y \cap (\xi_0 \neq 0) \cap (\eta_1 \neq 0)$ we have

$$d\eta_1 \wedge \cdots \wedge d\eta_{N-1} = \frac{1}{N \cdot \eta_1^{N-1}} dG \wedge d\eta_2 \wedge \cdots \wedge d\eta_{N-1},$$

so by the definition of the residue,

$$\omega_Y = \frac{d\eta_2 \wedge \cdots \wedge d\eta_{N-1}}{N \cdot \eta_1^{N-1}}$$

on $Y \cap (\xi_0 \neq 0) \cap (\eta_1 \neq 0)$. Thus $\theta|_Y = N \cdot \omega_Y$ there. Both sides are regular forms on $Y \cap (\xi_0 \neq 0)$, and $Y \cap (\xi_0 \neq 0) \cap (\eta_1 \neq 0)$ is a dense open subset of it because Y is irreducible and $Y \not\subset (\xi_1 = 0)$. Hence $\theta|_Y = N \cdot \omega_Y$ on $Y \cap (\xi_0 \neq 0)$, and

$$d(h\eta_1) \wedge \cdots \wedge d(h\eta_{N-1}) = h^{N-2} dh \wedge \theta = N \cdot h^{N-2} dh \wedge \omega_Y$$

on $Z \times \big(Y \cap (\xi_0 \neq 0)\big)$. □

5.3. **Smooth points.** In this subsection, we prove Proposition 5.9, which bounds the log discrepancy of F when p is not a point of Σ. Part (c) of the sketch of the proof in Section 1 outlines the proof.

Proposition 5.9. *Notation and conditions as in Set-up 5.5. Assume that $p \notin \Sigma$. Then*

$$a(F, X, 0) - \frac{r}{\ell} \operatorname{mult}_F \mathcal{M} \geq \frac{d-2}{2} - \frac{r}{b}.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we make ℓ divisible by r and push the valuation of F forward to $\mathbb{P}^2$.

By Definition-Lemma 5.6(1), $\mathcal{M}^{[r]} \subset |r\ell H|$ is a mobile linear system and

$$\frac{r}{r\ell} \operatorname{mult}_F \mathcal{M}^{[r]} = \frac{r}{\ell} \operatorname{mult}_F \mathcal{M}.$$

Possibly replacing $\mathcal{M}$ with $\mathcal{M}^{[r]}$, we may assume that $\ell = qr$ for a positive integer q. Let $\lambda := 1/q$. Then $\frac{r}{\ell} \operatorname{mult}_F \mathcal{M} = \lambda \operatorname{mult}_F \mathcal{M}$. Let v_F be the divisorial valuation over $\mathbb{P}^2$ of Definition-Lemma 5.6(2), and let

$$\tau := a(v_F, \mathbb{P}^2, 0).$$

Step 2. In this step we construct a finite cover $\widetilde{T}$ of a neighborhood T of $\pi(p)$ and compute log discrepancies on $\widetilde{T} \times Y_{d,-1}$ by Lemma 5.7.

Since $\Sigma = S \cap C$ and $p \in C \backslash \Sigma$, we have $p \notin S$. We use the coordinates $u_1, u_2, w_1, \ldots, w_{d-1}$ of $X \backslash S$ of Construction-Proposition 3.2(1), and we also denote by u_1, u_2 the affine coordinates $y_1/w, y_2/w$ of $(w \neq 0) \cong \mathbb{A}^2$, so that $\pi(u_1, u_2, w_1, \ldots, w_{d-1}) = (u_1, u_2)$. Let $c := 1 + u_1^{kb} + u_2^{kb}$. Then $C_{\mathbb{P}^2} \cap (w \neq 0) = (c = 0)$, and

$$\pi^* c = -(w_1^d + \cdots + w_{d-1}^d) \quad \text{on } X \backslash S. \tag{5.4}$$

Since $p \in C$, we have $\pi(p) \in C_{\mathbb{P}^2}$ (Construction-Proposition 3.2(3)). Since $C_{\mathbb{P}^2}$ is smooth (Construction-Proposition 3.2(4)), the differential of c does not vanish at $\pi(p)$. Possibly exchanging u_1 and u_2, we may assume that $\frac{\partial c}{\partial u_1}(\pi(p)) \neq 0$. Let $T \subset (w \neq 0)$ be an affine open neighborhood of $\pi(p)$ such that $\frac{\partial c}{\partial u_1}$ is nowhere zero on T.

Let $\widetilde{T} := (h^d = -c) \subset T \times \mathbb{A}^1$, where h is the coordinate of $\mathbb{A}^1$. Since $\frac{\partial c}{\partial u_1}$ is nowhere zero on T, $\widetilde{T}$ is smooth, u_2, h form a system of local coordinates at any point of $\widetilde{T}$, and $du_2 \wedge dh$ is a nowhere vanishing 2-form on $\widetilde{T}$. On $\widetilde{T}$ we have

$$d \cdot h^{d-1} dh + \frac{\partial c}{\partial u_1} du_1 + \frac{\partial c}{\partial u_2} du_2 = 0,$$

hence

$$du_1 \wedge du_2 = -d \cdot \Big(\frac{\partial c}{\partial u_1}\Big)^{-1} h^{d-1} dh \wedge du_2.$$

Thus the ramification divisor of the finite flat surjective morphism $\widetilde{T} \to T$ is $(d-1)(h=0)$. Since $c(\pi(p))=0$, the point $\pi(p)$ has exactly one preimage $\tilde{t}$ in $\widetilde{T}$, namely the point with $h=0$. Since $\widetilde{T} \to T$ is finite and flat, any connected component of $\widetilde{T}$ maps onto T and hence contains $\tilde{t}$. Thus $\widetilde{T}$ is connected, and $\widetilde{T}$ is a smooth affine surface.

Let

$$Y := Y_{d,-1} = (-\xi_0^d + \xi_1^d + \cdots + \xi_{d-1}^d = 0) \subset \mathbb{P}^{d-1}$$

be the hypersurface of Construction 4.2, where $d \geq 3b \geq 12$. Then Y is smooth and $\dim Y = d-2$. Let $\eta_j := \xi_j/\xi_0$ for $1 \leq j \leq d-1$. Let $g : \widetilde{T} \times Y \dashrightarrow X$ be the rational map given on $\widetilde{T} \times (Y \cap (\xi_0 \neq 0))$ by

$$(u_1, u_2, w_1, \ldots, w_{d-1}) = (u_1, u_2, h\eta_1, \ldots, h\eta_{d-1}),$$

where u_1, u_2, h are regarded as functions on $\widetilde{T}$. The image is contained in $X \backslash S$ because

$$1 + u_1^{kb} + u_2^{kb} + \sum_{j=1}^{d-1} (h\eta_j)^d = c + h^d = 0$$

on $\widetilde{T} \times (Y \cap (\xi_0 \neq 0))$. By construction, $\pi \circ g$ is the composition of pr_1 and $\widetilde{T} \to T$. The rational map g is dominant: for a point of $\pi^{-1}(T) \cap (X \backslash S)$ with $\pi^* c \neq 0$, take h with $h^d = -c$ and $\eta_j := w_j/h$.

Let X_F be a normal variety with a proper birational morphism $X_F \to X$ such that F is a prime divisor on X_F, and let X' be the normalization of the closure of the graph of the induced rational map $\widetilde{T} \times Y \dashrightarrow X_F$. Let $\varphi : X' \to \widetilde{T} \times Y$ and $\psi : X' \to X$ be the induced morphisms. Then φ is proper and birational, and $\psi = g \circ \varphi$. The morphism $X' \to X_F$ is dominant, and it is proper over the preimage of $\pi^{-1}(T)$ in X_F because $\widetilde{T} \times Y \to T$ is proper. Since F intersects the preimage of $\pi^{-1}(T)$, some irreducible component F' of the preimage of F in X' maps onto F, and F' is a prime divisor on X' (since F' maps onto F, $\dim F' \geq d-1$, and $F' \neq X'$ since $X' \to X_F$ is dominant). Since the local ring of X_F at the generic point of F is a discrete valuation ring, g_* maps the divisorial valuation given by F' to e times the divisorial valuation given by F, where e is the multiplicity along F' of the pullback of a local equation of F at the generic point of F. Since $\psi(F') = p$, and since $\pi \circ \psi$ and the composition of $\mathrm{pr}_1 \circ \varphi$ with $\widetilde{T} \to T$ coincide on $\psi^{-1}(X \backslash V)$, the center $\varphi(F')$ of F' on $\widetilde{T} \times Y$ is contained in $\{\tilde{t}\} \times Y$.

Let $U := \pi^{-1}(T) \cap (X \backslash S)$, an open neighborhood of p. On $X \backslash S$, the partial derivative of the equation of Construction-Proposition 3.2(1) with respect to u_1 is $\pi^* \frac{\partial c}{\partial u_1}$, which is nowhere zero on U. Thus $u_2, w_1, \ldots, w_{d-1}$ form a system of local coordinates at any point of U, and

$$\Omega := du_2 \wedge dw_1 \wedge \cdots \wedge dw_{d-1}$$

is a nowhere vanishing d-form on U. By Construction-Proposition 5.8(2) with $N := d$ and $Z := \widetilde{T}$, we have

$$g^*(dw_1 \wedge \cdots \wedge dw_{d-1}) = d \cdot h^{d-2} dh \wedge \omega_Y$$

on $\widetilde{T} \times (Y \cap (\xi_0 \neq 0))$. Therefore

$$g^*\Omega = d \cdot h^{d-2} \, \mathrm{pr}_1^*(du_2 \wedge dh) \wedge \mathrm{pr}_2^* \omega_Y$$

as rational d-forms on $\widetilde{T} \times Y$. Since $du_2 \wedge dh$ is nowhere vanishing on $\widetilde{T}$ and ω_Y is nowhere vanishing on Y by Construction-Proposition 5.8(1), we obtain

$$\mathrm{div}(g^*\Omega) = (d-2) \, \mathrm{div}(\mathrm{pr}_1^* h) \quad \text{on } \widetilde{T} \times Y.$$

Let v' be the pushforward under pr_1 of the divisorial valuation given by F'. Since $\varphi(F') \subset \{\tilde{t}\} \times Y$ and $h(\tilde{t}) = 0$, we have $v'(h) > 0$, and by Lemma 2.12, v' is a divisorial valuation over $\widetilde{T}$ with center $\tilde{t}$. By Lemma 5.7,

$$ea(F, X, 0) = a(F', \widetilde{T} \times Y, 0) + (d-2)v'(h). \tag{5.5}$$

Step 3. In this step we bound $a(F,X,0)-\lambda\,\mathrm{mult}_F\,\mathcal{M}$ from below by plane invariants of v_F, using Lemma 5.4.

Let $W_{\mathcal{M}}\subset H^0(X,\mathcal{O}_X(qrH))$ be the subspace of sections defining the members of $\mathcal{M}$. Then $\dim W_{\mathcal{M}}<+\infty$. For $f\in W_{\mathcal{M}}$, let

$$f=\sum_J x^{s_J}z^J P_{f,J}(x^r,y_1,y_2)$$

be the expression of Construction-Proposition 5.3(1), and let J_* be the maximum of $|J|$ over all $f\in W_{\mathcal{M}}$ and J with $P_{f,J}\neq 0$. By Construction-Proposition 5.3(2), $bJ_*\le qr$, so $\lambda J_*\le r/b$.

On $X\backslash S$, x^{qr} is a nowhere vanishing section of $\mathcal{O}_X(qrH)$. Since $z^J=x^{b|J|}w^J$, where $w^J:=w_1^{j_1}\cdots w_{d-1}^{j_{d-1}}$, and

$$P_{f,J}(x^r,y_1,y_2)=x^{qr-s_J-b|J|}P_{f,J}(1,u_1,u_2)$$

by homogeneity, we have

$$\frac{f}{x^{qr}}=\sum_J w^J P_{f,J}(1,u_1,u_2)\quad\text{on } X\backslash S.$$

Let

$$\tilde f:=\sum_J h^{|J|}P_{f,J}(1,u_1,u_2)\xi^J\xi_0^{J_*-|J|}\in H^0(\widetilde{T}\times Y,\mathrm{pr}_2^*\,\mathcal{O}_Y(J_*)),$$

where $\xi^J:=\xi_1^{j_1}\cdots\xi_{d-1}^{j_{d-1}}$. Then $g^*(f/x^{qr})=\tilde f/\xi_0^{J_*}$. The monomials $\xi^J\xi_0^{J_*-|J|}$ with $|J|\le J_*$ and $j_{d-1}<d$ are linearly independent in $H^0(Y,\mathcal{O}_Y(J_*))$: a homogeneous polynomial of degree J_* vanishing on Y is a multiple of $-\xi_0^d+\xi_1^d+\cdots+\xi_{d-1}^d$, which is monic of degree d in ξ_{d-1}, while a non-zero linear combination of such monomials has degree less than d in ξ_{d-1}. Let

$$W:=\{\tilde f\mid f\in W_{\mathcal{M}}\}\subset H^0(\widetilde{T}\times Y,\mathrm{pr}_2^*\,\mathcal{O}_Y(J_*)).$$

Since g is dominant, the linear map $f\to\tilde f$ is injective, so $W\neq 0$ and $\dim W<+\infty$. By the linear independence above, the ideal $\mathfrak{a}\subset\mathcal{O}(\widetilde{T})$ of Lemma 5.4 for W is generated by $h^{|J|}P_{f,J}(1,u_1,u_2)$ for all $f\in W_{\mathcal{M}}$ and J.

For a non-zero $f\in W_{\mathcal{M}}$ with divisor $D_f\in\mathcal{M}$, we have $D_f=\mathrm{div}(f/x^{qr})$ on $X\backslash S$. Since g_* maps the divisorial valuation given by F' to e times the divisorial valuation given by F,

$$e\,\mathrm{mult}_F D_f=\mathrm{mult}_{F'}\,\mathrm{div}\Big(\frac{\tilde f}{\xi_0^{J_*}}\Big)=\mathrm{mult}_{F'}\,\mathrm{div}(\tilde f)-J_*\,\mathrm{mult}_{F'}\,\mathrm{pr}_2^*(\xi_0=0),$$

where $\mathrm{div}(\tilde f)$ is the divisor of the section $\tilde f$ and $(\xi_0=0)$ is the hyperplane section of Y. Taking the minimum over all non-zero $f\in W_{\mathcal{M}}$, we obtain

$$e\,\mathrm{mult}_F\,\mathcal{M}=\min_{0\neq\sigma\in W}\mathrm{mult}_{F'}\,\mathrm{div}(\sigma)-J_*\,\mathrm{mult}_{F'}\,\mathrm{pr}_2^*(\xi_0=0).\tag{5.6}$$

By Set-up 3.1, we have

$$\lambda J_*\le\frac{r}{b}\le\frac{d-3}{d}.$$

By Lemma 5.4 with $N:=d$, $\widetilde{T}$ in place of T, $m:=J_*$, and v the divisorial valuation given by F', we obtain

$$a(F',\widetilde{T}\times Y,0)-\lambda\min_{0\neq\sigma\in W}\mathrm{mult}_{F'}\,\mathrm{div}(\sigma)\ge a(v',\widetilde{T},0)-\lambda v'(\mathfrak{a}).$$

Combining the last inequality with (5.5) and (5.6), and dropping the term $\lambda J_*\,\mathrm{mult}_{F'}\,\mathrm{pr}_2^*(\xi_0=0)\ge 0$, we obtain

$$e\big(a(F,X,0)-\lambda\,\mathrm{mult}_F\,\mathcal{M}\big)\ge a(v',\widetilde{T},0)-\lambda v'(\mathfrak{a})+(d-2)v'(h).\tag{5.7}$$

We compute the right-hand side of (5.7). Since $\pi\circ g$ is the composition of pr_1 and $\widetilde{T}\to T$, the pushforward of v' under $\widetilde{T}\to T$ is ev_F. Since the ramification divisor of $\widetilde{T}\to T$ is $(d-1)(h=0)$, Lemma 2.13 implies that

$$a(v',\widetilde{T},-(d-1)(h=0))=a(ev_F,T,0)=e\tau,$$

that is, $a(v',\widetilde{T},0)=e\tau-(d-1)v'(h)$. Let

$$m_2:=v_F(c).$$

Since $h^d=-c$ on $\widetilde{T}$, we have $d\cdot v'(h)=em_2$.

Let $I_{\pi(p)}$ be the ideal of Construction-Proposition 5.3(4) for $\mathcal{M}\subset|qrH|$, and let

$$m_0:=\frac{1}{q}v_F(I_{\pi(p)}).$$

Taking the linear form w, which does not vanish at $\pi(p)$, the ideal $I_{\pi(p)}$ is generated by $P_{f,J}(1,u_1,u_2)$ for all $f\in W_{\mathcal{M}}$ and J with $P_{f,J}\neq 0$. Choose such f and J with $v_F(P_{f,J}(1,u_1,u_2))=qm_0$. Since $|J|\le J_*\le qr/b$ and $\frac{r}{b}\cdot\frac{1}{d}=\frac{1}{kb}$,

$$\lambda v'(\mathfrak{a})\le\lambda v'\bigl(h^{|J|}P_{f,J}(1,u_1,u_2)\bigr)=\lambda|J|\frac{em_2}{d}+em_0\le e\Bigl(m_0+\frac{m_2}{kb}\Bigr).$$

Combining the last two computations with (5.7) and dividing by e, we obtain

$$a(F,X,0)-\lambda\operatorname{mult}_F\mathcal{M}\ge\tau-m_0-\Bigl(\frac{1}{d}+\frac{1}{kb}\Bigr)m_2. \tag{5.8}$$

Step 4. In this step we apply the plane estimates and conclude.

By Construction-Proposition 5.3(4), $I_{\pi(p)}$ is generated by the local equations at $\pi(p)$ of the members of a linear system $\Lambda_{\pi(p)}$ of plane curves of degree at most q without fixed components. Thus Lemma 5.2 implies that $\tau\ge\frac{2}{q}v_F(I_{\pi(p)})=2m_0$. Since $C_{\mathbb{P}^2}$ is smooth, the pair $(T,C_{\mathbb{P}^2}\cap T)$ is lc [KM98, Corollary 2.31(3)], and c is a local equation of $C_{\mathbb{P}^2}\cap T$. Thus $a(v_F,T,C_{\mathbb{P}^2}\cap T)=\tau-m_2\ge 0$. Hence

$$\tau-m_0\ge\frac{\tau}{2}\ge\frac{m_2}{2}.$$

Since $p\in C$, each w_j is a regular function on $X\setminus S$ vanishing at p, so $\operatorname{mult}_F\operatorname{div}(w_j)\ge 1$. By (5.4), $m_2=\operatorname{mult}_F\operatorname{div}(\pi^*c)\ge d$. Since $kb>kr=d\ge(r+2)r\ge 15$, we have $\frac{1}{2}-\frac{1}{d}-\frac{1}{kb}>\frac{1}{2}-\frac{2}{15}>0$. By (5.8),

$$a(F,X,0)-\lambda\operatorname{mult}_F\mathcal{M}\ge\Bigl(\frac{1}{2}-\frac{1}{d}-\frac{1}{kb}\Bigr)m_2\ge\Bigl(\frac{1}{2}-\frac{1}{d}-\frac{1}{kb}\Bigr)d=\frac{d-2}{2}-\frac{r}{b},$$

where we use $\frac{d}{kb}=\frac{r}{b}$. Since $\lambda\operatorname{mult}_F\mathcal{M}=\frac{r}{\ell}\operatorname{mult}_F\mathcal{M}$, Proposition 5.9 follows. □

5.4. **Quotient points.** In this subsection, we prove Proposition 5.10, which bounds the log discrepancy of F when p is one of the cyclic quotient points of Σ, and we complete the proof of Theorem 5.1. Part (d) of the sketch of the proof in Section 1 outlines the proof.

Proposition 5.10. *Notation and conditions as in Set-up 5.5. Assume that $p\in\Sigma$. Then*

$$a(F,X,0)-\frac{r}{\ell}\operatorname{mult}_F\mathcal{M}\ge\frac{k-1}{r}-\frac{1}{b}.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we make ℓ divisible by r and push the valuation of F forward to $\mathbb{P}^2$.

By Definition-Lemma 5.6(1), $\mathcal{M}^{[r]}\subset|r\ell H|$ is a mobile linear system and

$$\frac{r}{r\ell}\operatorname{mult}_F\mathcal{M}^{[r]}=\frac{r}{\ell}\operatorname{mult}_F\mathcal{M}.$$

Possibly replacing $\mathcal{M}$ with $\mathcal{M}^{[r]}$, we may assume that $\ell = qr$ for a positive integer q. Let $\lambda := 1/q$. Then $\frac{r}{\ell}\operatorname{mult}_F \mathcal{M} = \lambda \operatorname{mult}_F \mathcal{M}$. Let v_F be the divisorial valuation over $\mathbb{P}^2$ of Definition-Lemma 5.6(2), and let

$$\tau := a(v_F, \mathbb{P}^2, 0).$$

Step 2. In this step we construct a finite cover $\widetilde{T}$ of a neighborhood T of $\pi(p)$, branched along $(w = 0)$ and $C_{\mathbb{P}^2}$, and compute log discrepancies on $\widetilde{T} \times Y_{d,-1}$ by Lemma 5.7.

By Proposition 3.3(2), $p = [0 : 1 : t : 0 : \cdots : 0]$ for some $t \in \mathbb{C}$ with $t^{kb} = -1$. Thus $p \in U_1$ and $\pi(p) \in (y_1 \neq 0) \subset \mathbb{P}^2$. We denote by w, y_2 the affine coordinates $w/y_1, y_2/y_1$ of $(y_1 \neq 0) \cong \mathbb{A}^2$, and let $c := w^{kb} + 1 + y_2^{kb}$. Then $C_{\mathbb{P}^2} \cap (y_1 \neq 0) = (c = 0)$ and $w(\pi(p)) = c(\pi(p)) = 0$. By Construction-Proposition 3.2(4), $(w = 0)$ and $C_{\mathbb{P}^2}$ intersect transversally at $\pi(p)$, so $dw \wedge dc$ does not vanish at $\pi(p)$. Let $T \subset (y_1 \neq 0)$ be an affine open neighborhood of $\pi(p)$ such that $dw \wedge dc$ is nowhere zero on T. In particular, $(w, c) : T \to \mathbb{A}^2$ is étale, and $(w = 0) + C_{\mathbb{P}^2} \cap T$ is an snc divisor on T.

Let $\widetilde{T} := (x^r = w,\ h^d = -c) \subset T \times \mathbb{A}^2$, where x, h are the coordinates of $\mathbb{A}^2$. The morphism $\widetilde{T} \to \mathbb{A}^2$ given by (x, h) is the base change of the étale morphism $(w, c) : T \to \mathbb{A}^2$ by $(x, h) \to (x^r, -h^d)$, so it is étale. Thus $\widetilde{T}$ is smooth and $dx \wedge dh$ is a nowhere vanishing 2-form on $\widetilde{T}$. On $\widetilde{T}$ we have

$$dw \wedge dc = -r \cdot d \cdot x^{r-1} h^{d-1} dx \wedge dh,$$

so the ramification divisor of the finite flat surjective morphism $\widetilde{T} \to T$ is

$$(r - 1)(x = 0) + (d - 1)(h = 0).$$

The point $\pi(p)$ has exactly one preimage $\tilde{t}$ in $\widetilde{T}$, namely the point with $x = h = 0$. Since $\widetilde{T} \to T$ is finite and flat, any connected component of $\widetilde{T}$ maps onto T and hence contains $\tilde{t}$. Thus $\widetilde{T}$ is connected, and $\widetilde{T}$ is a smooth affine surface.

Let

$$Y := Y_{d,-1} = (-\xi_0^d + \xi_1^d + \cdots + \xi_{d-1}^d = 0) \subset \mathbb{P}^{d-1}$$

be the hypersurface of Construction 4.2, where $d \geq 3b \geq 12$. Then Y is smooth and $\dim Y = d-2$. Let $\eta_j := \xi_j/\xi_0$ for $1 \leq j \leq d-1$. Let $\tilde{g} : \widetilde{T} \times Y \dashrightarrow \widetilde{U}_1$ be the rational map given on $\widetilde{T} \times (Y \cap (\xi_0 \neq 0))$ by

$$(x, y_2, z_1, \ldots, z_{d-1}) = (x, y_2, h\eta_1, \ldots, h\eta_{d-1}),$$

where x, y_2, h are regarded as functions on $\widetilde{T}$. The image is contained in $\widetilde{U}_1$ because $bd = r \cdot kb$ and

$$x^{bd} + 1 + y_2^{kb} + \sum_{j=1}^{d-1} (h\eta_j)^d = w^{kb} + 1 + y_2^{kb} + h^d = c + h^d = 0$$

on $\widetilde{T} \times (Y \cap (\xi_0 \neq 0))$. Let $g : \widetilde{T} \times Y \dashrightarrow X$ be the composition of $\tilde{g}$ with the quotient map $\widetilde{U}_1 \to \widetilde{U}_1/\mu_r$, where $\widetilde{U}_1/\mu_r \cong U_1 \subset X$ (Construction-Proposition 3.2(5)). Since

$$\pi([x : 1 : y_2 : z_1 : \cdots : z_{d-1}]) = [x^r : 1 : y_2],$$

the composition $\pi \circ g$ is the composition of pr_1 and $\widetilde{T} \to T$. The rational map g is dominant: for a point of $\widetilde{U}_1$ over T with $c \neq 0$, take h with $h^d = -c$ and $\eta_j := z_j/h$.

Let X_F be a normal variety with a proper birational morphism $X_F \to X$ such that F is a prime divisor on X_F, and let X' be the normalization of the closure of the graph of the induced rational map $\widetilde{T} \times Y \dashrightarrow X_F$. Let $\varphi : X' \to \widetilde{T} \times Y$ and $\psi : X' \to X$ be the induced morphisms. Then φ is proper and birational, and $\psi = g \circ \varphi$. The morphism $X' \to X_F$ is dominant, and it is proper over the preimage of $\pi^{-1}(T)$ in X_F because $\widetilde{T} \times Y \to T$ is proper. Since F intersects the preimage of $\pi^{-1}(T)$, some irreducible component F' of the preimage of F in X' maps onto F, and F' is a prime divisor on X' (since F' maps onto F, $\dim F' \geq d-1$, and $F' \neq X'$ since $X' \to X_F$ is dominant). Since the local ring of X_F at the generic point of F is a discrete valuation ring, g_*

maps the divisorial valuation given by F' to e times the divisorial valuation given by F, where e is the multiplicity along F' of the pullback of a local equation of F at the generic point of F. Since $\psi(F') = p$, and since $\pi \circ \psi$ and the composition of $\mathrm{pr}_1 \circ \varphi$ with $\widetilde{T} \to T$ coincide on $\psi^{-1}(X \backslash V)$, the center $\varphi(F')$ of F' on $\widetilde{T} \times Y$ is contained in $\{\tilde{t}\} \times Y$.

Let $\widetilde{U}_1^\circ := \widetilde{U}_1 \cap (y_2 \neq 0)$, which is μ_r-invariant, and let $U := U_1 \cap (y_2 \neq 0)$ be the image of $\widetilde{U}_1^\circ$ in U_1, an open neighborhood of p. On $\widetilde{U}_1^\circ$, the partial derivative of the equation of $\widetilde{U}_1$ with respect to y_2 is $kby_2^{kb-1} \neq 0$. Thus $x, z_1, \dots, z_{d-1}$ form a system of local coordinates at any point of $\widetilde{U}_1^\circ$, and

$$\widetilde{\Omega} := dx \wedge dz_1 \wedge \cdots \wedge dz_{d-1}$$

is a nowhere vanishing d-form on $\widetilde{U}_1^\circ$. A non-trivial element of μ_r fixes a point of $\widetilde{U}_1$ only if $x = z_1 = \cdots = z_{d-1} = 0$ at that point, so μ_r acts freely outside the finitely many points of $\widetilde{U}_1$ over Σ, and $\widetilde{U}_1 \to U_1$ is étale in codimension one. Since $\zeta \in \mu_r$ multiplies x and each z_j by ζ, and $d = kr$, we have

$$\zeta^* \widetilde{\Omega} = \zeta^d \widetilde{\Omega} = \widetilde{\Omega}.$$

By Proposition 3.3(2), the smooth locus of U is $U \backslash \Sigma$, because $V \cap U_1 = \emptyset$. Hence $\widetilde{\Omega}$ descends to a nowhere vanishing d-form Ω on $U \backslash \Sigma$, and $g^*\Omega = \tilde{g}^* \widetilde{\Omega}$.

By Construction-Proposition 5.8(2) with $N := d$ and $Z := \widetilde{T}$, we have

$$\tilde{g}^*(dz_1 \wedge \cdots \wedge dz_{d-1}) = d \cdot h^{d-2} dh \wedge \omega_Y$$

on $\widetilde{T} \times (Y \cap (\xi_0 \neq 0))$. Therefore

$$g^*\Omega = \tilde{g}^* \widetilde{\Omega} = d \cdot h^{d-2} \, \mathrm{pr}_1^*(dx \wedge dh) \wedge \mathrm{pr}_2^* \, \omega_Y$$

as rational d-forms on $\widetilde{T} \times Y$. Since $dx \wedge dh$ is nowhere vanishing on $\widetilde{T}$ and ω_Y is nowhere vanishing on Y by Construction-Proposition 5.8(1), we obtain

$$\mathrm{div}(g^*\Omega) = (d-2) \, \mathrm{div}(\mathrm{pr}_1^* \, h) \quad \text{on } \widetilde{T} \times Y.$$

In particular, $\mathrm{div}(g^*\Omega)$ has no component along $\mathrm{pr}_1^*(x = 0)$. The ramification of $\widetilde{T} \to T$ along $(x = 0)$ enters only in **Step 3**, through Lemma 2.13.

Let v' be the pushforward under pr_1 of the divisorial valuation given by F'. Since $\varphi(F') \subset \{\tilde{t}\} \times Y$ and $h(\tilde{t}) = 0$, we have $v'(h) > 0$, and by Lemma 2.12, v' is a divisorial valuation over $\widetilde{T}$ with center $\tilde{t}$. By Lemma 5.7,

$$ea(F, X, 0) = a(F', \widetilde{T} \times Y, 0) + (d-2)v'(h). \tag{5.9}$$

Step 3. In this step we bound $a(F, X, 0) - \lambda \, \mathrm{mult}_F \mathcal{M}$ from below by plane invariants of v_F, using Lemma 5.4.

Let $W_{\mathcal{M}} \subset H^0(X, \mathcal{O}_X(qrH))$ be the subspace of sections defining the members of $\mathcal{M}$. Then $\dim W_{\mathcal{M}} < +\infty$. For $f \in W_{\mathcal{M}}$, let

$$f = \sum_J x^{s_J} z^J P_{f,J}(x^r, y_1, y_2)$$

be the expression of Construction-Proposition 5.3(1), and let J_* be the maximum of $|J|$ over all $f \in W_{\mathcal{M}}$ and J with $P_{f,J} \neq 0$. By Construction-Proposition 5.3(2), $bJ_* \leq qr$, so $\lambda J_* \leq r/b$.

On U_1, y_1^q is a nowhere vanishing section of $\mathcal{O}_X(qrH)$. The pullback of f/y_1^q to $\widetilde{U}_1$ is $\sum_J x^{s_J} z^J P_{f,J}(x^r, 1, y_2)$. Let

$$\tilde{f} := \sum_J x^{s_J} h^{|J|} P_{f,J}(w, 1, y_2) \xi^J \xi_0^{J_* - |J|} \in H^0(\widetilde{T} \times Y, \mathrm{pr}_2^* \, \mathcal{O}_Y(J_*)),$$

where $\xi^J := \xi_1^{j_1} \cdots \xi_{d-1}^{j_{d-1}}$ and $w = x^r$ on $\widetilde{T}$. Then $g^*(f/y_1^q) = \tilde{f}/\xi_0^{J_*}$. The monomials $\xi^J \xi_0^{J_* - |J|}$ with $|J| \leq J_*$ and $j_{d-1} < d$ are linearly independent in $H^0(Y, \mathcal{O}_Y(J_*))$: a homogeneous polynomial

of degree J_* vanishing on Y is a multiple of $-\xi_0^d+\xi_1^d+\cdots+\xi_{d-1}^d$, which is monic of degree d in ξ_{d-1}, while a non-zero linear combination of such monomials has degree less than d in ξ_{d-1}. Let

$$W := \{\tilde f \mid f \in W_{\mathcal M}\} \subset H^0(\widetilde T \times Y, \mathrm{pr}_2^*\,\mathcal O_Y(J_*)).$$

Since g is dominant, the linear map $f \to \tilde f$ is injective, so $W \neq 0$ and $\dim W < +\infty$. By the linear independence above, the ideal $\mathfrak a \subset \mathcal O(\widetilde T)$ of Lemma 5.4 for W is generated by $x^{s_J}h^{|J|}P_{f,J}(w,1,y_2)$ for all $f \in W_{\mathcal M}$ and J.

For a non-zero $f \in W_{\mathcal M}$ with divisor $D_f \in \mathcal M$, we have $D_f = \mathrm{div}(f/y_1^q)$ on U_1. Since g_* maps the divisorial valuation given by F' to e times the divisorial valuation given by F,

$$e\,\mathrm{mult}_F D_f = \mathrm{mult}_{F'}\,\mathrm{div}\Big(\frac{\tilde f}{\xi_0^{J_*}}\Big) = \mathrm{mult}_{F'}\,\mathrm{div}(\tilde f) - J_*\,\mathrm{mult}_{F'}\,\mathrm{pr}_2^*(\xi_0=0),$$

where $\mathrm{div}(\tilde f)$ is the divisor of the section $\tilde f$ and $(\xi_0=0)$ is the hyperplane section of Y. Taking the minimum over all non-zero $f \in W_{\mathcal M}$, we obtain

$$e\,\mathrm{mult}_F \mathcal M = \min_{0\neq\sigma\in W} \mathrm{mult}_{F'}\,\mathrm{div}(\sigma) - J_*\,\mathrm{mult}_{F'}\,\mathrm{pr}_2^*(\xi_0=0). \tag{5.10}$$

By Set-up 3.1, we have

$$\lambda J_* \le \frac{r}{b} \le \frac{d-3}{d}.$$

By Lemma 5.4 with $N := d$, $\widetilde T$ in place of T, $m := J_*$, and v the divisorial valuation given by F', we obtain

$$a(F', \widetilde T \times Y, 0) - \lambda \min_{0\neq\sigma\in W} \mathrm{mult}_{F'}\,\mathrm{div}(\sigma) \ge a(v', \widetilde T, 0) - \lambda v'(\mathfrak a).$$

Combining the last inequality with (5.9) and (5.10), and dropping the term $\lambda J_*\,\mathrm{mult}_{F'}\,\mathrm{pr}_2^*(\xi_0=0) \ge 0$, we obtain

$$e\big(a(F,X,0) - \lambda\,\mathrm{mult}_F\,\mathcal M\big) \ge a(v',\widetilde T,0) - \lambda v'(\mathfrak a) + (d-2)v'(h). \tag{5.11}$$

We compute the right-hand side of (5.11). Since $\pi\circ g$ is the composition of pr_1 and $\widetilde T \to T$, the pushforward of v' under $\widetilde T \to T$ is ev_F. Since the ramification divisor of $\widetilde T \to T$ is $(r-1)(x=0)+(d-1)(h=0)$, Lemma 2.13 implies that

$$a\big(v', \widetilde T, -(r-1)(x=0)-(d-1)(h=0)\big) = a(ev_F, T, 0) = e\tau,$$

that is, $a(v',\widetilde T,0) = e\tau - (r-1)v'(x) - (d-1)v'(h)$. Let

$$m_1 := v_F(w), \qquad m_2 := v_F(c).$$

Since $x^r = w$ and $h^d = -c$ on $\widetilde T$, we have $rv'(x) = em_1$ and $d\cdot v'(h) = em_2$.

Let $I_{\pi(p)}$ be the ideal of Construction-Proposition 5.3(4) for $\mathcal M \subset |qrH|$, and let

$$m_0 := \frac{1}{q} v_F(I_{\pi(p)}).$$

Taking the linear form y_1, which does not vanish at $\pi(p)$, the ideal $I_{\pi(p)}$ is generated by $P_{f,J}(w,1,y_2)$ for all $f \in W_{\mathcal M}$ and J with $P_{f,J} \neq 0$. Choose such f and J with $v_F(P_{f,J}(w,1,y_2)) = qm_0$. Since $0 \le s_J \le r-1$, $|J| \le J_* \le qr/b$, and $\frac{r}{b}\cdot\frac{1}{d} = \frac{1}{kb}$,

$$\lambda v'(\mathfrak a) \le \lambda v'\big(x^{s_J}h^{|J|}P_{f,J}(w,1,y_2)\big) = \lambda s_J\frac{em_1}{r} + \lambda|J|\frac{em_2}{d} + em_0 \le e\Big(m_0 + \frac{r-1}{rq}m_1 + \frac{m_2}{kb}\Big).$$

Combining the last two computations with (5.11) and dividing by e, we obtain

$$a(F,X,0) - \lambda\,\mathrm{mult}_F\,\mathcal M \ge \tau - m_0 - \Big(1-\frac{1}{r}\Big)m_1 - \Big(\frac{1}{d}+\frac{1}{kb}\Big)m_2 - \frac{r-1}{rq}m_1. \tag{5.12}$$

Step 4. In this step we apply the plane estimates.

By Construction-Proposition 5.3(4), $I_{\pi(p)}$ is generated by the local equations at $\pi(p)$ of the members of a linear system $\Lambda_{\pi(p)}$ of plane curves of degree at most q without fixed components.

The line $(w=0)$ passes through $\pi(p)$, and w is a local equation of $(w=0)$ at $\pi(p)$. Thus Lemma 5.2 implies that

$$\tau \geq 2m_0 \quad \text{and} \quad \tau - m_0 \geq m_1.$$

By the choice of T, the pair $(T, (w=0) + C_{\mathbb{P}^2} \cap T)$ is snc, hence lc [KM98, Corollary 2.31(3)], and w and c are local equations of $(w=0)$ and $C_{\mathbb{P}^2} \cap T$. Thus

$$a(v_F, T, (w=0) + C_{\mathbb{P}^2} \cap T) = \tau - m_1 - m_2 \geq 0.$$

Hence

$$\tau - m_0 \geq \max\Big\{m_1, \frac{\tau}{2}\Big\} \geq \max\Big\{m_1, \frac{m_1+m_2}{2}\Big\}.$$

By (5.12), we obtain

$$a(F,X,0) - \lambda \operatorname{mult}_F \mathcal{M} \geq \max\Big\{m_1, \frac{m_1+m_2}{2}\Big\} - \Big(1-\frac{1}{r}\Big)m_1 - \Big(\frac{1}{d}+\frac{1}{kb}\Big)m_2 - \frac{r-1}{rq}m_1. \tag{5.13}$$

Step 5. In this step we remove the last term of (5.13) and conclude.

Let N be a positive integer. By Definition-Lemma 5.6(1), $\mathcal{M}^{[N]} \subset |NqrH|$ is a mobile linear system and

$$\frac{1}{Nq} \operatorname{mult}_F \mathcal{M}^{[N]} = \lambda \operatorname{mult}_F \mathcal{M}.$$

The quantities τ, m_1, and m_2 depend only on F. Applying **Steps 3 and 4** to $\mathcal{M}^{[N]} \subset |NqrH|$ in place of $\mathcal{M}$ (**Step 2** does not depend on $\mathcal{M}$, and m_0 does not appear in (5.13)), we obtain (5.13) with q replaced by Nq, where the left-hand side is again $a(F,X,0) - \lambda \operatorname{mult}_F \mathcal{M}$. Letting N tend to infinity, we obtain

$$a(F,X,0) - \lambda \operatorname{mult}_F \mathcal{M} \geq \max\Big\{m_1, \frac{m_1+m_2}{2}\Big\} - \Big(1-\frac{1}{r}\Big)m_1 - \Big(\frac{1}{d}+\frac{1}{kb}\Big)m_2. \tag{5.14}$$

Case 1. Suppose that $m_1 \geq m_2$. Then $m_1 \geq \frac{m_1+m_2}{2}$, and the right-hand side of (5.14) is

$$\frac{m_1}{r} - \Big(\frac{1}{d}+\frac{1}{kb}\Big)m_2 \geq \Big(\frac{1}{r}-\frac{1}{d}-\frac{1}{kb}\Big)m_2.$$

Case 2. Suppose that $m_1 \leq m_2$. Then $\frac{m_1+m_2}{2} \geq m_1$, and the right-hand side of (5.14) is

$$\Big(\frac{1}{r}-\frac{1}{2}\Big)m_1 + \Big(\frac{1}{2}-\frac{1}{d}-\frac{1}{kb}\Big)m_2 \geq \Big(\frac{1}{r}-\frac{1}{d}-\frac{1}{kb}\Big)m_2,$$

because $r \geq 3$ implies $\frac{1}{r} - \frac{1}{2} < 0$, and $m_1 \leq m_2$.

In both cases,

$$a(F,X,0) - \lambda \operatorname{mult}_F \mathcal{M} \geq \Big(\frac{1}{r}-\frac{1}{d}-\frac{1}{kb}\Big)m_2, \tag{5.15}$$

and $\frac{1}{r} - \frac{1}{d} - \frac{1}{kb} > \frac{1}{r} - \frac{2}{kr} > 0$ because $kb > kr = d$ and $k \geq r+2 > 2$.

It remains to bound m_2 from below. For $1 \leq j \leq d-1$, the function z_j^r/y_1^b is regular on U_1 and vanishes at p, so $\operatorname{mult}_F \operatorname{div}(z_j^r/y_1^b) \geq 1$. On U_1 we have $\pi^*w = x^r/y_1$ and $\pi^*y_2 = y_2/y_1$. Since $bd = r \cdot kb$ and $d = kr$, dividing the equation of X by y_1^{kb} shows that

$$\pi^*c = \frac{x^{bd} + y_1^{kb} + y_2^{kb}}{y_1^{kb}} = -\sum_{j=1}^{d-1}\Big(\frac{z_j^r}{y_1^b}\Big)^k \quad \text{on } U_1.$$

Hence $m_2 = \operatorname{mult}_F \operatorname{div}(\pi^*c) \geq k$. The bound $m_2 \geq k$ is attained: if F is the exceptional divisor of the weighted blow-up of X at p with weights $\frac{1}{r}(1,\dots,1)$ on $x, z_1, \dots, z_{d-1}$, then $m_2 = k$. By (5.15),

$$a(F,X,0) - \lambda \operatorname{mult}_F \mathcal{M} \geq \Big(\frac{1}{r}-\frac{1}{d}-\frac{1}{kb}\Big)k = \frac{k-1}{r} - \frac{1}{b},$$

where we use $\frac{k}{d} = \frac{1}{r}$ and $\frac{k}{kb} = \frac{1}{b}$. Since $\lambda \operatorname{mult}_F \mathcal{M} = \frac{r}{\ell} \operatorname{mult}_F \mathcal{M}$, Proposition 5.10 follows. □

Proof of Theorem 5.1. Let F be an exceptional prime divisor over X, and suppose that

$$\frac{r}{\ell}\operatorname{mult}_F \mathcal{M} > a(F, X, 0) - 1.$$

By Proposition 4.1, $\operatorname{center}_X F$ is a closed point of C, so the conditions of Set-up 5.5 hold for ℓ, $\mathcal{M}$, and F. Since $d = kr \geq (r+2)r$ and $b = r+1$,

$$\frac{d-2}{2} - \frac{r}{b} \geq \frac{r^2+2r-2}{2} - \frac{r}{r+1} \geq \frac{23}{4} > 1,$$

where the middle expression is increasing in r and equals $\frac{23}{4}$ at $r = 3$. Since $k \geq r+2$,

$$\frac{k-1}{r} - \frac{1}{b} \geq \frac{r+1}{r} - \frac{1}{r+1} = 1 + \frac{1}{r(r+1)} > 1.$$

Thus Proposition 5.9 and Proposition 5.10 imply that $a(F, X, 0) - \frac{r}{\ell}\operatorname{mult}_F \mathcal{M} > 1$, a contradiction. Therefore $a(F, X, 0) - \frac{r}{\ell}\operatorname{mult}_F \mathcal{M} \geq 1$ for any exceptional prime divisor F over X, that is, $(X, \frac{r}{\ell}\mathcal{M})$ is canonical. □

6. An icosahedral quadric threefold

In this section, we prove Proposition 6.5, which Section 7 uses to prove Theorem 1.6.

Set-up 6.1. Let $2.\mathfrak{A}_5 \subset \mathrm{SL}_2(\mathbb{C})$ be the preimage of an icosahedral subgroup $\mathfrak{A}_5 \subset \mathrm{PGL}_2(\mathbb{C})$. Let $Q \subset \mathbb{P}(\mathrm{Mat}_2(\mathbb{C}) \oplus \mathbb{C})$ be the quadric threefold defined by $\det(M) = t^2$, where $[M : t]$ with $M \in \mathrm{Mat}_2(\mathbb{C})$ and $t \in \mathbb{C}$ denotes a point of $\mathbb{P}(\mathrm{Mat}_2(\mathbb{C}) \oplus \mathbb{C})$. Let $G := (2.\mathfrak{A}_5 \times 2.\mathfrak{A}_5)/\langle(-1,-1)\rangle$ act on $\mathbb{P}(\mathrm{Mat}_2(\mathbb{C}) \oplus \mathbb{C})$ by $(g_1, g_2) \cdot [M : t] := [g_1 M g_2^{-1} : t]$. Let H be a hyperplane section of Q, and let $S := Q \cap (t = 0)$.

Remark 6.2. The image in G of the diagonal subgroup $\{(g, g) \mid g \in 2.\mathfrak{A}_5\}$ fixes the point $[I : 1] \in Q$, where I is the identity matrix, and the projection from $[I : 1]$ is an equivariant birational map from Q to $\mathbb{P}^3$. For any non-zero column vector $u \in \mathbb{C}^2$, the image in G of $2.\mathfrak{A}_5 \times \{1\}$ preserves the line $L_u \subset S$ consisting of the points $[vu^T : 0]$ with $v \in \mathbb{C}^2 \backslash \{0\}$, and the projection from L_u induces an equivariant $\mathbb{P}^1$-bundle $\mathrm{Bl}_{L_u} Q \to \mathbb{P}^2$ (we omit the standard verifications, which use that Q is a smooth quadric threefold with $\mathrm{Cl}(Q) = \mathbb{Z}[H]$ by Lemma 6.3(1): the projection from a point of a smooth quadric threefold is birational onto $\mathbb{P}^3$, the blow-up of a smooth quadric threefold along a line is a $\mathbb{P}^1$-bundle over $\mathbb{P}^2$ given by the planes through the line, and both $\mathbb{P}^3 \to \mathrm{pt}$ and $\mathrm{Bl}_{L_u} Q \to \mathbb{P}^2$ are equivariant Mori fiber spaces in the sense of Definition 2.7, since their total spaces are smooth, $-K$ is relatively ample, and the respective subgroup acts trivially on $\mathrm{Cl}(\mathbb{P}^3)$, which is generated by a plane, and on $\mathrm{Cl}(\mathrm{Bl}_{L_u} Q)$, which is freely generated by the pullback of H and the exceptional divisor). Thus, for either subgroup, Q is equivariantly birational to another equivariant Mori fiber space. In particular, Q is not birationally superrigid with respect to the image of $2.\mathfrak{A}_5 \times \{1\}$, nor, by the symmetric construction with the lines $[uv^T : 0]$, with respect to the image of $\{1\} \times 2.\mathfrak{A}_5$, in the sense of Definition 2.7. Since the diagonal subgroup contains $(-1,-1)$, the image in G of the diagonal subgroup is isomorphic to $\mathfrak{A}_5$, and, by the projection from $[I : 1]$, Q is not birationally superrigid in the sense of Definition 2.7 with respect to it either.

Lemma 6.3. *Notation and conditions as in Set-up 6.1. Then the following hold.*

1. *Q is a smooth Fano threefold, $-K_Q \sim 3H$, and $\mathrm{Cl}(Q) = \mathbb{Z}[H]$.*
2. *The action of G on Q is faithful, and $|G| = 7200$.*
3. *S is a G-invariant smooth surface, and there exists an isomorphism $S \cong \mathbb{P}^1 \times \mathbb{P}^1$ under which $H|_S \sim \mathcal{O}(1,1)$ and the image of G in $\mathrm{Aut}(S)$ is $\mathfrak{A}_5 \times \mathfrak{A}_5$ acting factorwise via the icosahedral action on $\mathbb{P}^1$.*

Proof. (1) Write

$$M = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}.$$

Then Q is the hypersurface in

$$\mathbb{P}^4 = \mathbb{P}(\mathrm{Mat}_2(\mathbb{C}) \oplus \mathbb{C})$$

defined by

$$x_{11}x_{22} - x_{12}x_{21} - t^2 = 0.$$

Since the quadratic form $x_{11}x_{22} - x_{12}x_{21} - t^2$ in the five variables $x_{11}, x_{12}, x_{21}, x_{22}, t$ has rank 5, Q is a smooth quadric threefold. By the adjunction formula,

$$K_Q \sim (K_{\mathbb{P}^4} + Q)|_Q \sim (-5+2)H = -3H.$$

Thus $-K_Q \sim 3H$ is ample, and Q is a smooth Fano threefold. By [Har77, Chapter II, Exercise 6.5(c) and Corollary 6.16], $\mathrm{Cl}(Q) = \mathrm{Pic}(Q) = \mathbb{Z}[H]$.

(2) For any $g_1, g_2 \in \mathrm{SL}_2(\mathbb{C})$, we have $\det(g_1 M g_2^{-1}) = \det(M)$, and the action fixes the coordinate t. Thus the action of $2.\mathfrak{A}_5 \times 2.\mathfrak{A}_5$ on $\mathbb{P}(\mathrm{Mat}_2(\mathbb{C}) \oplus \mathbb{C})$ given by $(g_1, g_2) \cdot [M : t] = [g_1 M g_2^{-1} : t]$ preserves Q. Since $(-1)M(-1)^{-1} = M$, the element $(-1,-1)$ acts trivially, so G acts on Q.

Suppose that $(g_1, g_2) \in 2.\mathfrak{A}_5 \times 2.\mathfrak{A}_5$ acts trivially on Q. For any $M \in \mathrm{SL}_2(\mathbb{C})$, the point $[M : 1]$ belongs to Q, and $[g_1 M g_2^{-1} : 1] = [M : 1]$. Since the last coordinates of both sides are equal to 1, we have

$$g_1 M g_2^{-1} = M \quad \text{for any } M \in \mathrm{SL}_2(\mathbb{C}).$$

Taking for M the identity matrix, we get $g_1 = g_2$. Thus g_1 commutes with any $M \in \mathrm{SL}_2(\mathbb{C})$. Since g_1 commutes with

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

g_1 is a scalar matrix, and $\det(g_1) = 1$ implies that $g_1 = \pm 1$. Hence the kernel of the action of $2.\mathfrak{A}_5 \times 2.\mathfrak{A}_5$ on Q is $\langle (-1,-1) \rangle$, and G acts faithfully on Q. Since the natural map

$$\mathrm{SL}_2(\mathbb{C}) \to \mathrm{PGL}_2(\mathbb{C})$$

is surjective with kernel $\{\pm 1\}$ and $|\mathfrak{A}_5| = 60$, we have $|2.\mathfrak{A}_5| = 120$, so

$$|G| = \frac{120^2}{2} = 7200.$$

(3) Since the action of G fixes the coordinate t, S is G-invariant. The surface S is the quadric

$$(\det(M) = 0) \subset \mathbb{P}(\mathrm{Mat}_2(\mathbb{C})) \cong \mathbb{P}^3,$$

and the quadratic form $x_{11}x_{22} - x_{12}x_{21}$ has rank 4, so S is smooth. Let

$$J := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and define

$$\varphi : \mathbb{P}^1 \times \mathbb{P}^1 \to S, \quad [v] \times [w] \to [v(Jw)^T : 0],$$

where $v, w \in \mathbb{C}^2 \backslash \{0\}$ are column vectors. A non-zero matrix $M \in \mathrm{Mat}_2(\mathbb{C})$ satisfies $\det(M) = 0$ if and only if $M = vu^T$ for some non-zero column vectors $v, u \in \mathbb{C}^2$, and $[v]$, $[u]$ are determined by $[M]$. Thus the Segre embedding $[v] \times [u] \to [vu^T]$ is an isomorphism from $\mathbb{P}^1 \times \mathbb{P}^1$ onto S, and φ is the composition of the Segre embedding with the automorphism $[w] \to [Jw]$ of the second factor. Hence φ is an isomorphism. Since the entries of $v(Jw)^T$ are bilinear in (v, w), we have $\varphi^*(H|_S) \sim \mathcal{O}(1,1)$.

For any $g \in \mathrm{SL}_2(\mathbb{C})$, we have $(g^{-1})^T = JgJ^{-1}$. Thus, for any $g_1, g_2 \in 2.\mathfrak{A}_5$,

$$g_1 v(Jw)^T g_2^{-1} = g_1 v\big((g_2^{-1})^T Jw\big)^T = g_1 v\big(Jg_2 J^{-1} Jw\big)^T = (g_1 v)(Jg_2 w)^T,$$

that is,

$$(g_1, g_2) \cdot \varphi([v] \times [w]) = \varphi([g_1 v] \times [g_2 w]).$$

Hence, under φ, the element (g_1, g_2) acts on $\mathbb{P}^1 \times \mathbb{P}^1$ by the images of g_1 and g_2 in $\mathrm{PGL}_2(\mathbb{C})$ on the first and second factor respectively. Since the image of $2.\mathfrak{A}_5$ in $\mathrm{PGL}_2(\mathbb{C})$ is $\mathfrak{A}_5$, the image of G in $\mathrm{Aut}(S)$ is $\mathfrak{A}_5 \times \mathfrak{A}_5$ acting factorwise via the icosahedral action on $\mathbb{P}^1$. □

In the proof of Proposition 6.5, we exclude G-invariant curves $\Gamma \subset Q$ with no component contained in S and with $H \cdot \Gamma \leq 72$, G-invariant curves in S of bidegree (e_1, e_2) with $\max\{e_1, e_2\} \leq 6$, and non-empty G-invariant zero-dimensional subschemes of Q of length at most 55. By Lemma 6.4, the corresponding lower bounds are 144, 12, and 120.

Lemma 6.4. *Notation and conditions as in Set-up 6.1. Then the following hold.*

(1) Any G-orbit in Q consists of at least 120 points, and any G-orbit in S consists of at least 144 points.
(2) If $\Gamma \subset Q$ is a G-invariant reduced curve and no component of Γ is contained in S, then $H \cdot \Gamma \geq 144$.
(3) If $\Gamma \subset S$ is a G-invariant reduced curve and $\Gamma \sim \mathcal{O}(e_1, e_2)$ under the isomorphism in Lemma 6.3(3), then $\max\{e_1, e_2\} \geq 12$.

Proof. We first show that any orbit of the icosahedral action of $\mathfrak{A}_5$ on $\mathbb{P}^1$ consists of at least 12 points. Let $G_0 \subset \mathrm{PGL}_2(\mathbb{C})$ be a finite subgroup fixing a point $q \in \mathbb{P}^1$. We show that G_0 is cyclic. Choose an affine coordinate z on $\mathbb{P}^1$ such that $q = +\infty$. Then any element $h \in G_0$ acts on $\mathbb{P}^1 \backslash \{q\} = \mathbb{C}$ by $z \to a_h z + b_h$ for some $a_h \in \mathbb{C}^*$ and $b_h \in \mathbb{C}$. Let $\Omega \subset \mathbb{C}$ be a G_0-orbit, and let

$$z_1 := \frac{1}{|\Omega|} \sum_{z \in \Omega} z.$$

Since any $h \in G_0$ permutes Ω,

$$h(z_1) = \frac{1}{|\Omega|} \sum_{z \in \Omega} (a_h z + b_h) = \frac{1}{|\Omega|} \sum_{z \in \Omega} h(z) = z_1.$$

Thus G_0 fixes z_1 and ∞. In the coordinate $z - z_1$, any element of G_0 acts by $z \to a_h z$, so G_0 is isomorphic to a finite subgroup of $\mathbb{C}^*$, which is cyclic. Since the orders of the elements of $\mathfrak{A}_5$ are 1, 2, 3, and 5, the stabilizer in $\mathfrak{A}_5$ of any point of $\mathbb{P}^1$ is a cyclic group of order at most 5. Hence any $\mathfrak{A}_5$-orbit in $\mathbb{P}^1$ consists of at least $60/5 = 12$ points.

(1) Let $\varphi : \mathbb{P}^1 \times \mathbb{P}^1 \to S$ be the isomorphism in Lemma 6.3(3). By Lemma 6.3(3), the G-orbit of $\varphi(p_1 \times p_2)$ is $\varphi(\Omega_1 \times \Omega_2)$, where Ω_i is the $\mathfrak{A}_5$-orbit of $p_i \in \mathbb{P}^1$. Thus any G-orbit in S consists of at least $12 \cdot 12 = 144$ points.

Let $x \in Q \backslash S$. Then $x = [M : 1]$ for some $M \in \mathrm{SL}_2(\mathbb{C})$. The G-orbit of x coincides with the $2.\mathfrak{A}_5 \times 2.\mathfrak{A}_5$-orbit of x. If $(g_1, g_2) \in 2.\mathfrak{A}_5 \times 2.\mathfrak{A}_5$ fixes x, then $[g_1 M g_2^{-1} : 1] = [M : 1]$, so $g_1 M g_2^{-1} = M$ and $g_1 = M g_2 M^{-1}$. Thus the projection $(g_1, g_2) \to g_2$ is injective on the stabilizer of x, and the stabilizer has order at most 120. Hence the G-orbit of x consists of at least $120^2/120 = 120$ points.

(2) Since $S \sim H$ is ample, $(S \cdot \Gamma') = (H \cdot \Gamma') > 0$ for any component Γ' of Γ, so $\Gamma \cap S \neq \emptyset$. Since no component of Γ is contained in S, $S|_\Gamma \geq 0$ is a Cartier divisor on Γ with support $\Gamma \cap S$, and $\Gamma \cap S$ is finite. Since Γ and S are G-invariant, $\Gamma \cap S$ is a non-empty union of G-orbits in S, so $\#(\Gamma \cap S) \geq 144$ by (1). Thus

$$H \cdot \Gamma = S \cdot \Gamma = \deg(S|_\Gamma) \geq \#(\Gamma \cap S) \geq 144.$$

(3) We identify S with $\mathbb{P}^1 \times \mathbb{P}^1$ by Lemma 6.3(3), and let $\mathrm{pr}_1, \mathrm{pr}_2$ be the projections, so that

$$\mathcal{O}(e_1, e_2) = \mathrm{pr}_1^* \mathcal{O}(e_1) \otimes \mathrm{pr}_2^* \mathcal{O}(e_2).$$

For a general point $p \in \mathbb{P}^1$, the fiber $F_p := \{p\} \times \mathbb{P}^1$ is not a component of Γ, so $\Gamma|_{F_p} \geq 0$ is a divisor on $F_p \cong \mathbb{P}^1$ with

$$\deg(\Gamma|_{F_p}) = (\Gamma \cdot F_p) = e_2.$$

For a general point $p \in \mathbb{P}^1$, the curve $\mathbb{P}^1 \times \{p\}$ is not a component of Γ either, so

$$e_1 = (\Gamma \cdot (\mathbb{P}^1 \times \{p\})) = \deg(\Gamma|_{\mathbb{P}^1 \times \{p\}}) \geq 0.$$

Since $\Gamma \neq 0$, $(e_1, e_2) \neq (0,0)$. By symmetry, we may assume that $e_2 > 0$. The subgroup $\{1\} \times \mathfrak{A}_5$ of the image of G in $\mathrm{Aut}(S)$ preserves F_p and Γ, so it preserves $\Gamma|_{F_p}$, and it acts on $F_p \cong \mathbb{P}^1$ via the icosahedral action. Since $\deg(\Gamma|_{F_p}) = e_2 > 0$, the support of $\Gamma|_{F_p}$ is a non-empty $\mathfrak{A}_5$-invariant subset of F_p, so it contains an $\mathfrak{A}_5$-orbit, which consists of at least 12 points. Thus $e_2 \geq 12$. □

Proposition 6.5. *Notation and conditions as in Set-up 6.1. Let ℓ be a positive integer and $\mathcal{M} \subset |\ell H|$ a G-invariant mobile linear system. Then $(Q, \frac{6}{\ell}\mathcal{M})$ is klt.*

Proof. Let $c := 6/\ell$. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce Proposition 6.5 to the equality $\mathcal{J} = \mathcal{O}_Q$ for a G-invariant ideal sheaf $\mathcal{J}$ which is trivial outside a closed subset of codimension at least 2.

Let $\mu : Q' \to Q$ be a log resolution of $\mathcal{M}$, and write $\mu^*\mathcal{M} = \mathcal{M}' + \Phi$, where $\mathcal{M}'$ is a base point free linear system and Φ is the fixed divisor of $\mu^*\mathcal{M}$. Let

$$\mathcal{J} := \mathcal{J}(Q, c \cdot \mathcal{M}) = \mu_* \mathcal{O}_{Q'}(K_{Q'/Q} - \lfloor c\Phi \rfloor)$$

be the multiplier ideal of $c \cdot \mathcal{M}$ [Laz04, Definition 9.2.10]. Since $\mathcal{M}$ has no fixed component, the base locus of $\mathcal{M}$ has codimension at least 2 in Q, so $\mu(\mathrm{Supp}\,\Phi)$ has codimension at least 2 in Q, and $\mu(\mathrm{Exc}(\mu))$ has codimension at least 2 in Q since Q is normal [Har77, Chapter V, Lemma 5.1]. Thus $\mu(\mathrm{Exc}(\mu) \cup \mathrm{Supp}\,\Phi)$ has codimension at least 2 in Q. Since $K_{Q'/Q} - \lfloor c\Phi \rfloor$ is supported on $\mathrm{Exc}(\mu) \cup \mathrm{Supp}\,\Phi$, $\mathcal{J}$ coincides with $\mathcal{O}_Q$ outside a closed subset of Q of codimension at least 2. The ideal $\mathcal{J}$ does not depend on the log resolution μ [Laz04, Theorem 9.2.18], and $g^*\mathcal{M} = \mathcal{M}$ for any $g \in G$, so $g^*\mathcal{J} = \mathcal{J}$ for any $g \in G$.

Let $s > c$ be an integer which is large enough that Lemma 2.4 holds for $\mathcal{M}$, $\lambda = c$, and s, let $D_1, \dots, D_s \in \mathcal{M}$ be general members, and let

$$B := \frac{c}{s}(D_1 + \cdots + D_s) \sim_{\mathbb{Q}} \frac{c}{s} \cdot s\ell H = 6H.$$

Since Q is smooth, by [Laz04, Proposition 9.2.26], $\mathcal{J} = \mathcal{J}(Q, B)$, where $\mathcal{J}(Q, B)$ is the multiplier ideal of the pair (Q, B) [CS08, Definition 2.5]. Let $\mathcal{L}(Q, B)$ be the subscheme defined by $\mathcal{J}(Q, B)$, and let $\mathrm{LCS}(Q, B)$ be the locus of log canonical singularities of (Q, B) in the sense of [CS08, Section 2], which is the set of points of Q at which (Q, B) is not klt. For any open subset $U \subset Q$, $\mathcal{J}(Q, B)|_U = \mathcal{O}_U$ if and only if $(U, B|_U)$ is klt, by the equality $\mathrm{Supp}\,\mathcal{L}(Q, B) = \mathrm{LCS}(Q, B)$ following [CS08, Definition 2.5] and the characterization of klt pairs following [CS08, Definition 2.2]. By Lemma 2.4, (Q, B) is klt if and only if $(Q, c\mathcal{M})$ is klt. Thus it suffices to show that $\mathcal{J} = \mathcal{O}_Q$.

Step 2. In this step we show that $\mathrm{Supp}(\mathcal{O}_Q/\mathcal{J})$ contains no curve.

Suppose that $\mathrm{Supp}(\mathcal{O}_Q/\mathcal{J})$ contains an irreducible curve C_0. Let

$$m := \min\{\mathrm{mult}_{C_0} D \mid D \in \mathcal{M}\}.$$

The members $D \in \mathcal{M}$ with $\mathrm{mult}_{C_0} D \geq m + 1$ form a proper linear subsystem of $\mathcal{M}$. Since $\mathcal{J}$, hence C_0, does not depend on the choice of $D_1, \dots, D_s$, and $D_1, \dots, D_s$ are general members of $\mathcal{M}$, we may assume that $\mathrm{mult}_{C_0} D_i = m$ for any i.

We show that

$$cm \geq 1. \tag{6.1}$$

Suppose that $cm < 1$. Let $p \in C_0$ be a general closed point. Then $\mathrm{mult}_p D_i = \mathrm{mult}_{C_0} D_i = m$ for any i, so

$$\mathrm{mult}_p B = \frac{c}{s} \sum_{i=1}^{s} \mathrm{mult}_p D_i = cm < 1.$$

Since Q is smooth at p, by [Kol96, 8.10 Lemma, (8.10.1)] (see also [KOW18, Lemma 2.2]), (Q, B) is klt on a neighborhood U of p. By **Step 1**, $\mathcal{J}|_U = \mathcal{O}_U$, which contradicts $p \in C_0 \subset \mathrm{Supp}(\mathcal{O}_Q/\mathcal{J})$. Thus (6.1) holds.

Let Γ be the reduced union of the curves $g(C_0)$, $g \in G$. Then Γ is a G-invariant reduced curve, and G permutes the components of Γ transitively. Since $g^*\mathcal{M} = \mathcal{M}$ for any $g \in G$, we have

$$\mathrm{mult}_{\Gamma_1} D \geq m \quad \text{for any component } \Gamma_1 \text{ of } \Gamma \text{ and any } D \in \mathcal{M}.$$

Let $D', D'' \in \mathcal{M}$ be general members. Since $\mathcal{M}$ has no fixed component, D' and D'' have no common component, so the intersection cycle $D' \cdot D''$ is an effective 1-cycle on Q. Let Γ_1 be a component of Γ, let $H_0 \in |H|$ be a general member, and let $p \in \Gamma_1 \cap H_0$. Since H_0 is general, H_0 is a smooth surface, and the coefficient of Γ_1 in $D' \cdot D''$ is equal to the local intersection multiplicity $i(p, D'|_{H_0} \cdot D''|_{H_0}; H_0)$ of the curves $D'|_{H_0}$ and $D''|_{H_0}$ at p on H_0 (since H_0 is general, p is a smooth point of Γ_1 not contained in any other component of $D' \cap D''$, and H_0 intersects Γ_1 transversally at p by [Har77, Chapter III, Corollary 10.9 and Remark 10.9.2] applied to the smooth locus of Γ_1, and the equality then follows from [Ful98, Example 7.1.10(a) and Proposition 7.1(b)] and [Mat86, Theorems 14.7 and 17.11]). By [Ful98, Corollary 12.4],

$$\begin{aligned} i(p, D'|_{H_0} \cdot D''|_{H_0}; H_0) &\geq \mathrm{mult}_p(D'|_{H_0}) \cdot \mathrm{mult}_p(D''|_{H_0}) \\ &\geq \mathrm{mult}_p D' \cdot \mathrm{mult}_p D'' \geq (\mathrm{mult}_{\Gamma_1} D') \cdot (\mathrm{mult}_{\Gamma_1} D'') \geq m^2. \end{aligned}$$

Thus $D' \cdot D'' - m^2\Gamma$ is an effective 1-cycle. Since $Q \subset \mathbb{P}^4$ is a quadric, $(H^3) = 2$. Since H is ample,

$$2\ell^2 = \ell^2(H^3) = H \cdot D' \cdot D'' \geq m^2(H \cdot \Gamma).$$

By (6.1), $m \geq \ell/6$, so

$$H \cdot \Gamma \leq \frac{2\ell^2}{m^2} \leq 72. \tag{6.2}$$

Since S is G-invariant and G permutes the components of Γ transitively, either no component of Γ is contained in S, or $\Gamma \subset S$.

Case 1. No component of Γ is contained in S. By Lemma 6.4(2), $H \cdot \Gamma \geq 144$, which contradicts (6.2).

Case 2. $\Gamma \subset S$. Write $\Gamma \sim \mathcal{O}(e_1, e_2)$ under the isomorphism in Lemma 6.3(3). Since $\mathcal{M}$ has no fixed component, a general member $D \in \mathcal{M}$ does not contain S, so $D|_S \geq 0$ is a divisor on S with

$$D|_S \sim \ell H|_S \sim \mathcal{O}(\ell, \ell).$$

Since $\mathrm{mult}_{\Gamma_1} D \geq m$ for any component Γ_1 of Γ, the coefficient of Γ_1 in $D|_S$ is at least m, so $D|_S - m\Gamma \geq 0$ is a divisor on S with

$$D|_S - m\Gamma \sim \mathcal{O}(\ell - me_1, \ell - me_2).$$

Intersecting with general fibers of both projections $S \cong \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$, we get $\ell - me_1 \geq 0$ and $\ell - me_2 \geq 0$. By Lemma 6.4(3), $\max\{e_1, e_2\} \geq 12$, so $\ell \geq 12m$ and

$$cm = \frac{6m}{\ell} \leq \frac{1}{2},$$

which contradicts (6.1).

Thus $\mathrm{Supp}(\mathcal{O}_Q/\mathcal{J})$ contains no curve. By **Step 1**, $\mathrm{Supp}(\mathcal{O}_Q/\mathcal{J})$ contains no divisor either. Hence the closed subscheme $W \subset Q$ defined by $\mathcal{J}$ is either empty or zero-dimensional.

Step 3. In this step we show that $W = \emptyset$, which implies that $\mathcal{J} = \mathcal{O}_Q$.

By Lemma 6.3(1) and **Step 1**,

$$4H - (K_Q + B) \sim_{\mathbb{Q}} 4H - (-3H + 6H) = H,$$

which is ample, hence nef and big. By the Nadel–Shokurov vanishing theorem [CS08, Theorem 2.6], applied to the pair (Q, B) and the Cartier divisor $4H$,

$$H^1(Q, \mathcal{J} \otimes \mathcal{O}_Q(4H)) = 0.$$

From the exact sequence

$$0 \to \mathcal{J} \otimes \mathcal{O}_Q(4H) \to \mathcal{O}_Q(4H) \to \mathcal{O}_W(4H) \to 0,$$

the restriction map $H^0(Q, \mathcal{O}_Q(4H)) \to H^0(W, \mathcal{O}_W(4H))$ is surjective. Since W is empty or zero-dimensional, $h^0(W, \mathcal{O}_W(4H)) = \operatorname{length}(W)$. From the exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^4}(2) \to \mathcal{O}_{\mathbb{P}^4}(4) \to \mathcal{O}_Q(4H) \to 0$$

and $H^1(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(2)) = 0$, we get

$$h^0(Q, \mathcal{O}_Q(4H)) = h^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(4)) - h^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(2)) = 70 - 15 = 55.$$

Thus

$$\operatorname{length}(W) \leq 55.$$

Suppose that $W \neq \emptyset$. Since $\mathcal{J}$ is G-invariant, $\operatorname{Supp} W$ is a non-empty finite G-invariant subset of Q, so it contains a G-orbit. By Lemma 6.4(1), $\operatorname{length}(W) \geq \#\operatorname{Supp} W \geq 120$, a contradiction. Thus $W = \emptyset$ and $\mathcal{J} = \mathcal{O}_Q$. By **Step 1**, $(Q, \frac{6}{\ell}\mathcal{M})$ is klt. □

7. Proofs of the main theorems

In this section, we prove Theorem 1.4, Corollary 1.5, and Theorem 1.6.

Proof of Theorem 1.4. We use the notation of Set-up 3.1. The variety X, the class H, and the prime divisor S in Theorem 1.4 are the ones in Set-up 3.1. By Proposition 3.3(1), X is a terminal $\mathbb{Q}$-factorial Fano variety of Picard number one, $\dim X = d$, $\operatorname{Cl}(X) = \mathbb{Z}[H]$, and $-K_X \sim rH$.

Let $\mathcal{M}$ be a mobile linear system on X and $\lambda > 0$ a rational number such that $K_X + \lambda\mathcal{M} \sim_{\mathbb{Q}} 0$. Since $\operatorname{Cl}(X) = \mathbb{Z}[H]$ and $-K_X \sim rH$, we have $\mathcal{M} \subset |\ell H|$ for some integer $\ell \geq 1$, and $\lambda = \frac{r}{\ell}$. By Theorem 5.1, $(X, \frac{r}{\ell}\mathcal{M})$ is canonical. By Lemma 2.4, the pair

$$\left(X, \frac{r}{\ell s}(D_1 + \cdots + D_s)\right)$$

is canonical for any sufficiently large integer s and general members $D_1, \ldots, D_s \in \mathcal{M}$, that is, $(X, \frac{r}{\ell}\mathcal{M})$ is canonical in the sense of [SZ19, Section 2.1]. Since $\mathcal{M}$ is arbitrary, the Noether–Fano criterion [SZ19, Theorem 2.3] (see also [CS08, Theorem 1.26]) implies that X is birationally superrigid.

Finally, $rS \sim rH \sim -K_X$ since $S \sim H$, and $\operatorname{lct}(X; rS) = \frac{1}{r}$ by Proposition 3.3(3). Since $rS \geq 0$ and $rS \sim_{\mathbb{Q}} -K_X$, Definition 2.2 implies

$$\alpha(X) \leq \operatorname{lct}(X; rS) = \frac{1}{r}.$$ □

Proof of Corollary 1.5. (1) Let $r = 3$ and $k = 5$. Then $k \geq r + 2$, $b = 4$, $d = 15$, $kb = 20$, and $bd = 60$. Thus the hypersurface X of Theorem 1.4 is the hypersurface $X_{60} \subset \mathbb{P}(1, 3, 3, 4^{14})$ of (1), and by Theorem 1.4, X_{60} is a birationally superrigid Fano variety such that $\dim X_{60} = 15$ and $\alpha(X_{60}) \leq \frac{1}{3}$.

(2) Let $\epsilon > 0$ be a real number, let $r > \max\{3, \frac{1}{\epsilon}\}$ be an integer, and let X be the hypersurface of Theorem 1.4 for r and $k = r + 2$. By Theorem 1.4, X is a birationally superrigid Fano variety such that

$$\alpha(X) \leq \frac{1}{r} < \epsilon.$$

In [CSZ23, Section 1], where Question 1.2 is posed, a Fano variety X with terminal singularities and $\operatorname{rank} \operatorname{Cl}(X) = 1$ is birationally superrigid if X is not birational to any other Mori fiber space and $\operatorname{Bir}(X) = \operatorname{Aut}(X)$. Let X be the variety of (1) or of (2). Then $\operatorname{Cl}(X) = \mathbb{Z}[H]$ by Theorem 1.4, so $\operatorname{rank} \operatorname{Cl}(X) = 1$. By Proposition 3.3(1), the morphism from X to a point is a Mori fiber space

in the sense of Definition 2.6. Since X is birationally superrigid in the sense of Definition 2.6, any birational self-map of X is an isomorphism, hence $\mathrm{Bir}(X) = \mathrm{Aut}(X)$. Moreover, any birational map from X to a Mori fiber space is an isomorphism, hence X is not birational to any other Mori fiber space. Thus X is birationally superrigid in the sense of [CSZ23, Section 1].

Since $\alpha(X_{60}) \leq \frac{1}{3} < \frac{1}{2}$ by (1), Questions 1.1 and 1.2 have negative answers. □

Proof of Theorem 1.6. We use the notation of Set-up 6.1. In particular, H is a hyperplane section of Q and $S = Q \cap (t = 0)$. By Lemma 6.3(2), the action of G on Q is faithful and $|G| = 7200$. By Lemma 6.3(1), Q is a smooth Fano threefold, $-K_Q \sim 3H$, and $\mathrm{Cl}(Q) = \mathbb{Z}[H]$. In particular, Q is terminal [KM98, Corollary 2.31(2)]. Any element of G is an automorphism of Q, hence maps the ample generator of $\mathrm{Cl}(Q) \cong \mathbb{Z}$ to an ample generator, that is, fixes the class of H. Thus $\mathrm{Cl}(Q)^G = \mathrm{Cl}(Q) = \mathbb{Z}[H]$. Since Q is smooth, any Weil divisor on Q is Cartier [Har77, Chapter II, Proposition 6.11 and Remark 6.11.1A], so Q and G satisfy the assumptions of Definition 2.7.

Let $\mathcal{M}$ be a G-invariant mobile linear system on Q and $\lambda > 0$ a rational number such that $K_Q + \lambda\mathcal{M} \sim_{\mathbb{Q}} 0$. Since $\mathrm{Cl}(Q) = \mathbb{Z}[H]$ and $-K_Q \sim 3H$, we have $\mathcal{M} \subset |\ell H|$ for some integer $\ell \geq 1$, and $\lambda = \frac{3}{\ell}$. By Proposition 6.5, $(Q, \frac{6}{\ell}\mathcal{M})$ is klt. Let E be an exceptional prime divisor over Q. Since Q is smooth, $a(E, Q, 0) \geq 2$ [KM98, Corollary 2.31(2)]. Since $(Q, \frac{6}{\ell}\mathcal{M})$ is klt, $\frac{6}{\ell}\,\mathrm{mult}_E\,\mathcal{M} < a(E, Q, 0)$. Thus

$$\frac{3}{\ell}\,\mathrm{mult}_E\,\mathcal{M} < \frac{1}{2}a(E, Q, 0) \leq a(E, Q, 0) - 1.$$

Since E is arbitrary, $(Q, \frac{3}{\ell}\mathcal{M})$ is canonical (for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $B \geq 0$ on a terminal variety Y and an exceptional prime divisor F over Y with $c = a(F, Y, 0) - 1$, [CSZ23, Remark 3.6] shows that if $a(F, Y, B) < 1$, then $(Y, (1 + \frac{1}{c})B)$ is not lc). By Lemma 2.4, the pair

$$\Big(Q, \frac{3}{\ell s}(D_1 + \cdots + D_s)\Big)$$

is also canonical for any sufficiently large integer s and general members $D_1, \ldots, D_s \in \mathcal{M}$. Since $\mathcal{M}$ is arbitrary, $(Q, \frac{3}{\ell'}\mathcal{M}')$ is canonical for any positive integer ℓ' and any G-invariant mobile linear system $\mathcal{M}' \subset |\ell' H|$, where $\frac{3}{\ell'}\mathcal{M}' \sim_{\mathbb{Q}} -K_Q$. We deduce from the equivariant Noether–Fano inequality [PZ25, Theorem 2.1] that Q is G-birationally superrigid in the sense of Definition 2.7 (cf. [CS16, Theorems 3.2.1(ii) and 3.2.6(ii)], where in a G-Mori fiber space any G-invariant Weil divisor is Cartier and the condition on ranks uses Pic^G instead of Cl^G, a smaller class than in Definition 2.7). Let $f : V \to Z$ be a G-Mori fiber space and $\chi : Q \dashrightarrow V$ a G-birational map. For $V = Q$ and $Z = \mathrm{Spec}\,\mathbb{C}$, χ can be any G-birational self-map of Q. Since G acts faithfully on Q and χ is G-birational, G acts faithfully on V, hence generically freely, as the fixed locus of any non-trivial element of G is a proper closed subset of V. If $\dim Z = 0$, then V is a Fano variety with terminal singularities and $\mathrm{rank}\,\mathrm{Cl}(V)^G = 1$. If $\dim Z > 0$, then f has connected fibers since $f_*\mathcal{O}_V = \mathcal{O}_Z$ [Har77, Chapter III, Corollary 11.3], and a general fiber V_z of f is a Fano variety, since $-K_{V_z} = -K_V|_{V_z}$ by adjunction, $-K_V$ is f-ample, and V_z has terminal singularities by [KM98, Lemma 5.17(1)] applied $\dim Z$ times to general members of $f^*|A|$ for a very ample divisor A on Z. In both cases, the linear system $\mathcal{M}'$ on Q in [PZ25, Theorem 2.1] (denoted $\mathcal{M}_X$ there) is G-invariant, mobile, and contained in $|\ell' H|$ for some positive integer ℓ'. Hence $(Q, \frac{3}{\ell'}\mathcal{M}')$ is canonical, and [PZ25, Theorem 2.1] shows that χ is biregular. If $\dim Z > 0$, then, since χ is a G-equivariant isomorphism, $\mathrm{rank}\,\mathrm{Cl}(Z)^G = \mathrm{rank}\,\mathrm{Cl}(Q)^G - 1 = 0$, which is impossible since the sum of the G-translates of an ample Cartier divisor on Z is G-invariant and has a class of infinite order in $\mathrm{Cl}(Z)$. Hence $\dim Z = 0$, and χ is a G-equivariant isomorphism.

Finally, S is a G-invariant smooth surface by Lemma 6.3(3), and $3S \sim 3H \sim -K_Q$. Since S is a smooth prime divisor on the smooth variety Q, the pair (Q, S) is lc [KM98, Corollary 2.31(3)], and for any real number $\tau > 1$, the pair $(Q, \tau S)$ is not lc since $a(S, Q, \tau S) = 1 - \tau < 0$. Hence

$\mathrm{lct}(Q;3S) = \frac{1}{3}$. Since $3S \geq 0$ is G-invariant and $3S \sim_{\mathbb{Q}} -K_Q$, Definition 2.2 implies

$$\alpha_G(Q) \leq \mathrm{lct}(Q;3S) = \frac{1}{3}.$$

Since $\frac{1}{3} < \frac{1}{2}$ and $\dim Q = 3$, Question 1.3 has a negative answer in dimension three. □

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